

OPTIMISATION

IDRISS MAZARI-FOUQUER



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CONTENTS

General introduction, pre-requisites	3
References	3
Part 1. A review of basic concepts	4
1. Existence of optimisers, optimality conditions	4
2. Convexity, strict convexity	7
Exercises for Part 1	11
Solutions of the exercises of Part 1	13
Part 2. Unconstrained optimisation: approximation and gradient descent	18
3. An overview of gradient descent: basic principle & study of convex functions	18
4. Beyond convexity	25
Exercises for Part 2	27
Solutions of the exercises of Part 2	29
Part 3. Constrained optimisation: theory and approximation	32
5. The Lagrange multiplier rule I: equality constraints and first discussion	32
Exercises for the first part of Part 3	41
Solutions for the first part of Part 3	43
6. The Lagrange multiplier rule II: inequality constraints and the Karush-Kuhn-Tucker theorem	48
Exercises for the second part of Part 3	54
Solutions for the second part of Part 3	56
7. Numerical approximation of constrained problems	62
Exercises for the third part of Part 3	64
Solutions for the third part of Part 3	65
8. Duality: theory and practice	67
Exercises for the fourth part of Part 3	72
Solutions for the fourth part of Part 3	74
Part 4. Calculus of variations & Optimal control	79

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Optimisation-Part # 0

9. General introduction	79
10. The dynamic programming principle	80
Exercises for the first part of Part 4	87
Solutions for the first part of Part 4	89
11. A bit of calculus of variations	95
Exercises for the second part of Part 4	103
Solutions for the second part of Part 4	105
12. Optimal control of differential equations: Hamilton-Jacobi equations & the Pontryagin maximum principle	108
Exercises for the third part of Part 4	113
Solutions for the third part of Part 4	115
Midterms & Exams	124
References	129

GENERAL INTRODUCTION, PRE-REQUISITES

The goal of this course is to provide sound theoretical foundations for continuous optimisation. Optimisation problems arise everywhere, whether it be in medicine, in mechanics, in finance or in artificial intelligence. We will not be considering any specific application as we rather seek to provide a general overview of these problems and of the basic tools required to handle them.

We will consider optimisation problems that write

$$\min_{x \in X} f(x)$$

where X is a subset of a (possibly infinite dimensional) vector space and $f : X \rightarrow \mathbb{R}$ is the so-called “objective function”. The class will proceed along the following lines:

- (1) We will first review basic concepts in optimisation, the main goal being to provide standard tools to establish existence (and possibly uniqueness) of minimisers x^* of f . The key words are coercivity and convexity; a certain familiarity with differential calculus is required, as is a certain dexterity with linear algebra.
- (2) The second part of the class will be dealing with unconstrained finite-dimensional optimisation, that is, when $X = \mathbb{R}^d$ for some $d \geq 1$. The main emphasis of this part will be on approximation procedures, the key concept being that of gradient descent. A reasonable familiarity with differential equations will be expected in the last section of this part.
- (3) The third part of the class will be devoted to constrained optimisation problems, that is, when X is a strict subset of a vector space (still finite-dimensional at this stage). The key-concepts that the student should master at the end of this section are Lagrange multipliers and basic duality. We will also place a great emphasis on approximation methods, the two main algorithms being the projected gradient algorithm and the Uzawa algorithm.
- (4) The final part of this class should be seen as an introduction on more advanced topics in optimisation in the infinite dimensional setting: calculus of variations, optimal control and back-propagation, the latter being of central importance in the study of Neural Networks.

Parts of these lectures are inspired by classes I gave at other points, parts are inspired by the works of my predecessors (P. Cardaliaguet, O. Mula, Y. Viossat). The class should be self-contained and, as usual, there is no need to use other references: the lectures and exercise sessions should provide enough material. Nevertheless, some “standard” references are the following:

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Part 1. A review of basic concepts

1. EXISTENCE OF OPTIMISERS, OPTIMALITY CONDITIONS

Throughout this entire chapter, unless stated otherwise, we consider a fixed, $\mathcal{C}^2$ function $f : \mathbb{R}^d \rightarrow \mathbb{R}$.

1.0.1. *First definitions.* The goal of this section is to fix the terminology, as well as some notations.

Definition 1.1. A point $x^* \in \mathbb{R}^d$ is called:

- (1) A global minimiser of f if

$$\forall x \in \mathbb{R}^d, f(x^*) \leq f(x).$$

- (2) A local minimiser of f if

$$\exists \varepsilon > 0, \forall x \in \mathbb{R}^d, \|x - x^*\| \leq \varepsilon \Rightarrow f(x^*) \leq f(x).$$

In the remainder of this document, we will adopt the following notational conventions:

- (1) $M_d(\mathbb{R})$ denotes the set of $d \times d$ matrices, $M_{p,q}(\mathbb{R})$ denotes the set of $p \times q$ matrices.
- (2) $S_d(\mathbb{R})$ denotes the set of symmetric matrices in $M_d(\mathbb{R})$. The transpose of a matrix M is written M^T .
- (3) $S_d^+(\mathbb{R})$, resp. $S_d^{++}(\mathbb{R})$, resp. $S_d^-(\mathbb{R})$, resp. $S_d^{--}(\mathbb{R})$ denotes the set of symmetric positive, resp. definite positive, resp. negative, resp. symmetric negative, matrices.

1.1. **The optimisation problem under consideration.** The goal of this class is to study the optimisation problem

$$(1.1) \quad \inf_{x \in \mathbb{R}^d} f(x).$$

A first question is whether or not a solution x^* actually exists, in which case we are allowed to write

$$\min_{x \in \mathbb{R}^d} f(x).$$

To this end, let us recall the definition of *coercivity*:

Definition 1.2. We say that a continuous function f is coercive if for any $M \in \mathbb{R}$ the sub-level set $\{x \in \mathbb{R}^d : f(x) \leq M\}$ is bounded.

The main point of coercivity is the following proposition:

Proposition 1.1. Assume f is coercive. Then f has a global minimiser x^* . With a slight abuse of terminology, we will say that x^* solves (5.1), and dub it a minimiser of f in $\mathbb{R}^d$.

Proof of Proposition 1.1. Let $x_0 \in \mathbb{R}^d$ be arbitrary and consider the set

$$E_0 := \{x \in \mathbb{R}^d : f(x) \leq f(x_0)\}.$$

By assumption, E_0 is a compact set and it is clear that if x^* solves

$$(1.2) \quad \inf_{x \in E_0} f(x)$$

then it solves (5.1). However, the existence of a solution to (5.3) follows from the Weierstraßtheorem: every continuous function on a compact set reaches its extrema on this set. $\square$

1.2. Optimality conditions. Now, consider (5.1), and assume that x^* is a solution. Naturally, one would like to have either an explicit or a good enough numerical approximation of the minimiser x^* . Unless we are quite lucky and an easy comparison argument provides an explicit value, this is a hopeless endeavour. The best we can do is to rely on *optimality conditions*. Optimality conditions allow to reduce the search of a minimiser to the resolution of a non-linear system of equations.

There are two optimality conditions. The first-order optimality condition reads

$$(1.3) \quad \nabla f(x^*) = 0$$

while the second-order optimality condition writes

$$\nabla^2 f(x^*) \in S_d^+(\mathbb{R}).$$

Assume for simplicity that (1.3) has a finite number of solutions $x_1, \dots, x_N$, which have tractable expressions. This still does not provide any conclusion, and we need to compute the Hessian of f at each x_k , $k = 1, \dots, N$. There are several possibilities, summarised in the following proposition:

For the sake of future references, let us single out the following definition:

Definition 1.3. Let $f \in \mathcal{C}^1(\mathbb{R}^d; \mathbb{R})$. A point $x^* \in \mathbb{R}^d$ is called a critical point of f if

$$\nabla f(x^*) = 0.$$

Proposition 1.2. Assume f is $\mathcal{C}^2$ and let x^* be a critical point of f .

- (1) If $\nabla^2 f(x^*) \in S_d^{++}(\mathbb{R})$, then x^* is a strict local minimiser of f .
- (2) If $\nabla^2 f(x^*) \in S_d^{--}(\mathbb{R})$, then x^* is a strict local maximiser of f .
- (3) If $\nabla^2 f(x^*)$ has at least one negative and one positive eigenvalue, x^* is a saddle point: there exist two orthogonal directions $\vec{e}_1, \vec{e}_2$ such that $t^* = 0$ is a local minimiser of $t \mapsto f(x + t\vec{e}_1)$, and a local maximiser of $t \mapsto f(x + t\vec{e}_2)$.
- (4) If $\nabla^2 f(x^*) \in S_d^+(\mathbb{R})$, but not in $S_d^{++}(\mathbb{R})$, then we cannot conclude and further analysis is required.

In the first two cases, we say that x^* is a non-degenerate critical point.

Proof of Proposition 1.2. We only prove the first and third points. We first assume that

$$(1.4) \quad \nabla^2 f(x^*) \in S_d^{++}(\mathbb{R}).$$

There are two ways to prove that x^* is a local minimiser, both of which naturally rely on Taylor expansions, and on the following consequence of (1.4) (see Exercise 1.1): there exists a constant $\underline{\lambda} > 0$ such that, for any $z \in \mathbb{R}^d$,

$$(1.5) \quad \langle \nabla^2 f(x^*)z, z \rangle \geq \underline{\lambda} \|z\|^2.$$

- (1) First approach: a proof by contradiction Argue by contradiction and assume that there exists a sequence $\{x_k\}_{k \in \mathbb{N}}$ such that

$$\forall k \in \mathbb{N}, f(x_k) \leq f(x^*).$$

From the mean-value formula, we know that for any $k \in \mathbb{N}$ there exists $\xi_k \in [x_k; x^*] = \{(1-t)x_k + tx^*, t \in [0; 1]\}$ such that

$$f(x_k) = f(x^*) + \frac{1}{2} \langle \nabla^2 f(\xi_k)(x_k - x^*), x_k - x^* \rangle.$$

In particular, setting for any $k \in \mathbb{N}$ $z_k := \frac{x_k - x^*}{\|x_k - x^*\|}$,

$$\langle \nabla^2 f(\xi_k) z_k, z_k \rangle \leq 0.$$

As for any $k \in \mathbb{N}$ we have $\|z_k\| = 1$ we can (up to taking a subsequence) assume that $\{z_k\}_{k \in \mathbb{N}}$ converges to some z_∞ , $\|z_\infty\| = 1$. Since $x_k \xrightarrow[k \rightarrow \infty]{} x^*$, it follows that $\xi_k \xrightarrow[k \rightarrow \infty]{} x^*$. Passing to the limit in the previous inequality we obtain

$$\langle \nabla^2 f(x^*) z_\infty, z_\infty \rangle \leq 0,$$

in contradiction with (1.5).

- (2) Second approach: continuity of the Hessian From (1.5) and the fact that $\nabla^2 f$ is continuous it is possible to show the following fact (see Exercise 1.1): there exists $\underline{\lambda}' > 0$ and $\varepsilon > 0$ such that

$$\forall x \in \mathbb{B}(x^*; \varepsilon), \forall z \in \mathbb{R}^d, \langle \nabla^2 f(x) z, z \rangle \geq \underline{\lambda}' \|z\|^2.$$

We can conclude as before: fix such an $\varepsilon > 0$. Then, for any $x \in \mathbb{B}(x^*; \varepsilon)$, there exists $\xi \in \mathbb{B}(x^*; \varepsilon)$ such that

$$f(x) - f(x^*) = \frac{1}{2} \langle \nabla^2 f(\xi)(x - x^*), (x - x^*) \rangle \geq \frac{\underline{\lambda}'}{2} \|x - x^*\|^2$$

and the conclusion follows. □

The exercises of this chapter (in particular Exercise 1.3) contain several examples of optimisation problems that can be solved by hand. Such examples are usually limited to dimension 2 or 3, unless the problem has a very specific structure.

2. CONVEXITY, STRICT CONVEXITY

The second crucial notion is that of convexity, which has two main interests, as we shall explain:

- (1) The first one is when studying global properties of a functions: in that case, assuming convexity helps in either proving uniqueness of the optimiser (in the strongly convex case) or at least to have some geometric structure on the set of minimisers.
- (2) The second one is more local in nature: consider a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ that admits a non-degenerate local minimum $x^* \in \mathbb{R}^d$. Then, locally around x^* , by Taylor expansions, f can be suitably approximated by a (strongly) convex functions. Thus, we can hope that any global study of convex functions can translate to local studies around non-degenerate local minima (this is of particular importance when dealing with gradient descents).

2.1. Various definitions of convexity and basic properties.

Definition 2.1. A set $K \subset \mathbb{R}^n$ is said to be *convex* if for all x and y in K , $tx + (1-t)y \in K$ for all t in $[0, 1]$ (for any two points in K , the segment that unites them is in K).

Definition 2.2. Let $K \subset \mathbb{R}^n$ be a convex set and $f : K \rightarrow \mathbb{R}$ be a function.

- (1) f is **convex** on K if

$$\forall x, y \in K, t \in (0, 1), f(tx + (1-t)y) \leq tf(x) + (1-t)f(y).$$

- (2) f is **strictly convex** on K if

$$\forall x \neq y \in K, t \in (0, 1), f(tx + (1-t)y) < tf(x) + (1-t)f(y).$$

- (3) f is **strongly convex** on K if there exists $\alpha > 0$ such that

$$\forall x, y \in K, t \in [0, 1], f(tx + (1-t)y) \leq tf(x) + (1-t)f(y) - \frac{\alpha}{2}t(1-t)\|x - y\|^2.$$

- (4) f is said to be **concave** if $-f$ is convex (and similar definitions for strictly or strongly concave).

Of course, these definitions can be a tad annoying to work with. When f is more regular (say, $\mathcal{C}^1$ or $\mathcal{C}^2$), equivalent characterisations are available.

Proposition 2.1 (Equivalent characterisation of convexity in the $\mathcal{C}^1$ regime). Let $f \in \mathcal{C}^1(\mathbb{R}^d; \mathbb{R})$. Then the following propositions are equivalent:

- (1) f is convex.

- (2) (A convex function is above its tangent hyperplane) For any $x, y \in \mathbb{R}^d$,

$$(2.1) \quad f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle.$$

- (3) (The gradient of a convex function is monotone) For any $x, y \in \mathbb{R}^d$,

$$(2.2) \quad \langle \nabla f(y) - \nabla f(x), y - x \rangle \geq 0.$$

Proof of Proposition 2.1. Assume that f is convex and let $x, y \in \mathbb{R}^d$. Thus, for any $t \in [0, 1]$,

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y)$$

or, alternatively,

$$\frac{f(x + t(y - x)) - f(x)}{t} \leq f(y) - f(x).$$

As f is $\mathcal{C}^1$, passing to the limit $t \rightarrow 0$ gives

$$\langle \nabla f(x), y - x \rangle \leq f(y) - f(x),$$

which is exactly (2.1). Now, assuming (2.1), we obtain

$$\langle \nabla f(x), y - x \rangle \leq f(y) - f(x) \leq -\langle \nabla f(y), x - y \rangle$$

or, equivalently,

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq 0.$$

Now, assuming (2.2), let us show (2.1). Let $x, y \in \mathbb{R}^d$. Then, by the Taylor formula,

$$\begin{aligned} f(y) - f(x) &= \int_0^1 \langle \nabla f((1-t)x + ty), y - x \rangle dt \\ &= \int_0^1 \langle \nabla f((1-t)x + ty) - \nabla f(x), y - x \rangle dt + \langle \nabla f(x), y - x \rangle. \end{aligned}$$

However, setting $x_t := (1-t)x + ty$, we have

$$x_t - x = t(y - x).$$

In particular,

$$\langle \nabla f(x_t) - \nabla f(x), y - x \rangle \geq 0$$

whence

$$f(y) - f(x) \geq \langle \nabla f(x), y - x \rangle$$

so that (2.1) is satisfied. Finally, assume (2.1). Let us show that f is convex. Let $x, y \in \mathbb{R}^d$ and $t \in [0; 1]$. Consider the map

$$g : t \mapsto f((1-t)x + ty) - (1-t)f(x) - tf(y).$$

Then $g(0) = g(1) = 0$ and, retaining the notation $x_t := (1-t)x + ty$,

$$g'(t) = \langle \nabla f(x_t), y - x \rangle - (f(y) - f(x)).$$

Observe that for any $t_0, t_1 \in [0; 1]$ there holds

$$\begin{aligned} (t_0 - t_1)(g'(t_0) - g'(t_1)) &= \langle \nabla f(x_{t_0}) - \nabla f(x_{t_1}), (t_0 - t_1)(y - x) \rangle \\ &= \langle \nabla f(x_{t_0}) - \nabla f(x_{t_1}), x_{t_0} - x_{t_1} \rangle \\ &\geq 0 \end{aligned}$$

whence g' is non-decreasing. By the Rolle theorem, there exists $s \in (0; 1)$ such that $g'(s) = 0$. Consequently, g is non-increasing on $(0; s)$ and non-decreasing on $(s; 1)$, and is thus maximal at either $t = 0$ or $t = 1$, thereby concluding the proof of convexity of f . $\square$

We leave as an exercise the following proposition:

Proposition 2.2 (Equivalent characterisation of strong convexity in the $\mathcal{C}^1$ regime). *Let $f \in \mathcal{C}^1(\mathbb{R}^d; \mathbb{R})$. Then the following propositions are equivalent:*

- (1) f is α -strongly convex.
- (2) For any $x, y \in \mathbb{R}^d$,

$$(2.3) \quad f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\alpha}{2} \|y - x\|^2.$$

- (3) For any $x, y \in \mathbb{R}^d$, $x \neq y$

$$(2.4) \quad \langle \nabla f(y) - \nabla f(x), y - x \rangle \geq \alpha \|y - x\|^2.$$

To conclude these reminders, we recall some characterisation of convex $\mathcal{C}^2$ functions:

Proposition 2.3. *Let $f \in \mathcal{C}^2(\mathbb{R}^d; \mathbb{R})$. Then the following properties are equivalent:*

- (1) *f is convex.*
- (2) *For any $x \in \mathbb{R}^d$, $\nabla^2 f(x) \in S_d^+(\mathbb{R})$.*

There is also a nice characterisation of α -strongly convex functions:

Proposition 2.4. *Let $f \in \mathcal{C}^2(\mathbb{R}^d; \mathbb{R})$ and $\alpha > 0$. Then the following properties are equivalent:*

- (1) *f is α -strongly convex.*
- (2) *For any $x \in \mathbb{R}^d$, the lowest eigenvalue $\lambda_1(\nabla^2 f(x))$ of the Hessian of f satisfies*

$$\lambda_1(\nabla^2 f(x)) \geq \alpha.$$

Unfortunately, there is no such nice characterisation of strict convexity, but merely an implication (we also refer to Exercise 1.5):

Proposition 2.5. *Let $f \in \mathcal{C}^2(\mathbb{R}^d; \mathbb{R})$. If*

$$\forall x \in \mathbb{R}^d, \nabla^2 f(x) \in S_d^{++}(\mathbb{R})$$

then f is strictly convex.

Similarly, we leave the proofs of these propositions as exercises.

2.2. Convex function and minimisation. The main result of this section is the following:

Theorem 2.1. *Let f be a convex function, and assume that the problem*

$$(2.5) \quad \min_{x \in \mathbb{R}^d} f(x)$$

has a solution x^ . Then:*

- (1) *If f is strictly convex, (2.5) has a unique solution.*
- (2) *In general, the set of minimisers*

$$X := \{x \in \mathbb{R}^d : f(x) = f(x^*)\}$$

is a convex set.

Proof of Theorem 2.1. We begin with the fact that the set of minimisers X is convex: let x, y be such that $f(x) = f(y) = f(x^*)$. Then, by convexity of f ,

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y) = f(x^*),$$

whence $(1-t)x + ty \in X$. Now, suppose that f is strictly convex and assume by contradiction that f has two distinct minimisers x, y . By strict convexity, for any $t \in (0; 1)$,

$$f((1-t)x + ty) < (1-t)f(x) + tf(y) = f(x^*),$$

in contradiction with the minimality of x^* . □

2.3. Several (useful) inequalities related to convex functions. To conclude the first part, we will give and prove several inequalities related to convex functions—these might seem quite abstruse at first, but they will be coming in handy in later parts of the class, so that this specific section of the lecture notes should be taken as a reference point for later purposes. Furthermore, the proofs allow to get familiar with usual tricks when dealing with convex functions.

Proposition 2.6. *Let f be a convex function and assume that ∇f is μ -Lipschitz continuous. Then*

$$(2.6) \quad \forall x, y \in \mathbb{R}^d, f(x) - f(y) - \langle \nabla f(y), y - x \rangle \geq \frac{1}{2\mu} \|\nabla f(x) - \nabla f(y)\|^2.$$

In particular, if f admits a minimiser x^ , it follows that*

$$(2.7) \quad \forall x \in \mathbb{R}^d, f(x) - f(x^*) \geq \frac{1}{2\mu} \|\nabla f(x)\|^2.$$

This is sometimes referred to as “co-coercivity of the gradient”.

Proof of Proposition 2.6. The key is to introduce an auxiliary point z . Fix $x, y \in \mathbb{R}^d$. Then for any $z \in \mathbb{R}^d$

$$\begin{aligned} f(x) - f(y) &= f(x) - f(z) + f(z) - f(y) \\ &\leq \langle \nabla f(x), x - z \rangle \\ &\quad + f(z) - f(y) \\ &\quad \text{by convexity} \\ &\leq \langle \nabla f(x), x - z \rangle + \langle \nabla f(y), z - y \rangle + \frac{\mu}{2} \|z - y\|^2 \\ &\quad \text{by } \mu\text{-Lipschitzianity of the gradient.} \end{aligned}$$

We now minimise the right-hand side with respect to z . As $\varphi : z \mapsto \langle z, \nabla f(y) - \nabla f(x) \rangle + \frac{\mu}{2} \|y - z\|^2$ is a strictly convex function of z , if z^* is a critical point of φ , then it is a global minimiser of φ . As

$$\nabla \varphi(z^*) = 0 \Leftrightarrow z^* = y + \frac{1}{\mu} (\nabla f(x) - \nabla f(y))$$

we obtain

$$f(x) - f(y) \leq \langle \nabla f(x), x \rangle - \langle \nabla f(y), y \rangle + \varphi(z^*).$$

Expanding, we deduce

$$\begin{aligned} f(x) - f(y) &\leq \langle \nabla f(x), x - y \rangle - \frac{1}{\mu} \langle \nabla f(x), \nabla f(x) - \nabla f(y) \rangle \\ &\quad + \frac{1}{\mu} \langle \nabla f(y), \nabla f(x) - \nabla f(y) \rangle + \frac{1}{2\mu} \|\nabla f(x) - \nabla f(y)\|^2 \\ &\leq \langle \nabla f(x), x - y \rangle - \frac{1}{2\mu} \|\nabla f(x) - \nabla f(y)\|^2, \end{aligned}$$

which is the desired inequality. $\square$

The final inequality, the proof of which is to be found in Exercise 1.9, is the Polyak-Lojasiewicz inequality:

Proposition 2.7. *Let $f \in \mathcal{C}^1(\mathbb{R}^d; \mathbb{R})$ be an α -strongly convex function. Then*

$$\forall x \in \mathbb{R}^d, f(x) - \inf_{\mathbb{R}^d} f \leq \frac{1}{2\alpha} \|\nabla f(x)\|^2.$$

EXERCISES FOR PART 1

Exercise 1.1. Let $A \in S_d(\mathbb{R})$.

- (1) Letting $\lambda_1(A) \leq \lambda_2(A) \leq \dots \leq \lambda_d(A)$ be the eigenvalues of A , show that

$$\lambda_1(A) = \inf_{\|z\|^2=1} \langle Az, z \rangle.$$

- (2) Show that for any two $A, B \in S_d(\mathbb{R})$ there holds

$$|\lambda_1(A) - \lambda_1(B)| \leq \|A - B\|_{\text{op}}$$

where $\|\cdot\|_{\text{op}}$ stands for the standard operator norm on the set of matrices.

Exercise 1.2. Let $A \in S_d(\mathbb{R})$ and $b \in \mathbb{R}^d$. We consider

$$f : x \mapsto \frac{1}{2} \langle Ax, x \rangle - \langle b, x \rangle.$$

- (1) Show that f is coercive if, and only if $A \in S_d^{++}(\mathbb{R})$.
 (2) Show that f is convex if, and only if $A \in S_d^+(\mathbb{R})$.
 (3) Show that f is strictly convex if, and only if $A \in S_d^{++}(\mathbb{R})$.

Exercise 1.3. Classify the critical points (local minimisers, local maximisers, saddle points, indeterminate critical points) of the following functions:

- (1) $f_1 : (x, y) \mapsto (x - y)^2 + (x + y)^3$,
 (2) $f_2 : (x, y) \mapsto x^2 - 2y^2 + 3xy$,
 (3) $f_3(x, y) \mapsto x^4 + y^3 - 3y - 2$.

Exercise 1.4 (Distance between two sets). Let A and B be two closed, nonempty subsets of $\mathbb{R}^d$.

- (1) Show that if A is compact, then the problem

$$\min_{a \in A, b \in B} \|a - b\|$$

has a solution (at least one).

- (2) Show with a counter-example that this problem need not have a solution if neither A nor B is assumed compact, even if A and B are convex.

Exercise 1.5. Give an example of a strictly convex function $\varphi : \mathbb{R}^d \rightarrow \mathbb{R}$ such that the equation

$$\nabla^2 \varphi(x) = 0$$

has infinitely many solutions.

Exercise 1.6 (Carathéodory theorem). Let $\Omega \subset \mathbb{R}^d$. We call the convex hull of Ω the smallest convex set containing Ω . We denote it by $C(\Omega)$.

- (1) Show that

$$C(\Omega) = \left\{ \sum_{i=0}^N t_i x_i, N \in \mathbb{N}, \{t_i\}_{i=0, \dots, N} \in [0; 1]^{N+1}, \sum_{i=0}^N t_i = 1, \{x_i\}_{i=0, \dots, N} \in \Omega^{N+1} \right\}.$$

- (2) We now want to show the Carathéodory theorem: for any $x \in C(\Omega)$, there exist $t_0, \dots, t_d \in [0; 1], x_0, \dots, x_d \in \Omega$ such that

$$\sum_{i=0}^d t_i = 1, x = \sum_{i=0}^d t_i x_i.$$

- (a) Using an example, show why one needs at least $(d + 1)$ points.
- (b) Prove the Carathéodory theorem; you can argue by descending induction, starting (for instance) from a point $x \in C(\Omega)$ that writes $x = \sum_{i=0}^{d+1} t_i x_i$, and showing that one of the vectors x_i 's can be expressed using the others.
- (c) Deduce from the Carathéodory theorem that if Ω is compact, then so is $C(\Omega)$.

Exercise 1.7 (Extreme points I: projection on closed convex sets). Let $K \subset \mathbb{R}^d$ be a closed convex set. Show that there exists a unique $z \in K$, denoted by Π_K and dubbed the orthogonal projection of x on K , such that

$$\|x - \Pi_K(x)\| = \min_{z \in K} \|x - z\|$$

and that

$$\forall y \in K, \langle x - \Pi_K(x), y - \Pi_K(x) \rangle \leq 0.$$

Show that Π_K is 1-Lipschitz.

Exercise 1.8 (Extreme points II: The Krein-Milman theorem). (1) Give an example of a convex set $K \subset \mathbb{R}^d$ that has no extreme points.

- (2) We assume that K is compact. Prove that K has extreme points.
- (3) We now want to prove the (finite-dimensional) Krein-Milman theorem: any $x \in K$ is a convex combination of extreme points of K .
 - (a) Let $x \in \partial K$. Show that there exists a hyperplane (called the supporting hyperplane) $H = \{\varphi = 0\}$ where $\varphi \in (\mathbb{R}^d)'$, $\varphi \neq 0$ (the dual of $\mathbb{R}^d$) such that $x \in H$ and $\varphi(K) \subset (-\infty; 0]$ (hint: think of the projection Π_K).
 - (b) Let $x \in K$. Show that if $x \in H$ for some supporting hyperplane of K (i.e. associated with some $y \in K$) then x is an extreme point of K if, and only if, x is an extreme point of $H \cap K$.
 - (c) Show the Krein-Milman theorem proceeding by induction on the dimension.

Exercise 1.9 (Polyak-Lojasciewicz Inequality). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an α -strongly convex function and let x^* be a minimiser of f . First, prove that

$$\forall x \in \mathbb{R}^d, \|x - x^*\|^2 \leq \frac{2}{\alpha} (f(x) - f(x^*)).$$

Second, show the following inequality:

$$\forall x \in \mathbb{R}^d, f(x) - f(x^*) \leq \frac{1}{2\alpha} \|\nabla f(x)\|^2.$$

Finally, deduce that

$$\forall x \in \mathbb{R}^d, \|x - x^*\| \leq \frac{1}{\alpha} \|\nabla f(x)\|.$$

SOLUTIONS OF THE EXERCISES OF PART 1

Solution 1.1. Let $A \in S_d(\mathbb{R})$.

- (1) It suffices to diagonalise A in an appropriate orthonormal basis.
- (2) We let $\lambda_1(A)$, x_A be the lowest eigenvalue of A and an associate eigenvector.

Solution 1.2. Let $A \in S_d(\mathbb{R})$ and $b \in \mathbb{R}^d$. We consider

$$f : x \mapsto \frac{1}{2} \langle Ax, x \rangle - \langle b, x \rangle.$$

- (1) As $\nabla^2 f = A$, the coercivity is implied by $\lambda_1(A) > 0$, using the lower bound

$$2f(x) \geq \lambda_1(A) \|x\|^2 - 2\|b\| \cdot \|x\|.$$

Assume on the other-hand that $\lambda_1(A) \leq 0$ and let e_1 be an associated eigenvector. Then $f(te_1) = \frac{1}{2}\lambda_1(A)t^2 - t\langle b, e_1 \rangle$. If $\lambda_1(A) < 0$, $f(te_1) \xrightarrow[t \rightarrow \pm\infty]{} -\infty$.

If $\lambda_1(A) = 0$, then one of the limits $\lim_{t \rightarrow +\infty} f(\cdot e_1)$, $\lim_{t \rightarrow -\infty} f(\cdot e_1)$ is non-positive, a contradiction with the coercivity.

- (2) The convexity implies the non-negativity of $\nabla^2 f$, which concludes.
- (3) Similar arguments provide the conclusion.

Solution 1.3. (1) The critical points are the points (x, y) such that

$$\begin{cases} 2(x - y) + 3(x + y)^2 = 0, \\ -2(x - y) + 3(x + y)^2 = 0. \end{cases}$$

In particular, (x, y) is critical if and only if $x = y = 0$. The hessian at $(0, 0)$ is the matrix

$$\begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}$$

which has eigenvalues $\{0, 4\}$. In particular, we can not conclude immediately. Nevertheless, observe that $f(-\varepsilon, -\varepsilon) < 0$ for ε small enough while $f(-\varepsilon, \varepsilon) > 0$. Consequently, $(0, 0)$ is a saddle point.

- (2) Likewise, (x, y) is critical if, and only if,

$$\begin{cases} 2x + 3y = 0 \\ -4y + 3x = 0. \end{cases}$$

This system has $(0, 0)$ as a unique solution. Furthermore, the hessian at $(0, 0)$ is

$$\begin{pmatrix} 2 & 3 \\ 3 & -4 \end{pmatrix}$$

which has two eigenvalues with opposite signs. Thus $(0, 0)$ is a saddle point.

- (3) (x, y) is a critical point if, and only if,

$$\begin{cases} 4x^3 = 0, \\ y^2 - 1 = 0. \end{cases}$$

Similarly $(0, 1)$ and $(0, -1)$ are the only critical point. As the hessian at $(0, y)$ is $\begin{pmatrix} 0 & 0 \\ 0 & 6y \end{pmatrix}$ we deduce that $(0, -1)$ is a saddle point: it is a local minimiser in the x direction, and a local maximiser in the y direction. The situation at $(0, 1)$ is more delicate and we can either use a Taylor expansion

at order 4, or simply observe that $f(x, y) = \varphi(x) + \psi(y)$ to deduce that $(0, 1)$ is a local minimiser.

Solution 1.4 (Distance between two sets). (1) *The difficulty is that $A \times B$ is not necessarily compact. It is however closed since A and B are closed. Moreover, the map $f : A \times B \rightarrow \mathbb{R}$ defined by $f(a, b) = \|a - b\|$ is continuous. Let us show that f is coercive : let (a_n, b_n) be a sequence of elements of $A \times B$ such that $\|(a_n, b_n)\|$ goes to $+\infty$ (for the Euclidean distance on $\mathbb{R}^d \times \mathbb{R}^d$ for instance). Since a_n is in the bounded set A , there exists $M \geq 0$ such $\|a_n\| \leq M$ for all n . Thus, by the triangular inequality,*

$$\|b_n\| = \|(a_n, b_n) - (a_n, 0)\| \geq \|(a_n, b_n)\| - \|(a_n, 0)\| \geq \|(a_n, b_n)\| - M \rightarrow +\infty,$$

This shows that $\|b_n\| \rightarrow +\infty$. Therefore

$$f(a_n, b_n) = \|a_n - b_n\| \geq \|b_n\| - \|a_n\| \geq \|b_n\| - M \rightarrow +\infty,$$

and f is coercive. Therefore f is continuous and coercive on the closed set $A \times B$. Therefore, by a result of the course that we recalled above, f has a minimizer on $A \times B$.

(2) *For instance, in $\mathbb{R}^2$, take $A = \{(x, y) \in \mathbb{R}^d, y \leq 0\}$ and $B = \{(x, y) \in \mathbb{R}^d, y \geq e^x\}$. These sets are both closed and convex (since the exponential function is convex). Moreover, letting $a_n = (-n, 0) \in A$ and $b_n = (-n, e^{-n}) \in B$, we have that for all $n \in \mathbb{N}$,*

$$0 \leq \inf_{a \in A, b \in B} \|a - b\| \leq \|a_n - b_n\| \leq e^{-n} \rightarrow 0,$$

hence

$$\inf_{a \in A, b \in B} \|a - b\| = 0,$$

though $A \cap B$ is empty, since if $b = (x, y) \in B$, then $y > 0$.

Solution 1.5. *It suffices to take $\varphi(x) = \frac{x^2}{2} + \cos(x)$. Indeed, $\varphi''(x) = 1 - \cos(x)$, which has infinitely many zeroes. As these zero has zero Lebesgue measure the function φ' is strictly increasing, thereby giving the strict convexity of the function.*

Solution 1.6 (Carathéodory theorem). (1) *We introduce the set*

$$K(\Omega) := \left\{ \sum_{i=0}^N t_i x_i, N \in \mathbb{N}, \{t_i\}_{i=0, \dots, N} \in [0; 1]^{N+1}, \sum_{i=0}^N t_i = 1, \{x_i\}_{i=0, \dots, N} \in \Omega^{N+1} \right\}.$$

Our goal is to show that $C(\Omega) = K(\Omega)$. We proceed by double inclusion.

- *We begin with $K(\Omega) \subset C(\Omega)$. To this end, as we can equivalently define*

$$C(\Omega) = \cap_{Y \text{ convex}, \Omega \subset Y} Y,$$

we fix a convex set Y containing Ω . Observe that, as Y is convex, it follows that for any $\{t_i\}_{i=0, \dots, N} \in [0; 1]^{N+1}$, $\sum_i t_i = 1$ and any family $\{y_i\}_{i \in \mathbb{N}} \in \Omega^{\mathbb{N}}$, $\sum_{i=0}^N t_i y_i \in Y$. This can be proved by induction on N , and we only show it in the case $N = 2$ (the case $N = 1$ is the definition

of a convex set). Then

$$t_0 y_0 + t_1 y_1 + t_2 y_2 = \underbrace{(t_0 + t_1) \left(\underbrace{\frac{t_0}{t_0 + t_1} y_0 + \frac{t_1}{t_0 + t_1} y_1}_{\in Y \text{ by convexity of } Y} \right)}_{\in Y \text{ by convexity of } Y} + t_2 y_2.$$

In particular, as for any $x \in \Omega$ we have $x \in Y$, we deduce that $K(\Omega) \subset Y$ and so

$$K(\Omega) \subset C(\Omega).$$

- To show that $C(\Omega) \subset K(\Omega)$, it suffices to show that $K(\Omega)$ is convex. This is however trivial: let $x = \sum_{i=0}^N t_i x_i, y = \sum_{i=0}^M q_i y_i$ be two points of $K(\Omega)$ (with transparent notations for t_i 's, q_i 's, x_i 's, y_i 's...) and $t \in [0; 1]$. Then

$$(1-t)x + ty = \sum_{i=0}^N (1-t)t_i x_i + \sum_{i=N+1}^{M+N+1} t q_{i-(N+1)} y_{i-(N+1)} \in K(\Omega),$$

which concludes the proof.

- Consider three non-aligned points in $\mathbb{R}^2$. Then the set of convex combinations of two points is the border of the triangle, while the set of convex combinations of 3 points is the full triangle.
- Let $x \in C(\Omega)$ and assume that x writes

$$x = \sum_{i=0}^N t_i x_i, N > d.$$

Let us show that we can write

$$x = \sum_{i=0}^{N-1} t'_i x'_i, x'_i \in \Omega.$$

As $N > d$, the family $\{x_i - x_0\}_{i=1, \dots, N}$ is not free. In particular, there exists a non-trivial family of coefficients $\{\alpha_i\}_{i=1, \dots, N}$ such that

$$\sum_{i=1}^N \alpha_i x_i = \left(\sum_{j=1}^N \alpha_j \right) x_0.$$

Introduce the family

$$\mu_i = \alpha_i, i \geq 1, \mu_0 = - \sum_{j=1}^N \alpha_j$$

so that

$$\sum_{j=0}^N \mu_j = 0, \sum_{i=0}^N \mu_i x_i = 0.$$

We deduce that for any real number s ,

$$x = \sum_{i=0}^N (t_i + s \mu_i) x_i.$$

As $\sum t_i + s\mu_i = 1$, in order to conclude, it suffices to guarantee that for any i we have $t_i + s\mu_i \geq 0$. It suffices to take

$$s = \min \left\{ -\frac{t_i}{\mu_i}, \mu_i < 0 \right\}.$$

The conclusion follows.

- (c) It follows from the Carathéodory theorem that $C(\Omega) = \Phi([0; 1]^{d+1} \times \Omega^{N+1})$ where $\Phi((t_0, \dots, t_d), (x_0, \dots, x_d)) = \sum t_i x_i$. As the image of a compact set by a continuous function is compact, this concludes.

Solution 1.7 (Extreme points I: projection on closed convex sets). *The existence of a solution to*

$$\inf_{z \in K} \|x - z\|$$

is a consequence of the compactness of K . The uniqueness follows from the strict convexity of $z \mapsto \|x - z\|^2$: indeed, if z, z' are two distinct optimisers, it follows that for any $t \in [0; 1]$ so is $(1-t)z + tz'$. From equality in the triangle inequality

$$\|x - ((1-t)z + tz')\| \leq t\|x - z\| + (1-t)\|x - z'\|$$

we deduce, with $t = 2$, that

$$z = z'.$$

Alternatively you can use the strict convexity of $x \mapsto \|x\|^2$.

Let $z \in K$ be optimal. By convexity of K it follows that for any $y \in K$ and any $t \in [0; 1]$, we have

$$\|x - ty - (1-t)z\|^2 \geq \|x - z\|^2.$$

Passing to the limit as $t \rightarrow 0$ yields

$$\forall y \in K, \langle x - \Pi_K(x), y - \Pi_K(x) \rangle \leq 0.$$

Finally, to prove that Π_K is 1-Lipschitz, it suffices to observe that from the optimality conditions we deduce that, for any x, x' ,

$$\langle x - \Pi_K(x), \Pi_K(x') - \Pi_K(x) \rangle + \langle x' - \Pi_K(x'), \Pi_K(x) - \Pi_K(x') \rangle \leq 0.$$

Thus,

$$\|\Pi_K(x') - \Pi_K(x)\|^2 \leq \langle x - x', \Pi_K(x') - \Pi_K(x) \rangle.$$

The Cauchy-Schwarz inequality allows to conclude.

Solution 1.8 (Extreme points II: The Minkowski theorem). (1) *It suffices to take $H := \{(x, y) \in \mathbb{R}^2 : y \geq 0\}$.*

- (2) *A simple argument shows that, if we define*

$$x := \operatorname{argmax}_K \|\cdot\|^2$$

then x is an extreme point. Indeed, should it not be an extreme point, the strict convexity of $\|\cdot\|^2$ allows to conclude.

- (3) (a) *Let $x \in \partial K$. In particular, we can find $\{y_k\}_{k \in \mathbb{N}} \in K^c$ such that $y_k \xrightarrow[k \rightarrow \infty]{} x$. Let*

$$e_k := \frac{y_k - \Pi_K(y_k)}{\|y_k - \Pi_K(y_k)\|} \in \mathbb{S}^{d-1}$$

and e_∞ be a closure point of $\{e_k\}$. Define $\varphi_\infty := \langle e_\infty, \cdot \rangle$. Passing to the limit in

$$\forall z \in K, \langle e_k, z - \Pi_K(y_k) \rangle \leq 0$$

and keeping in mind that $\Pi_K(y_k) \xrightarrow{k \rightarrow \infty} x$ we deduce the result.

- (b) Assume that $x \in H$. Since $x \in H \cap K$, it is clear that if x is extreme in K , then it is so in $H \cap K$. Conversely, assume that $x \in H \cap K$ is an extreme point of $H \cap K$. Let φ be the linear form defining H . If $x = ta + (1-t)b$ with $a, b \in K$ then, as $\varphi \leq 0$ on K , we deduce that $\varphi(a) = \varphi(b) = 0$ whence $a, b \in H \cap K$. The conclusion follows by extremality.
- (c) We argue, as proposed, by induction on the dimension. In dimension 1, the situation is trivial. Assume the Krein-Milman theorem holds in dimension $d-1$ and let $K \subset \mathbb{R}^d$ be a convex, compact subset. Assume (up to reducing the dimension of the ambient space) that $\text{Vect}(K) = \mathbb{R}^d$ so that the interior of K is not empty. Now, assume that $x \in K$ and distinguish between two cases:
- Either $c \in \partial K$, in which case we consider the supporting hyperplane H at K in c . Then $c \in H \cap K$, which is a $(d-1)$ dimensional compact convex subset. We can apply the previous question and the induction assumption to conclude that c is a convex combination of extreme points of $H \cap K$ (and, consequently, of K).
 - Or c is an interior point of K . In that case, let D be any line passing through c and let $K' := D \cap K$. K' is a one-dimensional convex set, and $K' = [a; b]$ with $a, b \in \partial K$. In particular, applying the previous case, we can conclude: c is a barycenter of a and b , which themselves are barycenters of extreme points of K .

Solution 1.9 (Polyak-Lojasciewicz Inequality). As f is λ -convex we know that for any $x, y \in \mathbb{R}^d$ we have

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\lambda}{2} \|y - x\|^2.$$

- (1) We begin with the first inequality. Take $x = x^*$ in the previous inequality.
 (2) Letting x be fixed, minimise the right-hand side of the inequality in y . We obtain a minimum at

$$y^* = -\frac{\nabla f(x)}{\lambda} + x$$

thereby yielding

$$f(y) \geq f(x) - \frac{1}{2\lambda} \|\nabla f(x)\|^2.$$

This yields the conclusion.

- (3) The two previous inequalities combine to give the required result.

Part 2. Unconstrained optimisation: approximation and gradient descent

3. AN OVERVIEW OF GRADIENT DESCENT: BASIC PRINCIPLE & STUDY OF CONVEX FUNCTIONS

3.1. Goal of the part. In this part of the class we will study a first approximation method for optimisation problems. In other words, consider once again a problem of the form

$$(3.1) \quad \min_{x \in \mathbb{R}^d} f(x)$$

and assume for the sake of simplicity that a solution x^* exists. As we already mentioned, the first exploitable information on x^* we have is that it is a critical point of f :

$$\nabla f(x^*) = 0.$$

The goal of the gradient descent is to **find critical points of f** through an iterative method, that is, a method which can be written as

$$\begin{cases} \text{Start from an initial guess } x_0, \\ \text{Supposing } x_0, \dots, x_k \text{ are built, set } x_{k+1} = x_k + G_k(x_k) \end{cases}$$

for some function G_k , the definition of which might depend on the previous iterates $x_0, \dots, x_k$. Most of the time, this will not be the case, and the iteration map G_k will not depend on the index k . The goal is to obtain algorithms that produce sequences that converge at a “good enough” rate — most of the time, we will be satisfied with *linear convergence*, in the following sense:

Definition 3.1. Let $\{x_k\}_{k \in \mathbb{N}} \in (\mathbb{R}^d)^{\mathbb{N}}$ and $x^* \in \mathbb{R}^d$. We say that $\{x_k\}_{k \in \mathbb{N}}$ converges linearly, at rate $\alpha \in [0; 1)$, to x^* , if there exists a constant C such that

$$\forall k \in \mathbb{N}, \|x_k - x^*\| \leq C\alpha^k.$$

3.2. Definition of the gradient descent and basic properties.

3.2.1. First considerations. The gradient descent is a *local* algorithm that essentially relies on a Taylor expansion of the function f : assume that you are starting from an initial guess $x_0 \in \mathbb{R}^d$, and you want to solve (5.1). We look for a point that is close enough to x_0 , say at distance at most d_0 , and such that $f(x_1) < f(x_0)$ (if that is possible). In that case, a natural idea is to replace f by its first-order Taylor approximation

$$f(x_0 + z) = f(x_0) + \langle \nabla f(x_0), z \rangle + o_{z \rightarrow 0}(\|z\|)$$

so that, at first order, we are solving the minimisation problem

$$(3.2) \quad \min_{\|x - x_0\| \leq d_0} \langle \nabla f(x_0), x - x_0 \rangle.$$

At this stage, two things may happen:

- (1) Either the gradient vanishes ($\nabla f(x_0) = 0$), in which case we stop, as we are satisfied with what we already have. Now, if we wanted to go further in the analysis, we should note that two other possibilities arise: either f is convex, in which case this implies that x_0 is a global minimiser of the function f , or f is not convex and we would need to do something different

to investigate the local optimality of x_0 . This will very often not be the case.

- (2) Either the gradient does not vanish, so that the pseudo-optimisation problem (5.3) has a unique solution

$$x_1 = -\frac{d_0}{\|\nabla f(x_0)\|} \nabla f(x_0).$$

Now, the question remains of choosing the parameter d_0 . Of course, if we already know that the gradient is small enough in norm, it makes no sense to look for a point that would be far away, and this naturally leads to choosing d_0 as $d_0 = \tau \|\nabla f(x_0)\|$ for some $\tau > 0$.

Overall, we define the sequence of iterates of the gradient descents as follows:

$$\begin{cases} x_0 \in \mathbb{R}^d, \\ \forall k \in \mathbb{N}, x_{k+1} = x_k - \tau \nabla f(x_k). \end{cases}$$

The main questions under consideration from now on are:

- (1) The **convergence** of the generated sequence $\{x_k\}_{k \in \mathbb{N}}$.
- (2) The **convergence of the sequence of values** $\{f(x_k)\}_{k \in \mathbb{N}}$.
- (3) The **convergence of the gradient of the objective function** $\{\nabla f(x_k)\}_{k \in \mathbb{N}}$.

Of course, the convergence of $\{x_k\}_{k \in \mathbb{N}}$ implies the convergence of the values and of the gradient; the convergence of the values, on the other hand, does not imply the convergence of the sequence itself. It is also important to note that, in general, the presentation of gradient descent assumes, from the get-go, some strong convexity of f , which gives a positive answer to all the questions above. On the other hand, it is extremely important, both in practice and in theory, to distinguish these different steps and this is what we will do. At any rate, here is a simple result:

Proposition 3.1. *Assume that $f \in \mathcal{C}^1$ and that the gradient descent with fixed step size τ converges in the sense that $\{x_k\}_{k \in \mathbb{N}}$ converges to some x^* . Then $\nabla f(x^*) = 0$.*

Proof of Proposition 3.1. If the sequence converges then, passing to the limit in $x_{k+1} = x_k - \nabla f(x_k)$ yields $\nabla f(x^*) = 0$. $\square$

Of course the next question is, if we assume that $\{x_k\}_{k \in \mathbb{N}}$ converges to some x^* , is it true that x^* is, in fact, a minimiser of f ? The answer is no in general. Consider for instance the function

$$f : x \mapsto \frac{x^3}{3},$$

a fixed $1 > \tau > 0$ and an initialisation $x_0 = 1$. The sequence of iterates of the gradient descent is given by

$$\forall k \in \mathbb{N}, x_{k+1} = x_k(1 - \tau x_k).$$

Now, if $\tau \in (0; 1)$, a simple reasoning by induction shows that the sequence $\{x_k\}_{k \in \mathbb{N}}$ is positive and decreasing; in particular it is converging so that by Proposition 3.1 it converges to 0, which is not a minimiser of f . Of course, one might argue that this is cheating, as the function f is not coercive. Nevertheless, it is easy to adapt this example: simply modify f on $(-\infty; -1]$ to have a globally smooth, coercive function.

3.2.2. *Do the step-size and the regularity matter?* In this first paragraph, we investigate in a formal manner the constraints we should put on the step size and on the function f to obtain a converging sequence, where the parameters should be chosen uniformly with respect to the initial condition. We begin with the regularity of the function. Let us consider the case of a $\mathcal{C}^1$, but not $\mathcal{C}^{1,1}$ function, for instance, in two variables

$$f : (x, y) \mapsto \frac{2}{3} (x^2 + 2y^2)^{\frac{3}{4}}.$$

It is fairly easy to show that the function f is $\mathcal{C}^1$, but not $\mathcal{C}^{1,1}$ at 0 (this is left as an exercise): simply observe that

$$|f(x, y)| \leq C \|(x, y)\|^{\frac{3}{2}}.$$

Furthermore, 0 is the unique minimiser of f . For a given parameter $\tau > 0$, the sequence of iterates is given explicitly by

$$\begin{cases} x_{k+1} = x_k \left(1 - \frac{\tau}{(x_k^2 + 2y_k^2)^{\frac{1}{4}}} \right) \\ y_{k+1} = y_k \left(1 - \frac{2\tau}{(x_k^2 + 2y_k^2)^{\frac{1}{4}}} \right). \end{cases}$$

Observe that, at a formal level, if the sequence converges, then it must converge to 0. Thus, we “should” be able to write that

$$(x_{k+1}, y_{k+1}) \sim \frac{\tau}{(x_k^2 + 2y_k^2)^{\frac{1}{4}}} (-x_k, -2y_k).$$

Defining

$$z_k := x_k^2 + 2y_k^2$$

we deduce that (asymptotically)

$$z_{k+1} \geq C z_k^{\frac{1}{2}}, C = \tau^2.$$

Now, let us assume that this inequality is, in fact, satisfied for all $k \in \mathbb{N}$. This would give the lower bound

$$z_{k+1} \geq C C^{\frac{1}{2}} z_{k-1}^{\frac{1}{2^2}} \geq \dots \geq C^{\sum_{i=0}^k (\frac{1}{2})^i} z_0^{2^{-k}},$$

and cannot converge to 0.

We continue with an investigation of the step size; here the computations are much easier, as it suffices to consider, in the one-dimensional case, the function

$$f : x \mapsto \frac{\mu}{2} x^2.$$

Then, for any initialisation x_0 and any fixed step size $\tau > 0$, the sequence of iterates is given by

$$x_{k+1} = x_k(1 - \mu\tau) = x_0(1 - \mu\tau)^k.$$

Thus, the method converges if, and only if, $0 < \tau < \frac{1}{\mu}$. As μ quantifies the steepness of f' , or, put otherwise, the average variation of the gradient, we fairly easily understand that the wilder the gradient of a function, the smaller the step size needs to be.

3.2.3. *The gradient descent is a descent method.* In this section, we consider a function $f \in \mathcal{C}^1(\mathbb{R}^d)$ with a μ -Lipschitz gradient in the sense that

$$(3.3) \quad \forall x, y \in \mathbb{R}^d, \|\nabla f(x) - \nabla f(y)\| \leq \mu \|x - y\|.$$

We do not make any assumption on the coercivity of f , or on the existence of a minimiser. Our first result is the following:

Theorem 3.1. *For any $x_0 \in \mathbb{R}^d$, for any $\tau > 0$, the sequence generated by the gradient descent initialised at x_0 with step size τ satisfies*

$$\forall k \in \mathbb{N}, f(x_{k+1}) - f(x_k) \leq \tau(\tau\mu - 1) \|\nabla f(x_k)\|^2.$$

In particular, if $\tau \in (0; \frac{1}{\mu})$ then the sequence $\{f(x_k)\}_{k \in \mathbb{N}}$ is strictly decreasing unless it is stationary. Finally, for any $\tau \in (0; \frac{1}{2\mu})$ there holds

$$\forall k \in \mathbb{N}, f(x_{k+1}) - f(x_k) \leq -\frac{\tau}{2} \|\nabla f(x_k)\|^2.$$

Proof of Theorem 3.1. It suffices to write that for any $k \in \mathbb{N}$ there holds

$$f(x_{k+1}) = f(x_k - \tau \nabla f(x_k)).$$

From the mean-value theorem, there exists $\xi \in \mathbb{B}(x_k; \|x_{k+1} - x_k\|)$ such that

$$f(x_{k+1}) = f(x_k) + \langle \nabla f(\xi), -\tau \nabla f(x_k) \rangle.$$

This rewrites

$$\begin{aligned} f(x_{k+1}) - f(x_k) &= -\langle \nabla f(\xi) - \nabla f(x_k), \tau \nabla f(x_k) \rangle - \tau \|\nabla f(x_k)\|^2 \\ &\leq \tau \|\nabla f(\xi) - \nabla f(x_k)\| \cdot \|\nabla f(x_k)\| - \tau \|\nabla f(x_k)\|^2 \\ &\leq \tau \mu \|x_{k+1} - x_k\| \cdot \|\nabla f(x_k)\| - \tau \|\nabla f(x_k)\|^2 \\ &= \tau(\tau\mu - 1) \|\nabla f(x_k)\|^2. \end{aligned}$$

The conclusion follows. $\square$

We highlight once again that we did not require any information other than the regularity of ∇f . In the next section, we will illustrate several nice properties of gradient descent when the function f to be optimised is convex.

3.2.4. *Convergence of the gradient descent II: convex functions.* We now make one stronger assumption on the function f . Namely, we assume that f still satisfies (3.3) for some constant $\mu > 0$ and that f is convex.

Theorem 3.2. *Assume that f is convex and satisfies (3.3) for some $\mu > 0$. Finally, assume that f has a minimiser x^* . For any $\tau \in (0; \frac{1}{2\mu})$, for any initialisation x_0 , the gradient descent with fixed step size τ , initialised at x_0 , satisfies*

$$\forall k \in \mathbb{N}, f(x_{k+1}) - f(x^*) \leq \frac{\|x_0 - x^*\|^2}{2\tau(k+1)}.$$

Proof of Theorem 3.2. Recall that from Theorem 3.1 we have

$$\forall k \in \mathbb{N}, f(x_{k+1}) - f(x_k) \leq -\frac{\tau}{2} \|\nabla f(x_k)\|^2.$$

However, by convexity of f ,

$$f(x_k) \leq f(x^*) + \langle \nabla f(x_k), x_k - x^* \rangle.$$

Consequently,

$$\begin{aligned}
 f(x_{k+1}) &\leq f(x_k) - \frac{\tau}{2} \|\nabla f(x_k)\|^2 \\
 &\leq f(x^*) + \langle \nabla f(x_k), x_k - x^* \rangle - \frac{\tau}{2} \|\nabla f(x_k)\|^2 \\
 &= f(x^*) + \frac{2}{\tau} \left(\frac{\tau}{2} \langle \nabla f(x_k), x_k - x^* \rangle - \frac{\tau^2}{4} \|\nabla f(x_k)\|^2 \right) \\
 &= f(x^*) - \frac{2}{\tau} \left(\left\| \frac{\tau}{2} \nabla f(x_k) - \frac{1}{2} (x_k - x^*) \right\|^2 - \frac{1}{4} \|x_k - x^*\|^2 \right) \\
 &= f(x^*) - \frac{2}{\tau} \left(\frac{1}{4} \|x_{k+1} - x^*\|^2 - \frac{1}{4} \|x_k - x^*\|^2 \right).
 \end{aligned}$$

We thus deduce that

$$k(f(x_k) - f(x^*)) \leq \sum_{i=1}^k (f(x_i) - f(x^*)) \leq \frac{1}{2\tau} \|x_0 - x^*\|^2.$$

The conclusion follows. $\square$

3.2.5. Convergence of the gradient descent III: quadratic functions. We saw in the previous paragraph that, in the case of convex functions, we could get a convergence rate (algebraic, as it turns out) for the gradient descent. The goal of this section is to provide a finer convergence rate in the special case of quadratic functions.

Definition 3.2. We say that a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is quadratic if there exists $A \in M_d(\mathbb{R})$ and $b \in \mathbb{R}^d$ such that

$$\forall x \in \mathbb{R}^d, f(x) = \frac{1}{2} \langle Ax, x \rangle - \langle b, x \rangle.$$

When f is quadratic, we say that f is represented by (A, b) .

A straightforward computation shows that

$$\nabla f(x) = \frac{A + A^T}{2} x - b.$$

In particular, when A is symmetric,

$$\nabla f(x) = Ax - b$$

and x^* is a critical point of f if, and only if, x^* is a solution to $Ax^* = b$.

Theorem 3.3. Let $A \in S_d^{++}(\mathbb{R})$, $b \in \mathbb{R}^d$ and f be the quadratic function represented by (A, b) . Letting $0 < \lambda_1 \leq \lambda_d(A)$ be the eigenvalues of A , for any $\tau \in \left(0; \frac{2}{\lambda_d(A)}\right)$, for any $x_0 \in \mathbb{R}^d$, the gradient descent initialised at x_0 with fixed step size $\tau > 0$ converges linearly to the unique solution of $Ax^* = b$ and, more specifically,

$$\forall k \in \mathbb{N}, \|x_k - x^*\| \leq \alpha(\tau)^k \|x_0 - x^*\|$$

with $\alpha(\tau) = \max_{i=1, \dots, d} |1 - \tau \lambda_i(A)|$. Finally,

$$\min_{\tau \in \left(0; \frac{2}{\lambda_d(A)}\right)} \alpha(\tau) = \alpha \left(\frac{1}{\lambda_1(A) + \lambda_d(A)} \right) = \frac{\text{cond}(A) - 1}{\text{cond}(A) + 1} \text{ with } \text{cond}(A) = \frac{\lambda_d(A)}{\lambda_1(A)}.$$

Proof of Theorem 3.3. Observe that as $A \in S_d^{++}(\mathbb{R})$ all the eigenvalues of A are positive. Furthermore, A induces a bijection, whence x^* is uniquely defined. Additionally, as A is symmetric, $\nabla f(x) = Ax - b$. Now we explicitly obtain, for any $k \in \mathbb{N}$,

$$x_{k+1} = x_k - \tau Ax_k + \tau b$$

so that, defining $y_k := x_k - x^*$,

$$\forall k \in \mathbb{N}, y_{k+1} = y_k - \tau Ax_k + \tau b = y_k - \tau A(x_k - x^*) = (\text{Id} - \tau A)y_k.$$

The matrix $\text{Id} - \tau A$ has eigenvalues $1 - \tau\lambda_d(A) \leq \dots \leq 1 - \tau\lambda_1(A)$. We deduce that

$$\forall k \in \mathbb{N}, \|y_{k+1}\| = \|(\text{Id} - \tau A)y_k\| \leq \|\text{Id} - \tau A\|_{\text{op}} \cdot \|y_k\|.$$

Here, we used the operator norm on $\text{Id} - \tau A$. By a straightforward iteration argument we deduce that

$$\forall k \in \mathbb{N}, \|y_k\| \leq \|\text{Id} - \tau A\|_{\text{op}}^k \|y_0\|.$$

However,

$$\|\text{Id} - \tau A\|_{\text{op}} = \max_{i=1, \dots, d} |1 - \tau\lambda_i(A)|.$$

We refer to Exercise 1.1. In particular, if $\tau > 0$ is chosen so that

$$(3.4) \quad \alpha(\tau) := \max_{i=1, \dots, d} |1 - \tau\lambda_i(A)| < 1$$

we obtain

$$\forall k \in \mathbb{N}, \|x_k - x^*\| \leq \alpha(\tau)^k \|x_0 - x^*\|.$$

It remains to pick $\tau > 0$ so that $\alpha(\tau) < 1$. However

$$\alpha(\tau) < 1 \Leftrightarrow -1 < 1 - \tau\lambda_d(A) \leq 1 - \tau\lambda_1(A) < 1$$

which rewrites, in a compact form, as

$$\tau < \frac{2}{\lambda_d(A)}.$$

The conclusion follows. Finally, it is an easy exercise to see that α is minimised at τ^* such that

$$|1 - \tau^*\lambda_1(A)| = |1 - \tau^*\lambda_d(A)|.$$

Solving this equation explicitly in τ^* yields

$$\tau^* = \frac{2}{\lambda_1(A) + \lambda_d(A)}, \text{ whence } \alpha(\tau^*).$$

□

Let us observe that the convergence rate of gradient descent is quantified by the conditioning number of the matrix A : if $\text{cond}(A) \approx 1$ then the method converges extremely quickly if τ^* is chosen properly, while, if $\text{cond}(A) \gg 1$ (which means that A , as a linear map, dilates much more in certain directions than in others), the method will *a priori* converge extremely slowly. It is important to have basic reflexes regarding the conditioning number of matrix. We refer to Exercise 2.3.

3.2.6. *Convergence of the gradient descent IV: the case of strongly convex functions.* The purpose of this section is to generalise the results of the previous paragraph to the case of strongly convex functions:

Theorem 3.4. *Let f be a α -strongly convex, coercive, $\mathcal{C}^1$ function with a μ -Lipschitz gradient. Let x^* be the minimiser of f . Then, for any $x_0 \in \mathbb{R}^d$, for any $\tau \in \left(0; \frac{1}{2\mu}\right)$, the gradient descent initialised at x_0 with fixed step size τ converges linearly to x^* and, more precisely, we have*

$$\forall k \in \mathbb{N}, \|x_k - x^*\| \leq (1 - \alpha\tau)^{\frac{k}{2}} \|x_0 - x^*\|.$$

Proof of Theorem 3.4. We observe that, setting $y_k := x_k - x^*$, we have

$$\forall k \in \mathbb{N}, y_{k+1} = y_k - \tau (\nabla f(x_k) - \nabla f(x^*)).$$

Taking the squared norm on each side of this identity yields

$$\|y_{k+1}\|^2 = \|y_k\|^2 + \tau^2 \|\nabla f(x_k)\|^2 - 2\tau \langle \nabla f(x_k) - \nabla f(x^*), x_k - x^* \rangle$$

Now observe that

$$\frac{\alpha}{2} \|x_k - x^*\|^2 + \langle \nabla f(x_k), x^* - x_k \rangle \leq f(x^*) - f(x_k)$$

so that

$$\|y_{k+1}\|^2 \leq (1 - \alpha\tau) \|y_k\|^2 + 2\tau (f(x^*) - f(x_k)) + \tau^2 \|\nabla f(x_k)\|^2$$

From (2.7)

$$\tau^2 \|\nabla f(x_k)\|^2 \leq 2\tau^2 \mu (f(x_k) - f(x^*)).$$

Consequently

$$\|y_{k+1}\|^2 \leq (1 - \alpha\tau) \|y_k\|^2 + 2\tau (1 - \tau\mu) (f(x^*) - f(x_k)) \leq (1 - \alpha\tau) \|y_k\|^2.$$

□

4. BEYOND CONVEXITY

One might then wonder what happens when the function f to be optimised is no longer convex. We will be going over two phenomena: the first one is that, even when f is not convex, the gradient descent converges locally around non-degenerate minimisers—this is absolutely expected. The second phenomenon is different in nature, and amounts to investigating whether we can guarantee that the gradient descent at least converges. It is the case, provided f only has isolated critical points. This relies on the Zoutendijk theorem.

4.1. Convergence around non-degenerate local minimisers. The first theorem that we give, the proof of which is given in Exercise 2.4, is the following:

Theorem 4.1. *Let $f \in \mathcal{C}^2(\mathbb{R}^d; \mathbb{R})$ have a μ -Lipschitz gradient. Let x^* be a non-degenerate local minimiser of f in the sense that $\nabla f(x^*) = 0, \nabla^2 f(x^*) \in S_d^{++}(\mathbb{R})$. There exists $\varepsilon > 0$ such that, for any $\tau > 0$ small enough, for any $x_0 \in \mathbb{B}(x^*; \varepsilon)$, the gradient descent initialised at x_0 with step size τ converges linearly to x^* , in the sense that there exists a constant $\alpha \in (0; 1)$ and a constant C such that*

$$\forall k \in \mathbb{N}, \|x_k - x^*\| \leq C\alpha^k.$$

This is unsurprising: locally around x^* , one can write

$$f(x) \approx f(x^*) + \frac{1}{2} \langle \nabla^2 f(x^*)(x - x^*), x - x^* \rangle$$

so the situation “should” resemble the quadratic case. The only difficulty is in controlling the error term in the approximation above. The proof is carried out in Exercise 2.4.

4.2. Convergence to isolated critical points. A much more interesting extension of the analysis we carried out above is the case of non-convex functions. Observe that, in general, one can not obtain any convergence result, *per* the counter examples we already encountered. Nevertheless, if the function f is coercive and has only isolated critical points, it is possible to reach the following conclusion:

Theorem 4.2. *Let $f \in \mathcal{C}^1(\mathbb{R}^d; \mathbb{R})$ be coercive and have a μ -Lipschitz gradient. Assume that f only has isolated critical points. Then for any $\tau > 0$ small enough, for any $x_0 \in \mathbb{R}^d$, the sequence of iterates generated by the gradient descent initialised at x_0 with step-size τ converges, and its limit is a critical point of f .*

Proof of Theorem 4.2. The proof is quite lengthy, but elementary. We split it into several parts for the convenience of the reader:

- (1) The sequence $\{x_k\}_{k \in \mathbb{N}}$ remains bounded Recall that if $\tau \in (0; \frac{1}{2\mu})$ there holds

$$f(x_{k+1}) - f(x_k) \leq \frac{-\tau}{2} \|\nabla f(x_k)\|^2.$$

In particular, $\{x_k\}_{k \in \mathbb{N}} \subset \{f \leq f(x_0)\} = K_0$, which is, by assumption, a compact set. The conclusion follows.

- (2) There holds $\|x_{k+1} - x_k\| \xrightarrow[k \rightarrow \infty]{} 0$ Similarly, summing the previous estimates, we obtain

$$\frac{\tau}{2} \sum_{k=0}^{\infty} \|\nabla f(x_k)\|^2 \leq f(x_0) - f(x^*),$$

which implies

$$(4.1) \quad \|x_{k+1} - x_k\|^2 = \|\nabla f(x_k)\|^2 \xrightarrow{k \rightarrow \infty} 0.$$

- (3) The sequence $\{x_k\}_{k \in \mathbb{N}}$ has a unique closure point Of course, this property would suffice to conclude, as any sequence living in a compact set converges if, and only if, it has a unique closure point. However, observe that from (4.1) any closure point of $\{x_k\}_{k \in \mathbb{N}}$ is a critical point. As critical points are isolated, and as K_0 is compact, there are finitely many critical points $y_0, \dots, y_N$ in K_0 . By continuity of the gradient, we can fix $\varepsilon, \delta > 0$ such that

$$\forall i \neq j, \overline{\mathbb{B}(y_i; \varepsilon)} \cap \overline{\mathbb{B}(y_j; \varepsilon)} = \emptyset, \inf_{K_0 \setminus \bigcup_i \mathbb{B}(y_i; \varepsilon)} \|\nabla f\| \geq \delta > 0.$$

In particular, if $\{x_k\}_{k \in \mathbb{N}}$ had two distinct closure points, say y_0 and y_1 , it would follow from (4.1) that there exists a subsequence $\{x_{n_k}\}_{k \in \mathbb{N}} \subset \overline{K_0 \setminus \bigcup_i \mathbb{B}(y_i; \varepsilon)}$, which would contradict that any closure point is critical. Consequently, $\{x_k\}_{k \in \mathbb{N}}$ has a unique critical point, and thus converges. $\square$

EXERCISES FOR PART 2

- Exercise 2.1.** (1) We let $A = \begin{pmatrix} 1 & 3 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $b = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ and f be represented by (A, b) . Can the gradient descent initialised at a given $x_0 \in \mathbb{R}^d$ with fixed step size $\tau > 0$ converge?
- (2) Assume that A is symmetric and that for any $b \in \mathbb{R}^d$, for any $x_0 \in \mathbb{R}^d$ there exists $\tau > 0$ such that the gradient descent generated at x_0 with step size $\tau > 0$ converges. Show that $A \in S_d^{++}(\mathbb{R})$.

Exercise 2.2. We let $A \in S_d(\mathbb{R})$ be matrix with (at least) two eigenvalues of opposite signs. We let $b = 0$. Show that for any $\tau > 0$ the set $\{x_0 \in \mathbb{R}^d : \text{the gradient descent initialised at } x_0 \text{ with fixed step size } \tau \text{ converges}\}$ has measure zero.

Exercise 2.3. [Some basic properties of the conditioning number]

- (1) Show that, for any symmetric positive definite matrix M , $\text{cond}(M) \geq 1$.
- (2) Show that for any symmetric definite positive matrix $\text{cond}(M) = \|M\|_{\text{op}} \cdot \|M^{-1}\|_{\text{op}}$. We use this expression to define the conditioning number of any invertible matrix $M \in \text{Gl}_d(\mathbb{R})$.
- (3) Show that for any $M \in \text{Gl}_d(\mathbb{R})$ $\text{cond}(M) \geq 1$ and that, for any orthogonal matrix P , $\text{cond}(PM) = \text{cond}(M)$.
- (4) For any $M \in \text{Gl}_d(\mathbb{R})$ show that $\|M\|_{\text{op}} = \|M^T\|_{\text{op}}$.
- (5) Let $M \in \text{Gl}_d(\mathbb{R})$ be such that $\text{Cond}(M) = 1$. Show that there exists $x \in \mathbb{R}^*$ such that xM is an orthogonal matrix.

Exercise 2.4. Prove Theorem 4.1.

Exercise 2.5. Let $f \in \mathcal{C}^1(\mathbb{R}^d; \mathbb{R})$ be bounded from below, satisfy the Polyak-Lojasiewicz condition with constant α :

$$\forall x \in \mathbb{R}^d, f(x) - \inf_{\mathbb{R}^d} f \leq \frac{1}{2\alpha} \|\nabla f(x)\|^2.$$

Assume that ∇f is μ -Lipschitz. For any $\tau \in (0; \frac{1}{2\mu})$ any $x_0 \in \mathbb{R}^n$, let $\{x_k\}_{k \in \mathbb{N}}$ be the sequence generated by the gradient descent initialised at x_0 with fixed step size τ . Show that

$$\forall k \in \mathbb{N}, f(x_{k+1}) - \inf f \leq (1 - \tau\alpha)^{k+1} (f(x_0) - \inf f).$$

Exercise 2.6. The goal of this exercise is to show the convergence of the line-search gradient descent for quadratic functions.

- (1) Preliminary: Kantorovich inequality Let $A \in S_d^{++}(\mathbb{R})$ with eigenvalues $0 < \lambda_1 \leq \dots \leq \lambda_d$. Show that

$$\forall x \in \mathbb{R}^d \setminus \{0\}, \|x\|^4 \leq \langle Ax, x \rangle \cdot \langle A^{-1}x, x \rangle \leq \frac{\|x\|^4}{4} \cdot \frac{(\lambda_1 + \lambda_d)^2}{\lambda_1 \lambda_d}.$$

- (2) Let $A \in S_d^{++}(\mathbb{R})$ and $b \in \mathbb{R}^d$. Let $x \in \mathbb{R}^d$. Solve the optimisation problem¹

$$\min_{\tau > 0} f(x - \tau \nabla f(x)).$$

¹In particular, show existence and uniqueness of the optimiser

- (3) *We now consider the sequence generated by the line search algorithm. Using the explicit expression of the step size obtained at the previous question and defining, for any $k \in \mathbb{N}$, $y_k := A(x_k - x^*)$, show that*

$$\forall k \in \mathbb{N}, \langle y_{k+1}, x_{k+1} - x^* \rangle = \langle y_k, x_k - x^* \rangle \cdot \left(1 - \frac{\|y_k\|^4}{\langle Ay_k, y_k \rangle \langle A^{-1}y_k, y_k \rangle} \right).$$

- (4) *Conclude the proof.*

SOLUTIONS OF THE EXERCISES OF PART 2

Solution 2.1. (1) Let $M := \frac{A+A^T}{2}$. If the gradient with fixed step size $\tau > 0$ converges, it converges to a solution x^* of $Mx^* = b$. However, $b \in \text{Im}(M)^\perp \setminus \{0\}$, so that no such x^* exists. Thus, the gradient descent can not converge.

(2) Without loss of generality, we can assume that A is diagonal, $A = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_d \end{pmatrix}$.

If we assume that (for instance) $\lambda_1 < 0$, then, letting $x_0 = e_1$ be the associated first eigenvector and $b = 0$, we deduce that for any $k \in \mathbb{N}$ and any $\tau > 0$ there holds $x_k = (1 - \tau\lambda_1)^k x_0$, and it thus does not converge. Thus, for any i , $\lambda_i \geq 0$. If we assume, for instance, that $\lambda_1 = 0$, then taking $b = e_1$ and any x_0 , the same reasoning as in the previous question shows that the gradient descent can not converge. Thus, for any i , $\lambda_i > 0$ and so $A \in S_d^{++}(\mathbb{R})$.

Solution 2.2. Up to a change of basis, we can assume that $A = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_d \end{pmatrix}$

with $\lambda_1 < 0$. Letting $x_{k,1}$ be the first coordinate of the k -th iterate we observe that $x_{k,1} = (1 - \tau\lambda_1)^k x_{0,1}$. Thus, the set of initial conditions that guarantee convergence is included in $e_1^\perp$, which has Lebesgue measure zero (why? prove it), thereby concluding the proof.

Solution 2.3. (1) Trivial.

(2) It suffices to observe that, if M is positive definite, then $\lambda_d(M) = \|M\|_{\text{op}}$. Indeed, working in a diagonalisation basis,

$$\forall x \in \mathbb{R}^d, \|Mx\|^2 = \sum_{i=1}^d \lambda_i^2 x_i^2$$

and the conclusion follows.

(3) As $\|AB\|_{\text{op}} = 1$ we obtain $\text{cond}(M) \geq \|MM^{-1}\|_{\text{op}} = 1$. Likewise, as the operator norm is invariant by conjugation by an orthogonal matrix the second property follows.

(4)

$$\begin{aligned} \|M\|_{\text{op}}^2 &= \sup_{\|x\|^2=1} \langle x, M^T M x \rangle \\ &= \lambda_d(M^T M) \end{aligned}$$

by the Rayleigh quotient formulation of eigenvalues. Furthermore, $\det(AB - \text{Id}) = \det(BA - \text{Id})$ so that AB and BA have the same spectral radius. The conclusion follows.

(5) The conclusion is obvious if M is symmetric positive definite. In general let $A = MM^T$. By the previous questions we obtain $\text{cond}(A) = 1$. As A is symmetric, positive definite, $A = x^{-2}\text{Id}$ for some $x \in \mathbb{R}^*$, whence xM is orthogonal.

Solution 2.4. We assume in particular that

$$f(x^*) = 0, \quad \nabla^2 f(x^*) \in S_d^{++}(\mathbb{R}).$$

The first idea that comes to mind (and which works) to establish the convergence of the algorithm is as follows for any $x, y \in \mathbb{R}^d$ we have

$$\nabla f(y) = \nabla f(x) + \int_0^1 \nabla^2 f(x + t(y - x)) \cdot (y - x) dt.$$

In particular, with $x = x^*$ we obtain

$$f(y) = \int_0^1 \nabla^2 f(x^* + t(y - x^*)) \cdot (y - x^*) dt.$$

Thus, for any $k \in \mathbb{N}$,

$$x_{k+1} - x^* = (x_k - x^*) - \tau \nabla f(x_k) = (x_k - x^*) - \tau \int_0^1 \nabla^2 f(x^* + t(x_k - x^*)) \cdot (x_k - x^*) dt.$$

This expression can be rewritten in matrix form as

$$x_{k+1} - x^* = \left(I_d - \tau \int_0^1 \nabla^2 f(x^* + t(x_k - x^*)) dt \right) (x_k - x^*).$$

We introduce the matrix

$$M_k := I_d - \tau \int_0^1 \nabla^2 f(x^* + t(x_k - x^*)) dt.$$

We would like to use the information we have about $\nabla^2 f(x^*)$. We then use the fact that for all $A, B \in S_d(\mathbb{R})$, we have

$$\max(|\lambda_1(A) - \lambda_1(B)|, |\lambda_d(A) - \lambda_d(B)|) \leq \|A - B\|_{\text{op}}.$$

We refer to Exercise 1.2. Let

$$\ell_0(x^*) := \frac{\lambda_1(\nabla^2 f(x^*))}{2}, \quad \ell_1(x^*) := 2\lambda_d(\nabla^2 f(x^*)).$$

Thus, there exists $\epsilon > 0$ such that, for all $x \in B(x^*, \epsilon)$,

$$\ell_0(x^*) \leq \lambda_1(\nabla^2 f(x)) \leq \lambda_d(\nabla^2 f(x)) \leq \ell_1(x^*).$$

In particular, if we set

$$M(x) := I_d - \tau \int_0^1 \nabla^2 f(x^* + t(x - x^*)) dt,$$

we recover that, for the same $\epsilon > 0$, and for all $x \in B(x^*, \epsilon)$,

$$1 - \tau \ell_1(x^*) \leq \lambda_1(M(x)) \leq \lambda_d(M(x)) \leq 1 - \tau \ell_0(x^*),$$

and thus we deduce that, if $x_k \in B(x^*, \epsilon)$,

$$\|x_{k+1} - x^*\| \leq \max(|1 - \tau \ell_1(x^*)|, |1 - \tau \ell_0(x^*)|) \|x_k - x^*\|.$$

Thus, if $\tau < \frac{1}{\ell(x^*)}$, we indeed recover a linear convergence rate for gradient descent.

Of course, we can refine all this information as we did previously in the case of a quadratic function, and we expect, when optimizing the step size, to have a convergence rate of

$$\frac{4\lambda_d(\nabla^2 f(x^*)) - \lambda_1(\nabla^2 f(x^*))}{4\lambda_d(\nabla^2 f(x^*)) + \lambda_1(\nabla^2 f(x^*))}.$$

Once again, we encounter the problem of the conditioning of the Hessian $\nabla^2 f(x^*)$.

Solution 2.5. First observe that

$$\begin{aligned}
 f(x_{k+1}) &= f(x_k) + \int_0^1 \langle \nabla f((1-t)x_k + tx_{k+1}), x_{k+1} - x_k \rangle dt \\
 &\leq f(x_k) + \langle \nabla f(x_k), x_{k+1} - x_k \rangle + \frac{1}{2} \mu \|x_{k+1} - x_k\|^2 \\
 &= f(x_k) - \tau \|\nabla f(x_k)\|^2 + \frac{\mu}{2} \|x_{k+1} - x_k\|^2 \\
 &= f(x_k) - \tau \|\nabla f(x_k)\|^2 + \frac{\mu}{2} \tau^2 \|\nabla f(x_k)\|^2 \\
 &\leq f(x_k) + \tau(-1 + \frac{\mu}{2}) \|\nabla f(x_k)\|^2.
 \end{aligned}$$

Since $\tau\mu < 1$, $(-1 + \frac{\mu}{2}) \leq \frac{1}{2}$ whence

$$f(x_{k+1}) \leq f(x_k) - \frac{\tau}{2} \|\nabla f(x_k)\|^2.$$

By the Polyak-Lojasiewicz inequality this yields

$$f(x_{k+1}) \leq f(x_k) - \alpha\tau(f(x_k) - \inf f) = (1 - \alpha\tau)(f(x_k) - \inf f) + \inf f.$$

A straightforward iteration yields the conclusion.

Solution 2.6. (1) Assume $\|x\| = 1$ and that $A = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_d \end{pmatrix}$. We have

to show that for any $\{x_i\}_{i=1,\dots,d}^2$ such that $\sum x_k^2 = 1$ there holds

$$1 \leq \sum_{i=1}^d \lambda_i x_i^2 \sum_{j=1}^d \frac{x_j^2}{\lambda_j} \leq \frac{1}{4} \frac{(\lambda_1 + \lambda_d)^2}{\lambda_1 \lambda_d}.$$

The left-hand inequality is the Cauchy-Schwarz inequality. As for the right-hand inequality, observe that for any $\delta > 0$

$$2 \sqrt{\sum_{i=1}^d \lambda_i x_i^2 \sum_{j=1}^d \frac{x_j^2}{\lambda_j}} \leq \sum_{i=1}^d x_i^2 \left(\frac{1}{\delta \lambda_i} + \delta \lambda_i \right) = \sum_{i=1}^d x_i^2 \phi(\delta \lambda_i)$$

with $\phi(x) = x + \frac{1}{x}$. As ϕ is convex, it is maximal at λ_1 or λ_d whence

$$2 \sqrt{\sum_{i=1}^d \lambda_i x_i^2 \sum_{j=1}^d \frac{x_j^2}{\lambda_j}} \leq \max\{\phi(\delta \lambda_1), \phi(\delta \lambda_d)\}.$$

Choosing $\delta > 0$ such that $\phi(\delta \lambda_1) = \phi(\delta \lambda_d)$ provides the answer.

- (2) We obtain after explicit computations (the existence and uniqueness of $\tau^* > 0$ being guaranteed by the strict convexity of the underlying functional) that

$$\tau^* = \frac{\|Ax - b\|^2}{\langle A(Ax - b), Ax - b \rangle}.$$

- (3) Explicit computations.

- (4) A simple iterative argument allows to conclude.

Part 3. Constrained optimisation: theory and approximation

5. THE LAGRANGE MULTIPLIER RULE I: EQUALITY CONSTRAINTS AND FIRST DISCUSSION

In this part, we will be exploring constrained optimisation. Namely, we want to study optimisation problems of the form

$$\min_{x \in K} f(x)$$

where K is a strict subset of $\mathbb{R}^d$. What is important to notice is that the set K , called *the set of constraints* or *admissible set*, will sometimes be treated as an abstract object (so, merely a subregion assumed to satisfy certain topological or geometric properties like convexity or compactness) and will sometimes be described by a set of (in)equalities. As we will see when dealing with inequality constraints, it can sometimes be useful to go back and forth between the two presentations of the set K . To begin softly, we will begin by investigating the situation where K can be described by a single equality. Namely, in this section, we assume that there exists a coercive function φ such that

$$K = \{\varphi = 0\}.$$

Observe that the coercivity assumption implies that the set K is compact. We will mostly be focusing on first and second-order optimality conditions, so that the function f to be optimised is assumed to satisfy $f \in \mathcal{C}^2(\mathbb{R}^d)$. Observe that the optimality conditions $\nabla f = 0, \nabla^2 f \in S_d^+$ can no longer be satisfied: regarding the condition $\nabla f = 0$, this was derived using that, for any admissible x , there was an entire ball $\mathbb{B}(x; \varepsilon)$ of admissible perturbations. In other words, the set of admissible perturbations was d dimensional. Here, under the constraints $\varphi = 0$, we no longer have access to all directions but, provided φ is well-behaved enough, only to $(d-1)$ independent directions, so that the optimality condition will necessarily be weaker.

5.1. The Lagrange multiplier rule for one equality constraint: discussion and proof. Before discussing it in more details and providing examples, let us state the following theorem, which gives first-order optimality conditions for

$$(5.1) \quad \min_{x, \varphi(x)=0} f(x).$$

The Lagrange multiplier rule reads as follows:

Theorem 5.1. *Assume that x^* solves (5.1) and that $f, \varphi \in \mathcal{C}^1(\mathbb{R}^d; \mathbb{R})$. If*

$$\nabla \varphi(x^*) \neq 0$$

then there exists $\lambda \in \mathbb{R}$ (called the Lagrange multiplier at x^) such that*

$$(5.2) \quad \nabla f(x^*) + \lambda \nabla \varphi(x^*) = 0.$$

Remark 5.1. *It should be noted that the conclusion of Theorem 5.1 is a **local** optimality condition. In particular, the conclusion holds if x^* is only a **local** minimiser of f in $\{\varphi = 0\}$.*

We will give one proof of the Lagrange multiplier theorem that relies on the implicit function theorem, and which will be essentially similar for the case of multiple equality constraints, but we first discuss it geometrically: what does the condition (5.2) mean? To understand it, we work at a point $x \in \{\varphi \neq 0\}$ such that

$\nabla\varphi(x) \neq 0$, and we assume that, locally, $K = \{\varphi = 0\}$ is a $(d - 1)$ -dimensional submanifold of $\mathbb{R}^d$. We admit that, for a given point $x \in K$, the tangent plane to K at x is described algebraically by the equation

$$T_x K = \ker(z \mapsto \langle \nabla\varphi(x), z - x \rangle).$$

Observe that this indeed defines a $(d - 1)$ -dimensional hyperplane whose normal vector is $\nabla\varphi(x)$. We prove this (and give a proper definition of a tangent plane) in Exercise 3.4.

Now, assume x^* is a solution of the optimisation problem, and consider the level set

$$A_\varepsilon := \{f = f(x^*) - \varepsilon\}.$$

By definition, for any $\varepsilon > 0$, $A_\varepsilon \cap K = \emptyset$, but $\{x^*\} \subset A_0 \cap K$. Thus, we might interpret x^* as a “point of first contact” between A_ε and K . If we represent the situation graphically, this means that the normal vector to A_0 at x^* and the normal vector to K at x^* must coincide. As the former is given by $\nabla f(x^*)$ (assuming without loss of generality that $\nabla f(x^*) \neq 0$) and the latter by $\nabla\varphi(x^*)$, it means that the two vectors are co-linear or, in other words, that there exists $\lambda \in \mathbb{R}$ such that

$$\nabla f(x^*) + \lambda \nabla\varphi(x^*) = 0.$$

Intuitive as it is, this is not a proof, since we did not properly define what a tangent plane is, or characterised it as the orthogonal of the gradient.

There is another case that we can deal with directly: assume that the set of admissible points is a graph, namely,

$$K = \{\varphi = 0\} = \{(y, g(y)), y \in \mathbb{R}\} \subset \mathbb{R}^2.$$

The constrained optimisation problem becomes an unconstrained optimisation problem:

$$\min_{y \in \mathbb{R}} f(y, g(y)),$$

and the optimality conditions then read

$$\frac{\partial f}{\partial x} + g'(y) \frac{\partial f}{\partial y} = 0.$$

Furthermore, as

$$\varphi(y, g(y)) = 0$$

we deduce that

$$\frac{\partial \varphi}{\partial x} + g'(y) \frac{\partial \varphi}{\partial y} = 0$$

so that the optimality condition above becomes

$$\frac{\partial f}{\partial y} \frac{\partial \varphi}{\partial x} = \frac{\partial f}{\partial x} \frac{\partial \varphi}{\partial y}.$$

But if $\nabla\varphi \neq 0$, this is equivalent to the existence of a $\lambda \in \mathbb{R}$ such that

$$\nabla f + \lambda \nabla\varphi = 0.$$

Using the implicit function theorem, we will show that this simple computation actually holds in all generality.

5.1.1. *The implicit function theorem and the proof of Theorem 5.1.* Before we prove Theorem 5.1, we recall the following simple version of the implicit function theorem:

Theorem 5.2. *Let $\varphi : \mathbb{R}^{d-1} \times \mathbb{R} \rightarrow \mathbb{R}$ be of class $\mathcal{C}^1$, and let $(x_0, y_0) \in \mathbb{R}^{d-1} \times \mathbb{R}$ be such that*

$$\varphi(x_0, y_0) = 0, \partial_y \varphi(x_0, y_0) \neq 0.$$

Then, there exist two open sets $U_0 \subset \mathbb{R}^{d-1}$ and $U_1 \subset \mathbb{R}$ which are open neighbourhoods of x_0 and y_0 respectively, and a $\mathcal{C}^1$ function $g : U_0 \rightarrow U_1$ such that

$$\{(x, y) \in U_0 \times U_1, \varphi(x, y) = 0\} = \{(x, g(x)), x \in U_0\}.$$

This theorem is proved in Exercise 3.5, where we give a proof that (only) relies on the Banach fixed point theorem.

Proof of Theorem 5.1. Let x^* be a solution of (5.1) where $\nabla \varphi(x^*) \neq 0$. Up to relabelling coordinates, we can assume that

$$\frac{\partial \varphi}{\partial x_d}(x^*) \neq 0,$$

and we write $\mathbb{R}^d \ni x = (x', x_d) \in \mathbb{R}^{d-1} \times \mathbb{R}$. Up to a translation, we assume that $x^* = (0, 0)$. We are thus in a position to apply the implicit function theorem, and to find $\varepsilon > 0$ such that

$$\{\varphi = 0\} \cap \mathbb{B}_{\mathbb{R}^{d-1}}(0; \varepsilon) \times (-\varepsilon; \varepsilon) = \{(x', g(x')), x \in \mathbb{B}_{\mathbb{R}^{d-1}}(0; \varepsilon)\}$$

for some $\mathcal{C}^1$ function g . In particular, we deduce that $0 \in \mathbb{R}^{d-1}$ is an optimiser of

$$F : x' \mapsto f(x', g(x')).$$

Now, as 0 is an interior point of $\mathbb{B}(0; \varepsilon)$, we conclude that

$$\nabla F(x') = 0,$$

which is equivalent to

$$\forall i \in \{1, \dots, d-1\}, \frac{\partial f}{\partial x_i} + \frac{\partial g}{\partial x_i} \frac{\partial f}{\partial x_d} = 0.$$

But, from

$$\varphi(x', g(x')) = 0$$

we also obtain

$$\forall i \in \{1, \dots, d-1\}, \frac{\partial \varphi}{\partial x_i} + \frac{\partial g}{\partial x_i} \frac{\partial \varphi}{\partial x_d} = 0.$$

Combining these two equations, we deduce that

$$\forall i \in \{1, \dots, d-1\}, \frac{\partial f}{\partial x_i} = \frac{\frac{\partial f}{\partial x_d}}{\frac{\partial \varphi}{\partial x_d}} \frac{\partial \varphi}{\partial x_i}.$$

As this is also true for $i = d$, we deduce, with $\lambda = -\frac{\frac{\partial f}{\partial x_d}}{\frac{\partial \varphi}{\partial x_d}}$ that

$$\nabla f(x^*) + \lambda \nabla \varphi(x^*) = 0.$$

□

5.1.2. *What happens if $\nabla\varphi = 0$?* So one might ask: is the condition $\nabla\varphi(x^*) \neq 0$ necessary? As it turns out, yes. If we look for a counter-example in two dimensions, it means that we have to look for a set K with a “multiple point”, meaning that the admissible set branches out from the minimiser in several directions, and a function f with a non-zero gradient at that point. It suffices to take, for the set K ,

$$K := \{(x, y) : (y - x^{2k})(y - x^{2j}) = 0\}$$

and $f(x, y) = y$. Then, just by doing a drawing, it appears that

$$\min_K f = 0$$

and that this minimum is just reached at 0, and, setting $\varphi(x, y) := (y - x^2)(y - x^4)$, we observe that, if there were a Lagrange multiplier λ at $(0, 0)$, then we would have

$$0 = \nabla f(0, 0) + \lambda \nabla \varphi(0, 0) = \nabla f(0, 0) = (0, 1)^T,$$

which is absurd.

5.2. **An example.** Consider the following problem in $\mathbb{R}^2$:

$$\min_{x^2+y^2=1} 2x + y$$

Using the previous notation, we have $f(x, y) = 2x + y$ and $K = \{(x, y) : \varphi(x, y) = 0\}$, with $\varphi(x, y) = x^2 + y^2 - 1$. Since K is compact and non-empty (because $(1, 0) \in K$), and f is continuous, the problem has at least one solution $(x^*, y^*) \in K$.

Furthermore, $\nabla\varphi(x, y) = (2x, 2y)^T$ is non-zero for any $(x, y) \in K$. Theorem 5.1 states that if (x^*, y^*) is a minimiser of f on K , then there exists $\lambda \in \mathbb{R}$ such that

$$\nabla f(x^*, y^*) + \lambda \nabla \varphi(x^*, y^*) = 0 \quad \Leftrightarrow \quad \begin{cases} 2 + 2\lambda x^* = 0 \\ 1 + 2\lambda y^* = 0 \\ (x^*)^2 + (y^*)^2 = 1 \end{cases}$$

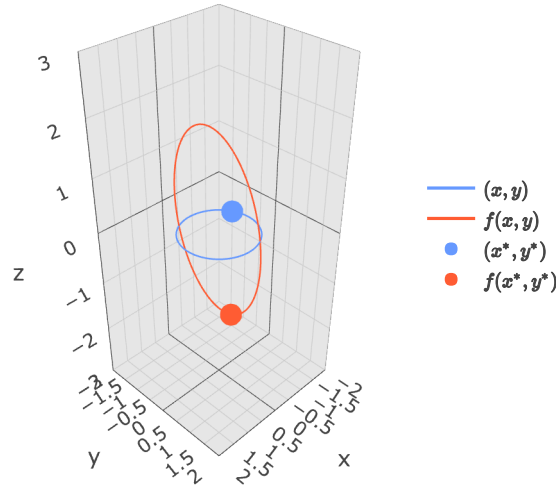
From the first two equations, we deduce that λ is non-zero, and that $x^* = -1/\lambda$, $y^* = -1/(2\lambda)$. Injecting this in the last equality, we obtain

$$(1/\lambda)^2 + (1/(2\lambda))^2 = 1$$

i.e. $\lambda = \sqrt{5}/2$ or $\lambda = -\sqrt{5}/2$. In the first case, $(x^*, y^*) = (-2\sqrt{5}/5, -\sqrt{5}/5)$, and in the second case $(x^*, y^*) = (2\sqrt{5}/5, \sqrt{5}/5)$. The set of points satisfying the necessary optimality conditions is $\{(2\sqrt{5}/5, \sqrt{5}/5), (-2\sqrt{5}/5, -\sqrt{5}/5)\}$. We know that the minimisers of the problem lie among these points. We thus compute the value of f to determine the minimisers:

$$f(2\sqrt{5}/5, \sqrt{5}/5) = \sqrt{5} \text{ and } f(-2\sqrt{5}/5, -\sqrt{5}/5) = -\sqrt{5}$$

So there is only one minimiser: the point $(-2\sqrt{5}/5, -\sqrt{5}/5)$. An illustration of this example is given in Figure II.1.


 FIGURE 1. Illustration of the problem $\min_{x^2+y^2=1} 2x + y$

5.3. The Lagrange multiplier rule: arbitrary number of equality constraints. The analysis of the previous paragraph can be readily extended to an arbitrary number of equality constraints with of course the limitation that we should be allowed to apply the implicit function theorem. Thus, we work with a finite family

$$\varphi_1, \dots, \varphi_m \in \mathcal{C}^1(\mathbb{R}^d; \mathbb{R})$$

where $m \leq d$, and we set

$$K = \cap_{i=1}^m \{\varphi_i = 0\}.$$

The problem is still

$$(5.3) \quad \min_K f$$

for a $\mathcal{C}^1$ function f . Our main assumption is the following: letting $x^* \in K$ be a solution of (5.3), we assume

$$(5.4) \quad \text{The family } \{\nabla \varphi_i(x^*)\}_{1 \leq i \leq m} \text{ is linearly independent.}$$

The main theorem is the following:

Theorem 5.3. *Let x^* be a solution of (5.3) where (5.4) is satisfied. There exist $\lambda_1, \dots, \lambda_m \in \mathbb{R}$ such that*

$$\nabla f(x^*) + \sum_{i=1}^m \lambda_i \nabla \varphi_i(x^*) = 0.$$

The same observation as before holds: it is not possible to bypass assumption (5.4).

Remark 5.2. *Remark 5.1 still holds: the conclusion of Theorem 5.3 still holds if x^* is only a local minimiser of f on K .*

The proof of Theorem 5.3 is slightly more complicated than that of Theorem 5.1, but similar. We recall the following, more general, version of the implicit function theorem:

Theorem 5.4. *Let $\Omega := \mathbb{B}_{\mathbb{R}^{d-m}}(0; \varepsilon) \times \mathbb{B}_{\mathbb{R}^m}(0; \varepsilon') \subset \mathbb{R}^{d-m} \times \mathbb{R}^m$ be an open neighbourhood of 0 in $\mathbb{R}^d$ (identified with $\mathbb{R}^{d-m} \times \mathbb{R}^m$). Let $F \in \mathcal{C}^k(\Omega; \mathbb{R}^m)$ ($k \geq 1$) and assume that $\nabla_y F(0, 0) \in \mathcal{L}(\mathbb{R}^m; \mathbb{R}^m)$ is invertible. Then there exist a neighborhood N of 0 in $\mathbb{R}^d$, $\varepsilon' > 0$ and $g_1, \dots, g_m : \mathbb{B}_{\mathbb{R}^{d-m}}(0; \varepsilon') \rightarrow \mathbb{R}$ of class $\mathcal{C}^k$ such that*

$$N \cap \{F = F(0, 0)\} = \{(x', g_1(x'), \dots, g_m(x')) , x' \in \mathbb{B}_{\mathbb{R}^{d-m}}(0; \varepsilon')\}.$$

Proof of Theorem 5.3. Owing to assumption (5.4) and Theorem 5.4, it suffices to treat the case where $x^* = 0$ and

$$K = \{(x_1, \dots, x_{d-m}, g_1(x_1, \dots, x_{d-m}), \dots, g_m(x_1, \dots, x_{d-m})), \\ (x_1, \dots, x_{d-m}) \in \mathbb{B}_{\mathbb{R}^{d-m}}(0; \varepsilon)\}$$

where $g_1, \dots, g_m$ are $\mathcal{C}^1$ functions. Indeed, Assumption (5.4) implies that the matrix $(\nabla \varphi_1(x^*), \dots, \nabla \varphi_m(x^*))$ has rank m and, up to relabelling coordinates, we can assume that the matrix $\left(\frac{\partial \varphi_i}{\partial x_{d-m+j}}\right)_{1 \leq i, j \leq m}$ is invertible

Let us first deal (to fix ideas) treat the case $d = 3$, $m = 2$. Then the unconstrained optimisation problem associated with (5.3) is

$$\min_{x \in (-\varepsilon; \varepsilon)} f(x, g_1(x), g_2(x)) = f(0, 0, 0)$$

and the optimality conditions at $x = 0$ are given by

$$(5.5) \quad \frac{\partial f}{\partial x} + g'_1(0) \frac{\partial f}{\partial y} + g'_2(0) \frac{\partial f}{\partial z} = 0.$$

On the other hand, differentiating $\varphi_1(x, g_1(x), g_2(x)) = \varphi_2(x, g_1(x), g_2(x)) = 0$ we obtain

$$(5.6) \quad \begin{cases} \frac{\partial \varphi_1}{\partial x} + g'_1(0) \frac{\partial \varphi_1}{\partial y} + g'_2(0) \frac{\partial \varphi_1}{\partial z} = 0, \\ \frac{\partial \varphi_2}{\partial x} + g'_1(0) \frac{\partial \varphi_2}{\partial y} + g'_2(0) \frac{\partial \varphi_2}{\partial z} = 0. \end{cases}$$

In particular, if we introduce the vector

$$X_1 := \begin{pmatrix} 1 \\ g'_1(0) \\ g'_2(0) \end{pmatrix}$$

we deduce that

$$\nabla \varphi_1(x^*), \nabla \varphi_2(x^*) \in X_1^\perp.$$

As $X_1^\perp$ is two-dimensional, we deduce from Assumption (5.4) that

$$X_1^\perp = \{\lambda_1 \nabla \varphi_1(x^*) + \lambda_2 \nabla \varphi_2(x^*), \lambda_1, \lambda_2 \in \mathbb{R}\}.$$

Now, observe that (5.5) can be rewritten

$$\nabla f(x^*) \in X_1^\perp.$$

The conclusion follows. In the case of m arbitrary constraints we obtain, by the same reasoning, the following optimality condition:

$$(5.7) \quad \forall i \in \{1, \dots, d-m\}, \frac{\partial f}{\partial x_i} + \sum_{k=d-m+1}^d \frac{\partial g_{k-d+m}}{\partial x_i} \cdot \frac{\partial f}{\partial x_d} = 0$$

while, differentiating

$$\forall j, \forall x_1, \dots, x_{d-m}, \varphi_j(x_1, \dots, x_{d-m}, g_1(x_1, \dots, x_{d-m}), \dots, g_m(x_1, \dots, x_{d-m})) = 0$$

we obtain

$$(5.8) \quad \forall i \in \{1, \dots, d-m\}, \forall j \in \{1, \dots, m\}, \frac{\partial \varphi_j}{\partial x_i} + \sum_{k=d-m+1}^d \frac{\partial g_{k-d+m}}{\partial x_i} \cdot \frac{\partial \varphi_j}{\partial x_d} = 0$$

Thus, if we introduce the $(d-m)$ vectors

$$X_i = \left(0, \dots, 0, 1, 0, \dots, 0, \frac{\partial g_1}{\partial x_i}, \dots, \frac{\partial g_m}{\partial x_i} \right)^T$$

we deduce

$$\forall j \in \{1, \dots, m\}, \langle \nabla \varphi_j(x^*), X_i \rangle = 0.$$

Letting $V := \text{Span}(X_1, \dots, X_{d-m})$, this gives (with a small dimension argument)

$$V^\perp = \text{Span}\{\nabla \varphi_j(x^*), j = 1, \dots, m\}.$$

As (5.7) reads $\nabla f(x^*) \in V^\perp$, the conclusion follows. $\square$

5.3.1. *An example of application.* Consider the following problem in $\mathbb{R}^3$:

$$\begin{aligned} \min \quad & x + z \\ \text{s.t.} \quad & x^2 + y^2 = 1 \\ & y^2 + z^2 = 4 \end{aligned}$$

Here, $f(x, y, z) = x + z$ and $K = \{(x, y, z) : \Phi(x, y, z) = 0\}$, with $\Phi(x, y, z) = (\varphi_1(x, y, z), \varphi_2(x, y, z))$ and $\varphi_1(x, y, z) = x^2 + y^2 - 1$, $\varphi_2(x, y, z) = y^2 + z^2 - 4$. The existence of an optimiser is straightforward.

The points $(x, y, z) \in K$ where Theorem 5.3 apply are those for which the vectors $\nabla \varphi_1(x, y, z), \nabla \varphi_2(x, y, z)$ are linearly independent. Observe that

$$\nabla \varphi_1(x, y, z) = \begin{pmatrix} 2x \\ 2y \\ 0 \end{pmatrix}, \quad \nabla \varphi_2(x, y, z) = \begin{pmatrix} 0 \\ 2y \\ 2z \end{pmatrix}$$

These two vectors are linearly dependent if and only if at least two coordinates among x, y, z are zero. But we cannot have $x = y = 0$ since $x^2 + y^2 = 1$, nor $y = z = 0$ since $y^2 + z^2 = 4$, nor $x = z = 0$ since otherwise we would have $y^2 = 1$ and $y^2 = 4$. So for all $(x, y, z) \in K$ the above vectors are linearly independent. Theorem 5.3 states that if (x^*, y^*) is a minimiser of f on K , then there exist $\lambda_1, \lambda_2 \in \mathbb{R}$ such that

$$\nabla f(x^*, y^*, z^*) + \lambda_1 \nabla \varphi_1(x^*, y^*, z^*) + \lambda_2 \nabla \varphi_2(x^*, y^*, z^*) = 0$$

That is,

$$\begin{cases} 1 + 2\lambda_1 x^* = 0 \\ 2\lambda_1 y^* + 2\lambda_2 y^* = 0 \\ 1 + 2\lambda_2 z^* = 0 \\ (x^*)^2 + (y^*)^2 = 1 \\ (y^*)^2 + (z^*)^2 = 4. \end{cases}$$

Note that λ_1 and λ_2 are non-zero and that

$$x^* = -1/(2\lambda_1) \text{ and } z^* = -1/(2\lambda_2)$$

Moreover either $\lambda_1 + \lambda_2 = 0$, or $y^* = 0$. The first case is impossible, because then $x^* = -z^*$ and

$$(y^*)^2 = 1 - (x^*)^2 = 1 - (z^*)^2 = 4 - (z^*)^2,$$

which leads to $1 = 4$ and a contradiction. So $y^* = 0$, which imposes $x^* = \pm 1$ and $z^* = \pm 2$. It is easy to see that the minimiser of the problem is $(-1, 0, -2)$.

5.4. A small comment: what happens when there are other constraints?

It can be the case, in applications, that one also has to take into account inequality constraints of the form

$$\psi(x) \geq 0.$$

In general, it is not possible to get rid of these constraints (and this will be the topic of the next chapter) but, in certain cases, it can be shown that at any optimiser one actually has $\psi(x^*) > 0$ so that one can simply be satisfied with considering the equality constraint. Some such examples will be discussed in the exercises sessions.

5.5. The Lagrange multiplier rule: second-order optimality conditions.

To conclude this section, we discuss second-order optimality conditions for equality constrained problems. Once again, we work on (5.3) and we assume that is satisfied. We further assume that

$$f, \varphi_1, \dots, \varphi_n \in \mathcal{C}^2(\mathbb{R}^d; \mathbb{R}).$$

The main theorem is the following:

Theorem 5.5. *Assume that x^* solves (5.3) and that (5.4) is satisfied at x^* . Let $\lambda_1, \dots, \lambda_m$ be the associated Lagrange multipliers. Then the matrix*

$$\nabla^2 f(x^*) + \sum_{i=1}^m \lambda_i \nabla^2 \varphi_i(x^*)$$

is non-negative on the tangent space

$$\cap_{i=1}^m (\nabla \varphi_i(x^*))^\perp.$$

Conversely, let $x^ \in \cap_{i=1}^m \{\varphi_i = 0\}$ be a point where (5.4) is satisfied, and assume that there exist $\lambda_1, \dots, \lambda_m$ such that $\nabla f(x^*) + \sum_{i=1}^m \lambda_i \nabla \varphi_i(x^*) = 0$. If, in addition, $\nabla^2 f(x^*) + \sum_{i=1}^m \lambda_i \nabla^2 \varphi_i(x^*)$ is positive definite on the tangent space*

$$\cap_{i=1}^m (\nabla \varphi_i(x^*))^\perp$$

then x^ is a strict local minimiser of f on $\cap_{i=1}^m \{\varphi_i = 0\}$.*

For the sake of clarity, we will only prove this theorem in the case of one-constraint, in dimension 2. The general case is proved in Exercise 3.4.

Sketch of proof of Theorem 5.5. As we said, we are only going to work in $\mathbb{R}^2$, with one constraint $\{\varphi = 0\}$ and, using the implicit function theorem, we actually assume that the set is locally given by a graph *i.e.*, in a small ball, $\{\varphi = 0\} = \{(y, g(y)), y \in (-\varepsilon; \varepsilon)\}$, $\partial_y \varphi \neq 0$. We assume, for further simplification, that $x^* = (0, g(0))$, and we let λ be the Lagrange multiplier associated with the constraint.

Observe (for future reference) that a tangent vector to any $x = (y, g(y)) \in \{\varphi = 0\}$ is $(1, g'(y))^T$.

By assumption $y = 0$ is a local minimiser of the map

$$F : y \mapsto f(y, g(y)).$$

We thus deduce that

$$\frac{\partial^2 f}{\partial x^2} + 2 \frac{\partial^2 f}{\partial x \partial y} g'(y) + g'(y)^2 \frac{\partial^2 f}{\partial y^2} + g''(y) \frac{\partial f}{\partial y} \geq 0.$$

We also have, differentiating $y \mapsto \varphi(y, g(y))$,

$$\frac{\partial^2 \varphi}{\partial x^2} + 2 \frac{\partial^2 \varphi}{\partial x \partial y} g'(y) + g'(y)^2 \frac{\partial^2 \varphi}{\partial y^2} + g''(y) \frac{\partial \varphi}{\partial y} = 0.$$

Now, this implies that

$$g''(x) = - \frac{\partial_{xx}^2 \varphi + 2 \partial_{xy}^2 \varphi g'(y) + g'(y)^2 \partial_{yy}^2 \varphi}{\partial_y \varphi}.$$

Keeping in mind that the Lagrange multiplier λ is given by

$$\lambda = - \frac{\partial_y f}{\partial_y \varphi}$$

we thus deduce

$$\frac{\partial^2 f}{\partial x^2} + 2 \frac{\partial^2 f}{\partial x \partial y} g'(y) + g'(y)^2 \frac{\partial^2 f}{\partial y^2} + \lambda \left(\frac{\partial^2 \varphi}{\partial x^2} + 2 \frac{\partial^2 \varphi}{\partial x \partial y} g'(y) + g'(y)^2 \frac{\partial^2 \varphi}{\partial y^2} \right) \geq 0.$$

Defining

$$\mathcal{L} := f + \lambda \varphi$$

this gives

$$\frac{\partial^2 \mathcal{L}}{\partial x^2} + 2 \frac{\partial^2 \mathcal{L}}{\partial x \partial y} g'(y) + g'(y)^2 \frac{\partial^2 \mathcal{L}}{\partial y^2} \geq 0.$$

The last quantity, however, is nothing but

$$\left\langle \nabla^2 \mathcal{L} \begin{pmatrix} 1 \\ g'(y) \end{pmatrix}, \begin{pmatrix} 1 \\ g'(y) \end{pmatrix} \right\rangle$$

and we thus deduce that $\nabla^2 \mathcal{L}$ is non-negative on the tangent plane at $(0, 0)$. The conclusion follows. It is possible—but tedious—to use this construction for an arbitrary number of variables. However, a more concise approach is possible once we have the proper notion of tangent planes.

The second part of the theorem is also doable with this approach, but the proof does not bring anything new. $\square$

EXERCISES FOR THE FIRST PART OF PART 3

Exercise 3.1. *The goal is to prove the spectral theorem using constrained optimisation. To be more specific, we consider a matrix $A \in S_d(\mathbb{R})$, and we want to show that there exists an orthonormal basis $\{e_i\}_{i \in \{1, \dots, d\}}$ of eigenvectors of A . To this end, we study the following variational problem:*

$$(5.9) \quad \min_{\|x\|=1} \langle Ax, x \rangle.$$

- (1) *Show that a minimiser exists for (5.10), and call it e_1 . Show that e_1 is an eigenvector of A .*
- (2) *Study the minimisation problem*

$$(5.10) \quad \min_{\|x\|=1, \langle x, e_1 \rangle = 0} \langle Ax, x \rangle.$$

- (3) *Conclude by iterating the procedure.*

Exercise 3.2 (Some applications of the Lagrange multiplier theorem). (1) *Solve the minimisation problem*

$$\min_{\frac{x^2}{2} + \frac{y^2}{4} + \frac{z^2}{6} = 1} x + y + z.$$

- (2) *Show that for any $x_1, \dots, x_n \in \mathbb{R}$ we have*

$$\prod_{i=1}^n |x_i|^{\frac{1}{n}} \leq \frac{1}{n} \sum_{i=1}^n |x_i|.$$

You can solve the optimisation problem

$$\max_{x_i \geq 0, \sum x_k = 1} \prod x_k.$$

- (3) *Let $p > 0$ be given. Justify that, among all rectangles with perimeter p , one has maximal area, and determine it.*

Exercise 3.3 (The Hölder inequality from the point of view of constrained optimisation). *The goal of this exercise is to prove the Hölder inequality. Let $p \in (1; +\infty)$ and q be such that*

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Let $x_1, \dots, x_d \geq 0, y_1, \dots, y_d \geq 0$. Show that

$$\sum_{i=1}^d x_i y_i \leq \left(\sum_{k=1}^d x_k^p \right)^{\frac{1}{p}} \left(\sum_{k=1}^d y_k^q \right)^{\frac{1}{q}}$$

by studying the optimisation problem

$$\max_{y_1, \dots, y_d \in \mathbb{R}, \sum_{k=1}^d |y_k|^q = 1} \sum_{i=1}^d x_i y_i.$$

Exercise 3.4. [Around tangent planes] *In this section, we give the rigorous arguments for the geometric interpretation of Lagrange multipliers by studying the tangent plane at submanifolds of $\mathbb{R}^d$. You can use the implicit function theorem. We consider*

$$K := \varphi^{-1}(\{0\})$$

where $\varphi \in \mathcal{C}^1(\mathbb{R}^d; \mathbb{R})$, and we let $x \in K$ be such that $\nabla\varphi(x) \neq 0$. We define the tangent plane to K at x as

$$T_x K := \{\gamma'(0), \gamma \in \mathcal{C}^1((-\varepsilon; \varepsilon); \mathbb{R}^d), \gamma(0) = x, \forall s \in (-\varepsilon; \varepsilon), \gamma(s) \in K\}.$$

The goal of this exercise is to show that

$$(5.11) \quad T_x K = \nabla\varphi(x)^\perp.$$

- (1) Let $f \in \mathcal{C}^1(\mathbb{R}^d; \mathbb{R})$, and assume (with the notations from above) that x solves

$$\min_K f.$$

Assuming that (5.11) holds, show the Lagrange multiplier theorem.

- (2) We now prove (5.11).
 (a) First of all, prove that $T_x K \subset \nabla\varphi(x)^\perp$.
 (b) Prove the converse inclusion (this is a bit more delicate: you need to construct a curve.... use the implicit function theorem to do that!).
 (3) Using this description of the tangent plane, recover Theorem 5.5.

Exercise 3.5 (A proof of the implicit function theorem). In this exercise, we give a proof of the implicit function theorem that relies on the following version of the Banach fixed point theorem:

Theorem 5.6 (Banach fixed point). Let X and Y be two closed subsets of Banach spaces and $g : X \times Y \rightarrow Y$ be continuous. We assume that there exists a constant $a < 1$ such that for any $x \in X$ the map $g(x, \cdot) : Y \rightarrow Y$ satisfies

$$\forall y, y' \in Y, \|g(x, y) - g(x, y')\| \leq a\|y - y'\|.$$

Then, for any $x \in X$, $g(x, \cdot)$ has a unique fixed point $f(x)$, and the map $x \mapsto f(x)$ is continuous.

Using this theorem, we want to show the following version of the implicit function theorem: let E_1, E_2, E_3 be three Banach spaces, $U \subset E_1 \times E_2$ be an open set, $(x_0, y_0) \in U$ and $F \in \mathcal{C}^1(U; E_3)$. We assume that $\nabla_y F$ is invertible (with continuous inverse). Then there exist two open sets U_1, U_2 with $U_i \subset E_i$ and such that $U_1 \times U_2 \subset U$ is an open neighbourhood of (x_0, y_0) , and a continuous map $f : U_1 \rightarrow U_2$ such that

$$f(x_0) = y_0, \{(x, y) \in U_1 \times U_2, F(x, y) = F(x_0, y_0)\} = \{(x, f(x)), x \in U_1\}.$$

We will admit that if F is $\mathcal{C}^k$, then so is f .

To prove the implicit function theorem, consider the map

$$g : (x, y) \mapsto y - \nabla_y F(x_0, y_0)^{-1} (F(x, y) - F(x_0, y_0)),$$

and try to show it has fixed points (for the sake of simplicity, you can assume F is $\mathcal{C}^2$).

Exercise 3.6 (Applications of the implicit function theorem). We let $P \in \mathbb{R}_n[X]$ be a polynomial with only simple roots $\lambda_1 < \dots < \lambda_n$. Show that there exists a neighbourhood V of P in $\mathbb{R}_n[X]$ and a $\mathcal{C}^\infty$ map $\lambda : V \rightarrow \mathbb{R}^n$ such that, for any $Q \in V$, Q only has simple roots, and its roots are given by $\lambda(Q)$.

SOLUTIONS FOR THE FIRST PART OF PART 3

Solution 3.1. (1) The existence of a minimiser comes from the compactness of $\mathbb{S}^{d-1}$.

- (2) Let $\varphi(x) := \|x\|^2 - 1 \in \mathcal{C}^1(\mathbb{R}; \mathbb{R})$ and $f(x) = \langle Ax, x \rangle$. First, observe that $\nabla \varphi(x) = 2x \neq 0$ on $\mathbb{S}^{d-1}$, and that $\nabla f(x) = 2Ax$. By the Lagrange multiplier theorem, we deduce that, e_1 being the minimiser, we have for some $\mu \in \mathbb{R}$

$$\nabla f(e_1) + \mu \nabla \varphi(e_1) = 0 \Leftrightarrow Ae_1 = (-\mu)e_1.$$

In particular, e_1 is an eigenvector associated with $-\mu$.

- (3) Similarly, we have two constraints and, letting e_2 solve this other minimisation problem, there exists μ_1, μ_2 such that

$$Ae_2 = (-\mu_1)e_2 + (-\mu_2)e_1, \langle e_1, e_2 \rangle = 0.$$

Let us show that e_2 is an eigenvector, that is, that $\mu_2 = 0$. Taking the scalar product with e_1 gives

$$(-\mu_2) = \langle Ae_2, e_1 \rangle = \langle Ae_1, e_2 \rangle = (-\mu) \langle e_1, e_2 \rangle = 0.$$

Thus e_2 is an eigenvector.

- (4) We can conclude by studying, for $k \leq d-1$,

$$\min_{x \in \mathbb{S}^{d-1}, \langle x, e_1 \rangle = \dots = \langle x, e_k \rangle = 0} \langle Ax, x \rangle.$$

This produces d orthogonal eigenvalues. As a free family of d vectors is a basis of $\mathbb{R}^d$, the conclusion follows.

Solution 3.2 (Some applications of the Lagrange multiplier theorem). (1) To solve

$$\min_{\frac{x^2}{2} + \frac{y^2}{4} + \frac{z^2}{6} = 1} x + y + z$$

we introduce

$$f(x, y, z) := x + y + z, \varphi(x, y, z) = \frac{x^2}{2} + \frac{y^2}{4} + \frac{z^2}{6} - 1.$$

Observe that a solution exists as the admissible set is compact. Besides,

$$\nabla \varphi(x, y, z) = \left(x, \frac{y}{2}, \frac{z}{3} \right)^T$$

which does not vanish on the set $\{\varphi = 0\}$. In particular, if (x^*, y^*, z^*) is optimal there exists λ such that

$$\begin{cases} 1 + \lambda x = 0 \\ 1 + \frac{\lambda}{2} y = 0 \\ 1 + \frac{\lambda}{3} z = 0. \end{cases}$$

We first deduce that $\lambda < 0$ and that

$$x^* = -\frac{1}{\lambda}, y^* = -\frac{2}{\lambda}, z^* = -\frac{3}{\lambda}.$$

Since $\varphi(x^*, y^*, z^*) = 0$ we obtain

$$\lambda^2 = \frac{1}{2} + 1 + \frac{3}{2} = \frac{6}{2} = 3,$$

and this yields the optimal point.

(2) We study the optimisation problem

$$\max_{x_i \geq 0, \sum x_k = 1} \prod x_k.$$

First of all, at an optimiser $x^* = (x_i^*)_{i=1, \dots, d}$ we have, for any i , $x_i^* > 0$ so that we do not care about the constraint $x_i^* \geq 0$. Introduce the function $\varphi(x) = \sum_{i=1}^d x_i$. As $\nabla \varphi(x) = (1, \dots, 1)^T$, the Lagrange multiplier rule applies and, at an optimiser, there exists λ such that

$$\forall i \in \{1, \dots, d\}, 1 + \lambda \prod_{k \neq i} x_k = 0.$$

This gives, first, $\lambda \neq 0$ and, second,

$$\forall k, x_j = x_k.$$

As $\varphi(x^*) = 1$, we deduce that

$$\forall k, x_k = \frac{1}{d}.$$

Thus, for any admissible x , we have

$$\prod x_k \leq \frac{1}{d^d}.$$

Put otherwise,

$$\forall x \geq 0, \sum_{k=1}^d x_k = 1 \Rightarrow \prod_{k=1}^d x_k^{\frac{1}{d}} \leq \frac{1}{d}.$$

This provides the conclusion.

(3) Let (x, y) be the sides of the rectangle, so that the constraint rewrites

$$x + y = \frac{p}{2}, x \geq 0, y \geq 0.$$

The optimisation problem becomes

$$\max_{0 \leq x, y, x+y=\frac{p}{2}} xy.$$

The first thing that needs to be observed is that the set of constraints is bounded and closed, so that a maximiser exists. Second, at a maximiser (x^*, y^*) , we necessarily have $x^*, y^* > 0$ so that we only need to take into account the equality constraint. Let $\varphi : (x, y) \mapsto x + y - \frac{p}{2}$. As $\nabla \varphi(x, y) = (1, 1)^T \neq 0$, by the Lagrange multiplier theorem we deduce that there exists μ such that

$$y = \lambda, x = \lambda.$$

In particular, $x = y$ and, from $\varphi(x, y) = 0$ we obtain $x = y = \frac{p}{4}$. Consequently, the optimal box is a square.

Solution 3.3 (The Hölder inequality from the point of view of constrained optimisation). We study the optimisation problem

$$\max_{y_1, \dots, y_d \in \mathbb{R}, \sum_{k=1}^d |y_k|^q = 1} \sum_{i=1}^d x_i y_i.$$

First of all, we observe by a simple argument that we can assume that $x_i > 0$ for any i , whence $y_i \geq 0$ for any i . As $q > 1$, $y \mapsto \sum_{k=1}^d |y_k|^q$ is $\mathcal{C}^1$. The existence

of a maximiser is immediate and, from the Lagrange multiplier theorem, we deduce that, at an optimal $y^* = (y_1^*, \dots, y_d^*)$ there exists λ such that

$$x_i + \lambda(y_i^*)^{q-1} = 0.$$

Thus, $\lambda < 0$, and we deduce that

$$y_i^* = -\frac{x_i^{\frac{1}{q-1}}}{\lambda}.$$

From the normalisation condition

$$\sum y_k^q = 1$$

we deduce that

$$|\lambda|^q = \sum_{k=1}^d x_k^{\frac{q}{q-1}}.$$

If $\frac{1}{p} + \frac{1}{q} = 1$, we have $p = \frac{q}{q-1}$ and so

$$-\lambda = \left(\sum x_k^p \right)^{\frac{1}{q}},$$

whence

$$y_i^* = \frac{x_i^{\frac{1}{q-1}}}{\left(\sum x_k^p \right)^{\frac{1}{q}}}.$$

In the end, we obtain

$$\max_{y_1, \dots, y_d \in \mathbb{R}, \sum_{k=1}^d |y_k|^q = 1} \sum_{i=1}^d x_i y_i = \frac{\sum_{i=1}^d x_i^{\frac{q}{q-1}}}{\left(\sum x_k^p \right)^{\frac{1}{q}}} = \left(\sum x_i^p \right)^{1-\frac{1}{q}} = \left(\sum x_i^p \right)^{\frac{1}{p}}.$$

This concludes the proof.

Solution 3.4. [Around tangent planes]

- (1) Let $f \in \mathcal{C}^1(\mathbb{R}^d; \mathbb{R})$, and assume (with the notations from above) that x solves

$$\min_K f.$$

Let $\gamma : (-\varepsilon; \varepsilon) \rightarrow K$ be any $\mathcal{C}^1$ curve satisfying $\gamma(0) = x$. Then $t = 0$ is a local (interior) minimum for $g : t \mapsto f(\gamma(t))$, so that

$$g'(0) = 0.$$

By the chain rule,

$$\langle \nabla f(x), \gamma'(0) \rangle = 0.$$

In particular,

$$\nabla f(x) \in \nabla \varphi(x)^{\perp\perp}.$$

Thus we deduce that there exists $c \in \mathbb{R}$ (exercise: why?) such that

$$\nabla f(x) = c \nabla \varphi(x),$$

which, with $\lambda = -c$, is just a rewriting of the Lagrange multiplier rule.

- (2) We now prove (5.11).

- (a) This follows once again from the chain rule, differentiating $t \mapsto \varphi(\gamma(t))$ (which should be constant if γ only takes values in K).

- (b) Now, let $z \in \nabla\varphi(x)^\perp$. We assume for notational simplicity that $\partial_{x_d}\varphi \neq 0$. By the implicit function theorem we know that, at least locally, we can describe K as

$$K = \{(x', g(x'))\}, x' \in \mathbb{B}_{\mathbb{R}^{d-1}}(0; \varepsilon)\}$$

for some $\mathcal{C}^1$ function g satisfying $g(0) = x$. Differentiating $x' \mapsto \varphi(x', g(x')) = 0$ we deduce that

$$\forall i \leq d-1, \frac{\partial\varphi}{\partial x_i} + \frac{\partial\varphi}{\partial x_d} \frac{\partial g}{\partial x_i} = 0.$$

We now consider $\Gamma : t \mapsto (th_1, \dots, th_{d-1})$ and we want to select $h_1, \dots, h_{d-1}$ so that the curve $\gamma : t \mapsto (\Gamma(t), g(\Gamma(t)))$ fits the bill. However, observe that

$$\gamma'(t) = (h_1, \dots, h_{d-1}, \langle \nabla g(0), h \rangle) \text{ with } h = (h_1, \dots, h_{d-1}).$$

Naturally, we want to set

$$h_i = z_i, i \leq d-1$$

and we thus only need to check that

$$\sum_{i=1}^{d-1} \partial_{x_i} g(0) z_i = z_d.$$

But

$$\forall i \leq d-1, \frac{\partial g}{\partial x_i} = -\frac{\partial_{x_i}\varphi}{\partial_{x_d}\varphi}.$$

Thus

$$\sum_{i=1}^{d-1} \partial_{x_i} g(0) z_i = z_d \Leftrightarrow \sum_{i=1}^{d-1} \partial_{x_i} \varphi(x) z_i + \partial_{x_d} \varphi(x) z_d = 0 \Leftrightarrow \langle \nabla \varphi(x), z \rangle = 0,$$

and the conclusion follows.

- (3) We consider once again a $\mathcal{C}^2$ path $\gamma : (-1; 1) \rightarrow K$ such that $\gamma(0) = x^*$, where x^* is a solution of the constrained minimisation problem. The second order optimality condition for $t = 0$ for the function $g : t \mapsto f(\gamma(t))$ reads:

$$0 \leq \langle \nabla^2 f(x^*) \dot{\gamma}(0), \dot{\gamma}(0) \rangle + \langle \nabla f(x^*), \ddot{\gamma}(0) \rangle.$$

Now, differentiating $t \mapsto \varphi(\gamma(t))$ we also have

$$\langle \nabla^2 \varphi(x^*) \dot{\gamma}(0), \dot{\gamma}(0) \rangle + \langle \nabla \varphi(x^*), \ddot{\gamma}(0) \rangle = 0.$$

From the first order optimality conditions, letting λ denote the Lagrange multiplier, we know that

$$\nabla f(x^*) + \lambda \nabla \varphi(x^*) = 0$$

and we thus obtain

$$\begin{aligned} 0 &\leq \langle \nabla^2 f(x^*) \dot{\gamma}(0), \dot{\gamma}(0) \rangle + \langle \nabla f(x^*), \ddot{\gamma}(0) \rangle \\ &= \langle \nabla^2 f(x^*) \dot{\gamma}(0), \dot{\gamma}(0) \rangle - \lambda \langle \nabla \varphi(x^*), \ddot{\gamma}(0) \rangle \\ &= \langle \nabla^2 f(x^*) \dot{\gamma}(0), \dot{\gamma}(0) \rangle + \lambda \langle \nabla^2 \varphi(x^*) \dot{\gamma}(0), \dot{\gamma}(0) \rangle. \end{aligned}$$

The conclusion follows.

Solution 3.5 (A proof of the implicit function theorem). *As indicated, we study the function g . First of all, observe that*

$$\nabla_y g(x, y) = \text{Id} - \nabla_y F(x_0, y_0)^{-1} \nabla_y F(x, y)$$

whence

$$\nabla_y g(x_0, y_0) = 0$$

so that, in a small neighbourhood of (x_0, y_0) , we have

$$\|\nabla_y g(x, y)\| \leq \frac{1}{2}.$$

To apply the Banach fixed point theorem, we need to find a neighbourhood $\mathbb{B}(x_0; \varepsilon) \times \mathbb{B}(y_0; \varepsilon')$ of (x_0, y_0) such that, for any $x \in \mathbb{B}(x_0; \varepsilon)$, $g(x, \cdot)$ maps $\mathbb{B}(y_0; \varepsilon')$ into itself. However, observe that

$$\begin{aligned} \|g(x, y) - y_0\| &= \|g(x, y) - g(x_0, y_0)\| \\ &\leq \|g(x, y) - g(x_0, y)\| + \|g(x_0, y) - g(x_0, y_0)\| \\ &\leq \|\nabla_y F(x_0, y_0)^{-1}\| \cdot \|F(x, y) - F(x_0, y)\| + \frac{1}{2} \|y - y_0\| \\ &\leq M \|x - x_0\| + \frac{1}{2} \|y - y_0\| \end{aligned}$$

where M only depends on $\nabla_x F, \nabla_x^2 F$. Thus, if we take ε small enough, the conclusion follows: we can apply the Banach fixed point theorem, which gives the existence of a continuous map f such that

$$g(x, f(x)) = f(x),$$

and $f(x)$ is the only fixed point of $g(x, \cdot)$ in a neighbourhood of y_0 . However, f being a fixed point is equivalent to $F(x, f(x)) = F(x_0, y_0)$, and the conclusion follows.

Solution 3.6 (Applications of the implicit function theorem). *As suggested by the title, we use the implicit function theorem applied to the map*

$$F : \mathbb{R}_n[X] \times \mathbb{R}^n \ni (Q, \mu) \mapsto (Q(\mu_1), \dots, Q(\mu_n)).$$

We observe that by definition $F(P, \lambda_1, \dots, \lambda_n) = 0$. Furthermore,

$$\nabla_\mu F(P, \lambda) = \text{Diag}(P'(\lambda_1), \dots, P'(\lambda_n))$$

which is invertible as every root is simple. In particular, the implicit function theorem applies.

6. THE LAGRANGE MULTIPLIER RULE II: INEQUALITY CONSTRAINTS AND THE KARUSH-KUHN-TUCKER THEOREM

6.1. Inequality constraints: first discussions. In this section, we will be discussing inequality constraints; this time, we will start straight ahead with an arbitrary number of constraints. In this section we will work (as this is now quite important) with N constraints

$$\forall i \in \{1, \dots, N\}, g_i(x) \leq 0.$$

For the time being, the running assumption is

$$(6.1) \quad \text{For any } i, g_i \text{ is convex and coercive.}$$

We refer to Section 6.3 for a loosening of these assumptions.

We let

$$K := \cap_{i=1}^N \{g_i \leq 0\}$$

and we assume that

$$(6.2) \quad \exists x_0, \forall i \in \{1, \dots, N\}, g_i(x_0) < 0.$$

We consider

$$(6.3) \quad \min_{x \in K} f(x)$$

where f is, as usual, a $\mathcal{C}^1$ function (observe that for the time being we do not require that f be convex). Regarding terminology, when $x \in K$ and $g_i(x) = 0$, we say that the constraint is *active*. When $g_i(x) < 0$, we say that the constraint is *inactive*.

Our goal here is to prove the following version of the Lagrange multiplier rule, which is due to Karush, Kuhn & Tucker:

Theorem 6.1. *Under the standing assumptions (6.1)-(6.2), if x^* solves (8.3) there exist $\lambda_1, \dots, \lambda_N \geq 0$ such that*

$$\begin{cases} \nabla f(x^*) + \sum_{i=1}^N \lambda_i \nabla g_i(x^*) = 0, \\ \sum_{i=1}^N \lambda_i g_i(x^*) = 0. \end{cases}$$

Proof of Theorem 6.2. The core of the proof amounts to determining the cone of admissible perturbations at the point x^* or, in other words, the set

$$C(x^*) := \{0\} \cup \left\{ y \in \mathbb{R}^d \setminus \{0\}, \text{ there exists } \{x_k\}_{k \in \mathbb{N}} \in K^{\mathbb{N}}, x_k \xrightarrow[k \rightarrow \infty]{} x^*, \frac{x_k - x^*}{\|x_k - x^*\|} \xrightarrow[k \rightarrow \infty]{} \frac{y}{\|y\|} \right\}.$$

Assume that, up to relabelling, $g_1(x^*), \dots, g_n(x^*) < 0, g_{n+1}(x^*) = \dots = g_N(x^*) = 0$. Indeed, if no constraints are active, there is nothing to prove. Observe that

$$(6.4) \quad \forall y \in C(x^*), \langle \nabla f(x^*), y \rangle \geq 0.$$

Remark 6.1. *Be careful: that a direction y is admissible does not mean that for any $t \geq 0$ small enough $x^* + ty \in K$. However, it means that there exist two sequences $\{t_k\}_{k \in \mathbb{N}} \in (0; +\infty)^{\mathbb{N}}, \{y_k\}_{k \in \mathbb{N}}$ such that*

$$t_k \xrightarrow[k \rightarrow \infty]{} 0, y_k \xrightarrow[k \rightarrow \infty]{} y \text{ and for any } k \in \mathbb{N}, x^* + t_k y_k \in K.$$

Indeed, letting $y \in C(x^*) \setminus \{0\}$ and choosing any sequence $\{x_k\}_{k \in \mathbb{N}} \in K^{\mathbb{N}}$ in the conditions of the definition, we have

$$f(x^*) \leq f(x_k)$$

whence

$$\langle \nabla f(x^*), x_k - x^* \rangle \geq 0.$$

Let us now prove that under (6.1)–(6.2) we have the explicit description of the tangent cone as

$$(6.5) \quad C(x^*) = \{y \in \mathbb{R}^d, \forall i \in \{1, \dots, n\}, \langle \nabla g_i(x^*), y \rangle \leq 0\}.$$

Define $\tilde{C} := \{y \in \mathbb{R}^d, \forall i \in \{1, \dots, n\}, \langle \nabla g_i(x^*), y \rangle \leq 0\}$. The fact that $C(x^*) \subset \tilde{C}$ is immediate. Conversely, let $y \in \tilde{C} \setminus \{0\}$. Assume, up to relabelling, that

$$\langle \nabla g_i(x^*), y \rangle = 0, 1 \leq i \leq n, \langle \nabla g_i(x^*), y \rangle < 0, i \geq n.$$

As the g_i are convex Assumption (6.2) guarantees that $\nabla g_1(x^*), \dots, \nabla g_n(x^*) \neq 0$ and, x_0 being a point provided by (6.2), that is, such that

$$\forall i \in \{1, \dots, N\}, g_i(x_0) < 0$$

we obtain

$$\forall i \in \{1, \dots, N\}, \langle \nabla g_i(x^*), x_0 - x^* \rangle < 0.$$

Indeed, convexity would otherwise imply $g_i(x_0) > 0$, which is absurd. Define, then, $w := x_0 - x^*$ and set, for $\varepsilon > 0$ small enough, $y_\varepsilon := y + \varepsilon w$. Thus, for any $t > 0$ small enough, $x_\varepsilon := x^* + ty_\varepsilon \in K$. Finally, observe that $C(x^*)$ is closed, as a consequence of a simple Cantor diagonal argument. Letting $\varepsilon \rightarrow 0$ we deduce that $y \in C(x^*)$, as claimed.

In particular, the optimality condition (6.4) rewrite

$$(6.6) \quad \{y \in \mathbb{R}^d, \forall i \in \{1, \dots, n\}, \langle \nabla g_i(x^*), y \rangle \leq 0\} \subset \{y \in \mathbb{R}^d, \langle \nabla f(x^*), y \rangle \geq 0\}.$$

We now show that this implies the existence of a family $\lambda_1, \dots, \lambda_n \geq 0$ such that

$$\nabla f(x^*) + \sum_{i=1}^n \lambda_i \nabla g_i(x^*) = 0.$$

In fact, we are going to prove it under the more restrictive assumption that the family $\{\nabla g_k(x^*)\}_{k=1, \dots, n}$ is linearly independent²The key is to consider the set

$$C := \left\{ \sum_{i=1}^n \mu_i \nabla g_i(x^*), \mu_i \leq 0. \right\}$$

We claim that C is convex (this is immediate), that it is a cone (meaning that for any $\lambda \geq 0$ and any $x \in C$ we have $\lambda x \in C$) and, finally, that it is closed. The fact that C is closed is far from immediate. However, in the case where $\{\nabla g_i(x^*)\}_{i=1, \dots, n}$ is linearly independent, this is quite easy to obtain, as it suffices to observe that, in this case, if a sequence $\{x_k = \sum \mu_{i,k} \nabla g_i(x^*)\}_{k \in \mathbb{N}}$ is bounded, then each of the sequence $\{\mu_{i,k}\}_{k \in \mathbb{N}}$ is bounded, so that we might extract a subsequence. Now, argue by contradiction and assume that $\nabla f(x^*) \notin C$. As C is a convex closed subset, this implies that there exists $z \in \mathbb{R}^d$ and $\alpha \in \mathbb{R}$ such that

$$\langle \nabla f(x^*), z \rangle < \alpha < \inf_{x \in C} \langle x, z \rangle.$$

²Even when the family is not linearly independent, this remains true, but the proof is much more involved—this is the Farkas-Minkowski lemma.

To prove the existence of z , we can either directly invoke the Hahn-Banach theorem, or more simply rely on the projection theorem: there exists $x_0 \in C$, the orthogonal projection of $\nabla f(x^*)$ on C , such that

$$\forall x \in C, \langle x - x_0, x_0 - \nabla f(x^*) \rangle \geq 0.$$

In particular,

$$\langle \nabla f(x^*), x_0 - \nabla f(x^*) \rangle < \langle x_0, x_0 - \nabla f(x^*) \rangle \leq \langle x, x_0 - \nabla f(x^*) \rangle.$$

Setting $z = x_0 - \nabla f(x^*)$ allows to conclude. Now, as $0 \in C$, we obtain $\alpha < 0$ and as, for any $\lambda \geq 0$ we have $\lambda x \in C$ we deduce that

$$0 = \lim_{\lambda \rightarrow \infty} \frac{\alpha}{\lambda} \leq \inf_{x \in C} \langle x, z \rangle.$$

Setting $\lambda_{n+1} = \dots = \lambda_N = 0$ concludes the proof. $\square$

6.2. Some examples.

6.2.1. *One inequality constraint.* Now consider the following problem in $\mathbb{R}^2$:

$$\min_{x^2 + y^2 \leq 1} 2 + xy$$

Here, $f(x, y) = 2 + xy$ and $K = \{(x, y) : g(x, y) \leq 0\}$, with $g(x, y) = x^2 + y^2 - 1$. Since K is compact and non-empty (because $(0, 0) \in K$), the problem has a solution. Furthermore, g is strictly convex. Consequently, from Theorem 6.2, if (x^*, y^*) is a minimiser of f on K , there exists $\mu \geq 0$ such that

$$\nabla f(x^*, y^*) + \lambda \nabla g(x^*, y^*) = 0 \quad \text{and} \quad \lambda g(x^*, y^*) = 0.$$

This system rewrites

$$\begin{cases} y^* + 2\lambda x^* = 0 \\ x^* + 2\lambda y^* = 0 \\ \lambda((x^*)^2 + (y^*)^2 - 1) = 0. \end{cases}$$

There are five possible solutions: $(\lambda = 0, x^* = y^* = 0)$, $(\lambda = 1/2, x^* = -y^* = \sqrt{2}/2)$, $(\lambda = 1/2, x^* = -y^* = -\sqrt{2}/2)$, $(\lambda = -1/2, x^* = y^* = \sqrt{2}/2)$, $(\lambda = -1/2, x^* = y^* = -\sqrt{2}/2)$. The last two solutions are not admissible because the associated multiplier λ is negative. So only the first three solutions remain. We then evaluate the criterion at these points:

$$f(\sqrt{2}/2, -\sqrt{2}/2) = f(-\sqrt{2}/2, \sqrt{2}/2) = 3/2 \text{ and } f(0, 0) = 2.$$

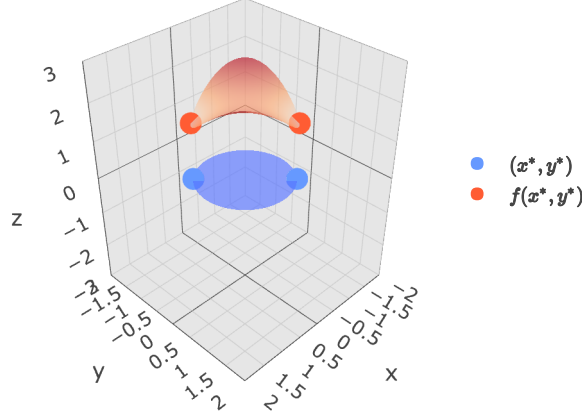
We have thus found two solutions: $(\sqrt{2}/2, -\sqrt{2}/2)$ and $(-\sqrt{2}/2, \sqrt{2}/2)$. An illustration of this example is given in Figure 6.2.1.

6.2.2. *Several inequality constraints.* Consider the following problem in $\mathbb{R}^3$:

$$\begin{aligned} \min \quad & x + y + z \\ \text{s.t.} \quad & x^2 + y^2 + z^2 \leq 1 \\ & x \geq 0 \end{aligned}$$

Here, $f(x, y, z) = x + y + z$ and $K = \{(x, y, z) : G(x, y, z) \leq 0\}$, with $G(x, y, z) = (g_1(x, y, z), g_2(x, y, z))$ and $g_1(x, y, z) = x^2 + y^2 + z^2 - 1$, $g_2(x, y, z) = -x$.

The constraint K being compact and non-empty (because $(0, 0, 0) \in K$) and f being continuous, the problem has a solution (x^*, y^*, z^*) .


 FIGURE 2. Illustration of the problem $\min_{x^2+y^2 \leq 1} 2 + xy$.

Let us show that the constraint is qualified at any point. Let (x, y, z) be an admissible point, and let $I(x, y, z)$ denote the set of active constraints. If $I(x, y, z) = \{1\}$, then $\nabla g_1(x, y, z)$ is non-zero because $x^2 + y^2 + z^2 - 1 = 0$ and $\nabla g_1(x, y, z)$ only cancels for $x = y = z = 0$. If $I(x, y, z) = \{2\}$, then $\nabla g_2(x, y, z) \neq 0$. Now assume that $I(x, y, z) = \{1, 2\}$. Let us choose $v = (1, -y, -z)$. Then

$$\langle \nabla g_1(x, y, z), v \rangle = -2y^2 - 2z^2 = -2 < 0 \text{ and } \langle \nabla g_2(x, y, z), v \rangle = -1 < 0; .$$

So the constraint is qualified at every point. At the point (x^*, y^*, z^*) , the following necessary condition is satisfied: there exist $\lambda_1 \geq 0$ and $\lambda_2 \geq 0$ such that

$$\begin{cases} 1 + 2\lambda_1 x^* - \lambda_2 = 0 \\ 1 + 2\lambda_1 y^* = 0 \\ 1 + 2\lambda_1 z^* = 0 \end{cases}$$

with the exclusion condition

$$\begin{cases} \lambda_1((x^*)^2 + (y^*)^2 + (z^*)^2 - 1) = 0 \\ \lambda_2(-x^*) = 0. \end{cases}$$

According to the previous equations, $\lambda_1 > 0$. So $(x^*)^2 + (y^*)^2 + (z^*)^2 - 1 = 0$, and $y^* = z^* = -1/(2\lambda_1)$. Assume $x^* > 0$. Then $\lambda_2 = 0$ and $x^* = -1/(2\lambda_1)$, which is impossible because $x^* > 0$ by assumption. So $x^* = 0$, which imposes

$$0 + (-1/(2\lambda_1))^2 + (-1/(2\lambda_1))^2 = 1$$

i.e. $\lambda_1 = \sqrt{2}/2$ because $\lambda_1 > 0$. Hence $x^* = 0$ and $y^* = z^* = -\sqrt{2}/2$. Finally, note that $\lambda_2 = 1 > 0$.

6.3. Bringing everything together: (in)equality constraints and qualifications.

6.3.1. *The Karush-Kuhn-Tucker theorem.* We are now in a position to state the optimality conditions for a general problem of the form

$$(6.7) \quad \min_{x \in \mathbb{R}^d, \varphi_1(x) = \dots = \varphi_m(x) = 0, g_1(x), \dots, g_n(x) \leq 0} f(x)$$

where $f, \varphi_1, \dots, \varphi_m, g_1, \dots, g_n$ are $\mathcal{C}^1$ functions, and we assume that at least one of them is coercive to ensure existence of an optimiser x^* . We call the set $\{x \in \mathbb{R}^d, \varphi_1(x) = \dots = \varphi_m(x) = 0, g_1(x), \dots, g_n(x) \leq 0\}$ the set of admissible points.

Now, if we look at what we needed in order to derive Theorems 5.3–6.2, did we really use strong properties of the $\varphi_1, \dots, \varphi_m, g_1, \dots, g_n$? No. In fact, the only two facts we relied on were the following:

- (1) For equality constraints, the fact that $\{\nabla \varphi_i(x^*)\}_{i=1, \dots, m}$ is a linearly independent family.
- (2) For inequality constraints, examining the proof of Theorem 6.2 carefully, we see that we only used convexity to guarantee the existence of x_0 such that

$$\langle g_i(x^*), x_0 - x^* \rangle < 0,$$

and that was used to identify the cone of admissible perturbations.

This leads to defining, in the general case, the notion of qualified constraints, which is slightly subtler:

Definition 6.1. *[Qualified constraints] We say that the constraints are qualified at an admissible point x^* if:*

- (1) *The family $\{\nabla \varphi_i(x^*)\}_{i=1, \dots, m}$ is linearly independent.*
- (2) *There exists a vector w such that*
 - (a) *For any $i \in \{1, \dots, m\}$, $\langle \nabla \varphi_i(x^*), w \rangle = 0$,*
 - (b) *For any $j \in \{1, \dots, n\}$ such that $g_j(x^*) = 0$ (in other words, for every active constraint), $\langle \nabla g_j(x^*), w \rangle < 0$.*

Remark 6.2. *It is an easy exercise to see that, if there are only inequality constraints, and if these inequality constraints are affine, then the constraints are automatically qualified.*

The (general version of the) Karush-Kuhn-Tucker is the following:

Theorem 6.2. *Let x^* be a solution of (6.7) and assume that the constraints are qualified at x^* . Then there exist Lagrange multipliers $\mu_1, \dots, \mu_n \in \mathbb{R}$ and $\lambda_1, \dots, \lambda_m \geq 0$ such that*

$$(6.8) \quad \nabla f(x^*) + \sum_{i=1}^n \lambda_i \nabla g_i(x^*) + \sum_{i=1}^m \mu_j \nabla \varphi_i(x^*) = 0, \quad \lambda_i \geq 0, \quad \lambda_i g_i(x^*) = 0, \forall i = 1, \dots, n.$$

Methodology: From a methodological point of view, when tackling an optimisation problem, you should, for every admissible point x , introduce the set $I(x)$ of active constraints, and check by hand that the constraints are qualified before computing necessary optimality conditions.

6.3.2. *An example.* Now we are mixing up the difficulties. Consider the problem in $\mathbb{R}^3$:

$$\begin{aligned} \min \quad & x + 2y + 3z \\ \text{subject to} \quad & x^2 + y^2 + z^2 = 1 \\ & x + y + z \leq 0 \end{aligned}$$

The admissible set is compact, non-empty (because $(-1, 0, 0) \in K$) and the criterion is continuous, so the problem has at least one solution. The criterion is $f(x, y, z) = x + 2y + 3z$, there is an equality constraint $h(x, y, z) = x^2 + y^2 + z^2 - 1 = 0$ and an inequality constraint $g(x, y, z) = x + y + z \leq 0$. Let us check that the constraint is qualified. First of all, $\nabla h(x, y, z) = 0$ if and only if $x = y = z = 0$, which is impossible because $x^2 + y^2 + z^2 = 1$. Moreover, for a point $(x, y, z) \in K$ such that $g(x, y, z) = 0$, if we take $v = (-1, -1, -1)$, we have

$$\langle \nabla h(x, y, z), v \rangle = -2x - 2y - 2z = -2(x + y + z) = 0 \text{ and } \langle \nabla g(x, y, z), v \rangle = -3 < 0.$$

So the constraint is qualified at every point. If (x^*, y^*, z^*) is a minimiser of the problem, the necessary optimality conditions are

$$\begin{cases} 1 + \lambda + 2\mu x^* = 0 \\ 2 + \lambda + 2\mu y^* = 0 \\ 3 + \lambda + 2\mu z^* = 0 \\ (x^*)^2 + (y^*)^2 + (z^*)^2 = 1 \\ \lambda(x^* + y^* + z^*) = 0 \end{cases}$$

where $\lambda \geq 0$ and $\mu \in \mathbb{R}$. We deduce from the first three equations that μ is non-zero and that

$$x^* = -\frac{1 + \lambda}{2\mu}, \quad y^* = -\frac{2 + \lambda}{2\mu}, \quad z^* = -\frac{3 + \lambda}{2\mu},$$

so that, according to the exclusion condition, we have

$$\lambda\left(\frac{1 + \lambda}{2\mu} + \frac{2 + \lambda}{2\mu} + \frac{3 + \lambda}{2\mu}\right) = 0$$

Hence $\lambda(6 + 3\lambda) = 0$. Since $\lambda \geq 0$, this imposes $\lambda = 0$. We then plot the expressions of x^* , y^* and z^* as a function of μ in the equality constraint to find $\mu = \sqrt{14}/2$ or $\mu = -\sqrt{14}/2$. There are thus two points satisfying the necessary optimality conditions. It is easily seen that the solution is $\frac{-1}{\sqrt{14}}(1, 2, 3)$.

6.3.3. *Introducing the Lagrangian.* Under the assumptions of (and with the notations of) Theorem 6.2, we can introduce the Lagrangian

$$\mathcal{L} : x \mapsto f(x) + \sum_{i=1}^n \mu_i \varphi_i(x) + \sum_{j=1}^N \lambda_j g_j(x)$$

and observe that, if x^* solves (6.7), then

$$\nabla \mathcal{L}(x^*) = 0.$$

In other words, through the introduction of the Lagrangian (which will be at the core of duality methods) we can say that a solution x^* (where constraints are qualified) of a constrained problem is a critical point of a function $\mathcal{L} : \mathbb{R}^d \rightarrow \mathbb{R}$.

EXERCISES FOR THE SECOND PART OF PART 3

Exercise 3.7. We investigate a situation where constraints are automatically qualified. Throughout, we consider the set

$$K = \{x : g_i(x) \leq 0, i = 1, \dots, m\}$$

where each of the g_i is a convex function.

- (1) Show that if there exists $x_0 \in K$ such that for any i $g_i(x_0) < 0$, then the constraints are qualified at any point $x \in K$.
- (2) We now add affine equality constraints. Namely, we consider

$$K' := K \cap (\cap_{i=1}^n \{\varphi_i = 0\})$$

where each of the φ_i is an affine function. We assume that there exists $x_0 \in K'$ such that $g_i(x_0) < 0$ for any i , and that the gradients of φ_i 's form a linearly independent family. Show that the constraints are qualified at any point $x \in K'$.

Exercise 3.8. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by $f(x, y) = x^2 + 2y^2$ for all $(x, y) \in \mathbb{R}^2$. Let $K = \{(x, y) \in \mathbb{R}^2, x \geq 0, y \geq 0, x + y \leq 1\}$. Consider the problem: (P) $\max_K f$.

- (1) Show that if (x, y) is solution of (P), then $x + y = 1$. Then solve Problem (P).
- (2) What is the method if you do not notice that solutions of (P) must satisfy $x + y = 1$?

Exercise 3.9. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by $f(x, y) = xy^2$ for all $(x, y) \in \mathbb{R}^2$. Let $K = \{(x, y) \in \mathbb{R}^2, x \geq 0, y \geq 0, x + y \leq 1\}$. Solve the problem $\max_K f$.

Exercise 3.10. Consider the problem

$$\max_{x^3 + y^3 - 3xy + 1 = 0} (x + y)$$

Compute the solution of this problem (we admit that a maximum exists).

Exercise 3.11. Consider the following problem:

$$\begin{aligned} & \max (x_1 + x_2) \\ & 0 \leq x_i \leq 42 \\ & x_1 + 2x_2 + 2x_3 \leq 72 \end{aligned}$$

Compute the solution of this problem.

Exercise 3.12 (Some examples of qualification failures). Consider the following problem:

$$\begin{aligned} & \max (3x + y) \\ & 0 \leq x \\ & 0 \leq y \leq (1 - x)^3 \end{aligned}$$

- (1) Show that point $(0, 1)$ is the only point satisfying the necessary optimality conditions
- (2) Show that point $(1, 0)$ is the minimizer of the problem.

Exercise 3.13. Let M denote the matrix: $M = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$ and S the convex set

$$S = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 + 2x_2 + 3x_3 \leq 1, x_i \geq 0 \text{ pour } i = 1, 2, 3\}$$

Solve the problem

$$\max_{X \in S} X^T M X.$$

Hint: the inverse of M is $M^{-1} = \frac{1}{4} \begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix}.$

Exercise 3.14. Consider the following problem:

$$(P) \quad \min_{(x,y) \in K} (x-2)^2 + y^2 \quad \text{where} \quad K = \{(x,y) \in \mathbb{R}^2 \mid 2x - y^2 \leq 1 \text{ et } x \geq 0\}.$$

- (1) Show that this problem has a solution (at least one).
- (2) Show that the constraint is qualified at all points.
- (3) Write the necessary optimality conditions (KKT) for this problem.
- (4) Find all points satisfying these conditions.
- (5) Compute the solution(s) of problem (P).

SOLUTIONS FOR THE SECOND PART OF PART 3

- Solution 3.7.** (1) As each of the g_i is convex, we know that, if $g_i(x^*) = 0$, then $\langle \nabla g_i(x^*), x_0 - x^* \rangle < 0$, where x_0 is any interior point of K . In particular, the constraints are qualified at any point $x \in K$.
- (2) Let us now take $x \in K$ and x_0 such that $g_i(x_0) < 0$ for any i . Let $I(x) := \{i, g_i(x) = 0\}$ and set $\omega := x - x_0$. Then, by convexity of the g_i 's, we know that

$$\forall i \in I(x), \langle \nabla g_i(x), \omega \rangle < 0.$$

On the other hand, for any $i \in \{1, \dots, n\}$,

$$\langle \nabla \varphi_i(x), \omega \rangle = \varphi_i(x) - \varphi_i(x_0) = 0.$$

Finally, $\{\nabla \varphi_i(x)\}_{i \in \{1, \dots, n\}}$ is, by assumption, a linearly independent family. This concludes the proof.

Solution 3.8. Here, there are no equality constraints, and three inequality constraints. We write these constraints under the form $g_i(x, y) \geq 0$, with $i = 1, 2, 3$, with $g_1(x, y) = x$, $g_2(x, y) = y$, $g_3(x, y) = 1 - x - y$. The Lagrangian is given by $L(x, y, \lambda_1, \lambda_2, \lambda_3) = x^2 + 2y^2 + \lambda_1 x + \lambda_2 y + \lambda_3(1 - x - y)$.

- (1) Let $X = (x, y)$ be a solution of (P) . If $x + y < 1$, then we let the reader check that the point $\tilde{X} = (x, 1 - x)$ is in K and that $f(\tilde{X}) > f(X)$. Therefore X cannot be a minimiser of f on K , a contradiction. Therefore $x + y = 1$. Let us now solve Problem (P) .

First, K (which is the closed triangle with vertices $(0, 0)$, $(1, 0)$ and $(0, 1)$), is non-empty, compact, and f is continuous, so a solution exist.

Second, the constraints are all affine, so the constraint is qualified everywhere (this is a simple adaptation of the first exercise).

Third, let $X = (x, y)$ be a solution. Since the constraint is qualified at X , it satisfies the KKT conditions. That is: there exists $\lambda_i \geq 0$, $i = 1, 2, 3$, such that

$$2x + \lambda_1 - \lambda_3 = 0$$

$$4y + \lambda_2 - \lambda_3 = 0$$

and $\lambda_1 x = 0$, $\lambda_2 y = 0$, $\lambda_3(1 - x - y) = 0$. Moreover, we saw before that necessarily, $x + y = 1$.

Case 1: if $x = 0$. Then $y = 1$ and $(x, y) = (0, 1)$.

Case 2: if $x > 0$. Then $\lambda_1 = 0$.

Subcase 2.1: if $y = 0$. Then $x = 1$, and $(x, y) = (1, 0)$.

Subcase 2.2: if $y > 0$. Then $\lambda_2 = 0$. Since we also have $\lambda_1 = 0$, the equations become:

$$2x - \lambda_3 = 0$$

$$4y - \lambda_3 = 0$$

$$x + y = 1$$

which leads to $(x, y) = (2/3, 1/3)$.

We thus have three candidates: $(0, 1)$, $(1, 0)$ and $(2/3, 1/3)$, which, as the reader will check, are all in K . Since $f(1, 0) = 1$, $f(0, 1) = 2$ and $f(2/3, 1/3) = 2/3$, there is a unique maximiser which is $(0, 1)$, and the value of this maximisation problem is $f(0, 1) = 2$.

Remark: firstly, we did not check that the three candidates were associated to nonnegative multipliers (i.e. that we could find non-negative values of $\lambda_1, \lambda_2, \lambda_3$ such that all conditions are satisfied). This is not essential as if we have extra-candidates, they will be eliminated in the end, as long as they are in K . Secondly, once we know that $x = Y = 1$, it is possible to proceed by substitution and reduce the problem to the maximisation of a function of x only on an appropriate interval. We did not do this as we wanted to illustrate the use of the KKT conditions.

- (2) *If we do not notice that any solution must satisfy $x = y = 1$ more case will need to be examined, as we will need to discuss whether $x + y = 1$ (in which case we proceed as we did above), or $x + y < 1$ (in which case $\lambda_3 = 0$, and then we discuss various subcases until we get all candidates). Doing so would have led to more cases and more candidates (for instance, $(0, 0)$ associated to $\lambda_i = 0$ for all i satisfied the KKT conditions, and would thus be an additional candidates). This would be longer.*

Solution 3.9. *Let (x, y) be a solution. Firstly, $x > 0$ and $y > 0$. Indeed, if $x = 0$ or $y = 0$, $f(x, y) = 0 < f(1/2, 1/2)$. Since $(1/2, 1/2) \in K$, this implies that (x, y) is not a maximizer of f on K , a contradiction. Secondly, $x + y = 1$. Indeed, otherwise, for $\varepsilon > 0$ small enough, $(x + \varepsilon, y + \varepsilon) \in K$ and $f(x, y) < f(x + \varepsilon, y + \varepsilon)$, a contradiction. To sum up, $x > 0$, $y > 0$ and $x + y = 1$.*

Let us now use Kuhn and Tucker theorem. Set K is nonempty, compact, and f is continuous, so a solution exists. Moreover, the constraints are affine so the constraint is qualified everywhere. Therefore, if (x, y) is a solution, then it satisfies the KKT conditions. That is (as the reader will check):

$$\begin{aligned} y^2 + \lambda_1 - \lambda_3 &= 0 \\ 2xy + \lambda_2 - \lambda_3 &= 0 \\ \lambda_1 x &= 0 \\ \lambda_2 y &= 0 \\ \lambda_3(1 - x - y) &= 0 \end{aligned}$$

with $\lambda_i \geq 0$, $i = 1, 2, 3$, and $(x, y) \in K$. But as we saw, we must have $x > 0$, $y > 0$ (so $\lambda_1 = \lambda_2 = 0$) and $x + y = 1$. So (x, y) must satisfy:

$$\begin{aligned} y^2 - \lambda_3 &= 0 \\ 2xy - \lambda_3 &= 0 \\ x &> 0 \\ y &> 0 \\ x + y &= 1 \end{aligned}$$

The first two lines lead to $y^2 = 2xy$, hence $y = 2x$ since $y \neq 0$. As $x + y = 1$, we obtain $(x, y) = (1/3, 2/3)$. We know that there is a solution, and there is a unique candidate to be solution, so this must be the solution. We conclude that there is a unique maximizer on K , $(x, y) = (1/3, 2/3)$, and that the value of this optimization problem is $f(1/3, 2/3) = 4/27$.

Solution 3.10. • *(Qualification) Let $h(x, y) = x^3 + y^3 - 3xy + 1$. Since there is only one constraint, the qualification condition boils down to $\nabla h(x, y) \neq 0$. But:*

$$\nabla h(x, y) = \begin{pmatrix} 3x^2 - 3y \\ 3y^2 - 3x \end{pmatrix}$$

If this quantity is zero, then $x^4 = y^2 = x$, so $(x, y) = (0, 0)$ or $(x, y) = (1, 1)$. Since $(0, 0)$ does not satisfy the constraint, the only point of K where the constraint is not qualified is $(1, 1)$, and then $x + y = 2$.

- (Necessary condition) The Lagrangian is: $L((x, y), \mu) = x + y + \mu h(x, y)$. Let $(x, y) \in K$. If (x, y) is solution and $(x, y) \neq (1, 1)$, then:

$$\begin{cases} 1 + 3\mu x^2 - 3\mu y &= 0 \\ 1 + 3\mu y^2 - 3\mu x &= 0 \\ x^3 + y^3 - 3xy + 1 &= 0 \end{cases}$$

Let us first note that $\mu \neq 0$, otherwise the first equality could not hold. Subtracting the second line to the first, then dividing by 3μ , we get: $x^2 - y^2 + x - y = 0$. Therefore $(x - y)(x + y + 1) = 0$.

Case 1 : $x + y + 1 = 0$. Then $x + y = -1 < 2$. Points of this type are suboptimal as they are worse than point $(1, 1)$.

Case 2 : $x + y - 1 \neq 0$, so $x = y$. Then $h(x, x) = 0$, so $2x^3 - 3x^2 + 1 = 0$. Since 1 is an obvious root, of multiplicity 2, we get: $(x - 1)^2(x + 1/2) = 0$.

If $x = 1$, then $(x, y) = (1, 1)$, the unqualified point; but we assumed $(x, y) \neq (1, 1)$. Therefore $x = -1/2$ and $(x, y) = (-1/2, -1/2)$, so $x + y = -1$, a contradiction.

- To sum up, the unique solution of this maximization problem is the unqualified point $(1, 1)$, and the value of the problem is 2.

Remark: we did not prove that a maximum exists. One way to do so is to show that when the norm of (x, y) is large enough, larger than a given M which we will not try to compute explicitly, then first y^3 is of the order of $-x^3$, and moreover, taking a closer look, $x + y \leq 1 < 2$. It follows that the maximization problem is equivalent to the problem obtained by adding the constraint $\|(x, y)\| \leq M$. The new constraint is non-empty compact, and the objective function is continuous. Therefore, the equivalent problem has a solution, hence also the initial problem.

Solution 3.11. • (Existence of a solution) The constraint is non-empty, compact, and the objective function is continuous. Thus, a solution exists.

- (Qualification) Constraints are affine hence qualified everywhere.
- (Necessary condition) We use here the remark made after Exercice ?? concerning the necessary optimality conditions for a maximization problem. Let λ_0 denote the multiplier associated to the constraint $72 - (x_1 + 2x_2 + 2x_3) \geq 0$, and λ_i, λ_{i+3} the multipliers associated to constraints $x_i \geq 0$ and $42 - x_i \geq 0$, respectively, for $i = 1, 2, 3$. The Lagrangian is:

$$L(x, \lambda_0, \dots, \lambda_6) = x_1 + x_2 + \lambda_0(72 - x_1 - 2x_2 - 2x_3) + \sum_{i=1}^3 [\lambda_i x_i + \lambda_{i+3}(42 - x_i)].$$

The necessary conditions are thus:

$$\begin{cases} 1 - \lambda_0 + \lambda_1 - \lambda_4 &= 0 \\ 1 - 2\lambda_0 + \lambda_2 - \lambda_5 &= 0 \\ -2\lambda_0 + \lambda_3 - \lambda_6 &= 0 \end{cases}$$

with $\lambda_i x_i = 0$, $\lambda_{i+3}(42 - x_i) = 0$, for $i = 1, 2, 3$, $\lambda_0[72 - x_1 - 2x_2 - 2x_3] = 0$, and $\lambda_i \geq 0$, $i = 0, 1, \dots, 6$.

Since $x_1 + 2x_2 + 2x_3 \leq 72$ and the x_i are nonnegative, we must have $x_2 < 42$ and $x_3 < 42$, therefore $\lambda_5 = 0$ and $\lambda_6 = 0$. The second equation then gives $2\lambda_0 = 1 + \lambda_2$, therefore $\lambda_0 > 0$, hence $x_1 + 2x_2 + 2x_3 = 72$. The third equation gives $\lambda_3 = 2\lambda_0$, or $\lambda_0 > 0$, therefore $\lambda_3 > 0$, hence $x_3 = 0$. Since $x_1 + 2x_2 + 2x_3 = 72$ and $x_3 = 0$, we get $x_1 + 2x_2 = 72$. Since $x_1 \leq 42$, it follows that $x_2 > 0$, therefore $\lambda_2 = 0$. The system becomes:

$$\begin{cases} 1 - \lambda_0 + \lambda_1 - \lambda_4 &= 0 \\ 1 - 2\lambda_0 &= 0 \\ -2\lambda_0 + \lambda_3 &= 0 \end{cases}$$

The second equation gives $\lambda_0 = 1/2$, hence due to the first equation $\lambda_4 = 1/2 + \lambda_1 > 0$, therefore $x_1 = 42$. But $x_1 + 2x_2 = 72$, hence $x_2 = 15$.

- (Summary) The problem has a solution. Moreover, constraints are qualified everywhere and there is a unique point satisfying Kuhn and Tucker necessary optimality conditions. Therefore this point is the unique solution. The value of the problem is thus 57, obtained for $x_1 = 42$, $x_2 = 15$, and $x_3 = 0$.³

Solution 3.12. First, this problem has a solution. Indeed, inequalities $0 \leq y \leq (1-x)^3$ imply $x \leq 1$. Moreover, since $x \geq 0$, the constraint $y \leq (1-x)^3$ implies that $y \leq 1$. The constraint (meaning: “the set K of points satisfying the constraints”) is thus bounded. Since it is also closed, it is compact, and also non-empty. Moreover, the objective is continuous. Thus a solution exists.

- (1) Let λ_1 , λ_2 , λ_3 denote the multipliers associated to the constraints $0 \leq x$, $0 \leq y$ and $y \leq (1-x)^3$, respectively. The Lagrangian is:

$$L((x, y), \lambda) = 3x + y + \lambda_1 x + \lambda_2 y + \lambda_3[(1-x)^3 - y].$$

The necessary conditions are thus:

$$\begin{cases} 3 + \lambda_1 - 3\lambda_3(1-x)^2 &= 0 & (1) \\ 1 + \lambda_2 - \lambda_3 &= 0 & (2) \end{cases}$$

with $\lambda_i \geq 0$, $i = 1, 2, 3$, the exclusion conditions: $\lambda_1 x = 0$, $\lambda_2 y = 0$, $\lambda_3[(1-x)^3 - y] = 0$, as well as the problem's constraints $((x, y) \in K)$.

The first equation implies that $\lambda_3 \neq 0$. Therefore $y = (1-x)^3$. Moreover, if $x = 0$, then $y = 1$, so $(x, y) = (0, 1)$. Similarly, if $y = 0$, then $x = 1$, therefore due to the first equation: $3 + \lambda_1 = 0$, which cannot be since $\lambda_1 \geq 0$. Finally, if $x > 0$, $y > 0$, then $\lambda_1 = \lambda_2 = 0$, therefore $\lambda_3 = 1$ by (2), therefore (1) leads to $(1-x)^2 = 1$, hence $x = 0$ or $x = 2$. This contradicts the fact that $x > 0$ and $x \leq 1$. There is thus a unique point potentially satisfying the necessary conditions: $(0, 1)$. For $\lambda_2 = 0$, $\lambda_3 = 1$, and $\lambda_1 = 0$, all conditions are satisfied, so $(0, 1)$ indeed satisfies the necessary conditions.

³This solution can also be found through the following economic interpretation. This is a utility maximization problem for a consumer who can buy three kinds of goods, with unit prices $p_1 = 1$ and $p_2 = p_3 = 2$, and spend at most 72 euros, with an additional availability constraint: at most 42 units of each of these goods may be bought. The consumer is not interested by good 3, and likes good 1 as much as he likes good 2 (with no satiety, and no complementary between goods 1 and 2). Since good 1 is cheaper than good 2, the consumer buys as much of good 2 as he can, and use whatever money remains on good 2.

(2) The gradients of the constraint functions are:

$$\nabla g_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \nabla g_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \nabla g_3 = \begin{pmatrix} -3(1-x)^2 \\ -1 \end{pmatrix}$$

Let us consider all possible cases for the set $I(x, y)$ of binding constraints.

If $I(x, y) = \{i\}$, the qualification condition boils down to $\nabla g_i(x, y) \neq 0$, which is true.

If $I(x, y) = \{1, 2\}$, take $v = (-1, -1)$.

If $I(x, y) = \{1, 3\}$, then $(x, y) = (0, 1)$. We thus need $v_1 < 0$ and $v_2 > -3v_1$. This holds for instance for $v = (-1, 4)$.

If $I(x, y) = \{2, 3\}$, then $(x, y) = (1, 0)$. We must have $v_2 < 0$ and $v_2 > 0$, which cannot be. This point is not qualified.

If $I(x, y) = \{1, 2, 3\}$: impossible as this implies $x = y = 0$ and $y = (1 - x)^3$, hence $y = 0$ and $y = 1$.

To sum up, the constraint is qualified everywhere except at $(1, 0)$. The candidates to be solution of the problem are thus, on the one hand, the unqualified point $(1, 0)$, and on the other hand, the point satisfying the necessary conditions $(0, 1)$. Since the objective is better at $(1, 0)$ than at $(0, 1)$, the maximizer is $(1, 0)$, and the value is 3.

Remarks. This exercise demonstrates the relevance of the qualification condition. Let us also note that, in question 1, the exclusion condition $\lambda_1 x = 0$ does not imply that $\lambda_1 \neq 0$ if $x = 0$: it may be that both $x = 0$ and $\lambda_1 = 0$. This is the case here for the unique point satisfying the necessary conditions.

Solution 3.13. First of all observe that since the constraints are affine the constraints are qualified at every admissible point. Second, observe that the matrix M only has positive eigenvalues, so that the objective function is strongly convex and, in particular, coercive. Thus, the problem admits a solution $X = (x_1, x_2, x_3)$. By the Karush-Kuhn-Tucker theorem, there exists $\lambda, \lambda_1, \lambda_2, \lambda_3 \geq 0$ such that

$$2MX + \lambda_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \lambda_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

This rewrites

$$2X + \lambda_1 M^{-1} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \lambda_2 M^{-1} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \lambda_3 M^{-1} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \lambda M^{-1} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

This system rewrites

$$\begin{cases} 8x_1 = -2\lambda - 3\lambda_1 + \lambda_2 + \lambda_3, \\ 8x_2 = 2\lambda + \lambda_1 - 3\lambda_2 + \lambda_3, \\ 8x_3 = 6\lambda + \lambda_1 + \lambda_2 - 3\lambda_3. \end{cases}$$

In particular, we deduce that

$$6\lambda = 8(x_1 + x_2 + x_3) + (\lambda_1 + \lambda_2 + \lambda_3).$$

Consequently, if $\lambda = 0$, as the other multipliers are non-negative, we deduce that $\sum x_i = 0$, which entails $X = 0$, which is trivially not a solution.

Thus, $\lambda > 0$, which in turn yields $x_1 + 2x_2 + 3x_3 = 1$. As $\lambda_2 + \lambda_3 \geq 2\lambda > 0$ we deduce that λ_2 or λ_3 (or both) is positive. If $\lambda_2, \lambda_3 > 0$ then $x_2, x_3 = 0$ and so $x_1 = 1$.

If $\lambda_2 = 0$ and $\lambda_3 > 0$ then $x_3 = 0$ and we are left with the system

$$8x_1 = -2\lambda - 3\lambda_1 + \lambda_3 \quad 8x_2 = 2\lambda + \lambda_1 + \lambda_3 \quad 0 = 6\lambda + \lambda_1 - 3\lambda_3.$$

This gives (multiplying the first equation by 3 and adding the third one)

$$24x_1 = -8\lambda_1,$$

giving $x_1 = 0$ and thus $x_2 = \frac{1}{2}$.

Finally, if $\lambda_2 > 0$ and $\lambda_3 = 0$ we deduce in a similar fashion that

$$8x_1 = -2(\lambda_1 + \lambda_2),$$

but this is in contradiction with $\lambda_2 > 0$.

There are thus two candidates to be optimisers, and it is readily checked that the optimal value is reached at $(1, 0, 0)$.

Solution 3.14. (1) The objective function is continuous and coercive, and the admissible set is closed and non-empty. The problem thus has a solution.

(2) Let $g_1(x, y) := 2x - y^2$, $g_2(x, y) := -x$. Let us denote by $I(x)$ the set of active constraints at a given point x . If $I(x)$ is empty, or reduced to $\{1\}$ or $\{2\}$ the qualification is trivially obtained. Assuming that $I(x) = \{1, 2\}$, we obtain $-y^2 = 1$, which is impossible. Thus, the constraints are qualified at every point.

(3) Thus, the necessary optimality conditions are

$$\begin{cases} 2(x - 2) + 2\lambda_1 - \lambda_2 = 0, \\ 2y - 2y\lambda_1 = 0, \\ g_i \leq 0, \\ \lambda_i \geq 0, \lambda_i g_i = 0. \end{cases}$$

(4) From the second equation we see that either $y = 0$ or $\lambda_1 = 1$. If $y = 0$ then $2x \leq 1$, which in turn gives $x - 2 < 0$, so that $\lambda_1 > 0$. This gives $x = \frac{1}{2}$, giving a first candidate, $(1/2, 0)$.

If $\lambda_1 = 1$ then $2x = 1 + y^2$ so that $x > 0$ and so $\lambda_2 = 0$. Thus, $x = 1$ and so $y^2 = 1$. This gives the other candidates $(1, \pm 1)$.

(5) Consequently, the optimal value is 2, and it is attained at $(1, \pm 1)$.

7. NUMERICAL APPROXIMATION OF CONSTRAINED PROBLEMS

In this brief section, we will review the standard numerical methods to approximate solutions of constrained optimisation problems. Most of the section will be descriptive, and several of the proofs will rather be presented in the exercise sessions.

7.1. The projected gradient algorithm. Throughout, we will be working on constrained optimisation problems of the form

$$(7.1) \quad \min_{x \in K} f(x),$$

where K is a compact, convex subset of $\mathbb{R}^d$ and f is a $\mathcal{C}^1$ function. This typically corresponds to an inequality constrained optimisation problem of the form

$$K = \cap_{i=1}^m \{g_i \leq 0\}$$

where the g_i 's are convex, continuous functions and at least one of them is coercive. It should be noted that writing it under the form (7.1) might not seem the most natural way to tackle the problem, as we would rather like to have a method relying explicitly on the set of constraints rather than on K itself—this will rather be done when discussing penalty methods.

At any rate, we assume that we know the set K itself well enough to have its orthogonal projector Π_K under control. Let us recall the following fact:

Proposition 7.1. *For any closed, convex non-empty set $C \subset \mathbb{R}^d$, for any $x \in \mathbb{R}^d$, there exists a unique $\Pi_C(x) \in C$ such that*

$$\|\Pi_C(x) - x\| = \min_{y \in C} \|y - x\|.$$

The operator Π_C is called the orthogonal projection on C , and is uniquely characterised as the solution of the variational inequality

$$\forall y \in C, \langle x - \Pi_C(x), x - y \rangle \geq 0.$$

We refer to Exercise 1.7 for a proof of this Proposition.

We thus devise a gradient descent in the following way: fix an initialisation $x_0 \in K$ and a step size $\tau \geq 0$. For any $k \in \mathbb{N}$, define

$$(7.2) \quad x_{k+1} = \Pi_K(x_k - \tau \nabla f(x_k)).$$

As always, there are two questions:

- (1) First, does the sequence $\{x_k\}_{k \in \mathbb{N}}$ converge?
- (2) If so, can we specify the rate of convergence?

Naturally, since if, x^* being a minimiser, we have that x^* is an interior point of K , and if $\tau > 0$ is small enough, this is nothing but the gradient descent in the unconstrained case, we can hope—at best—to get a linear convergence rate.

The main result is the following:

Theorem 7.1. *Assume that f is α -strongly convex and let ∇f be μ -Lipschitz in K . Then:*

- (1) (7.1) has a unique solution x^* .
- (2) For any $\tau \in \left(0; \frac{2\alpha}{\mu^2}\right)$, for any initialisation $x_0 \in K$, the sequence generated by the projected gradient algorithm converges linearly to x^* .

Just as in gradient descent, the strong convexity assumption simplifies our queries, but can be weakened. It is a good exercise to try and prove Theorem 7.1 as it is merely an adaptation of similar theorems on unconstrained gradient descent.

We note, for further reference, that although this is a neat (and well behaved algorithm), the main difficulty lies in the computation of the projection operator Π_K ; this observation is the motivation for the duality methods which we will investigate in the next chapter.

7.2. Penalisation methods. For the sake of clarity, let us now assume that

$$K = \{g \leq 0\}$$

with $g \in \mathcal{C}^1$, convex and coercive. Introduce, the function g being given, the map

$$h : x \mapsto \max(g(x), 0)^2.$$

Then

$$K = \{h = 0\},$$

and the function h is both $\mathcal{C}^1$ and convex. We can thus replace (7.1) with

$$(7.3) \quad \min_{h=0} f.$$

Now, let $\varepsilon > 0$ be a small parameter, and replace (7.1) with

$$(7.4) \quad \min_{x \in \mathbb{R}^d} F_\varepsilon(x) := \left(f(x) + \frac{1}{\varepsilon} h(x) \right).$$

As g is coercive, so is h and, provided f is bounded from below, this implies the coercivity of F_ε so that the optimisation problem above is well-posed. We can thus apply the algorithms we saw previously to provide a good (linearly converging) approximation of the minimiser x_ε^* of F_ε . Formally, we can expect that

$$x_\varepsilon^* \xrightarrow{\varepsilon \rightarrow 0} x^*,$$

a minimiser of f on K . It is in fact the case, and can be used to derive the multiplier rule, but that does not give a reasonable convergence rate. We refer to the exercise sheet.

EXERCISES FOR THE THIRD PART OF PART 3

Exercise 3.15 (Interior penalty method). Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ and $g : \mathbb{R}^d \rightarrow \mathbb{R}$ be two continuous functions on $\mathbb{R}^d$, strictly convex with g coercive. Assume that there exists a point x_0 such that $g(x_0) < 0$.

- (1) Show that the constrained optimisation problem

$$\min_{x \in K} f(x) \quad \text{where } K := \{x \in \mathbb{R}^d : g(x) \leq 0\}$$

has a unique solution $\bar{x}$.

The aim of this exercise is to approach $\bar{x}$ by a penalty method (that is, by introducing a series of problems defined on the interior of the constraint by adding a term to the objective and whose solutions converge to $\bar{x}$). For all $\varepsilon > 0$, let

$$J_\varepsilon(x) := f(x) - \frac{\varepsilon}{g(x)} \quad \forall x \in \text{Int}(K) = \{x \in \mathbb{R}^d : g(x) < 0\}.$$

- (2) Show that J_ε has a unique minimiser x_ε in $\text{Int}(K)$.
- (3) Let $x \in \text{Int}(K)$. Check that $J_\varepsilon(x) \geq J_\varepsilon(x_\varepsilon) \geq f(x_\varepsilon)$. Show that, if $\tilde{x}$ is an accumulation point of (x_ε) as $\varepsilon \rightarrow 0$, then $f(x) \geq f(\tilde{x})$.
- (4) Conclude that (x_ε) goes to $\bar{x}$ as $\varepsilon \rightarrow 0$.
- (5) Assume f and g of class C^1 on $\mathbb{R}^d$. Write the optimality condition for x_ε and prove the existence of a Lagrange multiplier $\lambda \geq 0$ for $\bar{x}$.
- (6) Suggest a numerical method to approximate $\bar{x}$.

Exercise 3.16. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ and $g : \mathbb{R}^d \rightarrow \mathbb{R}$ be two strictly convex C^1 functions on $\mathbb{R}^d$, with f coercive. Let $\bar{x}$ be the unique solution of the constrained problem

$$\min_{x \in K} f(x) \quad \text{where } K := \{x \in \mathbb{R}^d : g(x) \leq 0\}$$

Assume that constraint K is qualified. Let λ denote the multiplier in the necessary optimality condition for $\bar{x}$.

For all $\varepsilon > 0$, let

$$J_\varepsilon(x) := f(x) + \frac{1}{\varepsilon} (\max\{0, g(x)\}) \quad \forall x \in \mathbb{R}^d.$$

- (1) Show that J_ε has a unique minimiser x_ε in $\mathbb{R}^d$.
- (2) Show that if $\varepsilon \in]0, 1/\lambda[$, then $x_\varepsilon = \bar{x}$.
- (3) In practice, the value of λ is unknown, and we need to estimate it. Show that $x_\varepsilon = \bar{x}$ if $\varepsilon \in]0, M^{-1}[$ with $M := \max_{x \in K} \|\nabla f(x)\| / \min_{x \in \partial K} \|\nabla g(x)\|$.

SOLUTIONS FOR THE THIRD PART OF PART 3

- Solution 3.15.** (1) *The constraint K is nonempty, compact and convex (since g is convex and coercive) and the objective strictly convex and continuous, therefore the problem has a unique solution $\bar{x}$.*
- (2) *Let (x_n) be a minimising sequence for J_ε on $\text{Int}(K)$. Since K is compact, there exists a subsequence (which, to simplify, we still denote by (x_n)) which converges to $x \in K$. Let us show that $x \in \text{Int}(K)$. Indeed, otherwise, $g(x) = 0$ and therefore $g(x_n) \rightarrow 0^-$, which shows that*

$$\inf_{x \in \text{Int}(K)} J_\varepsilon(x) = \lim_{n \rightarrow +\infty} J_\varepsilon(x_n) = f(x_n) - \frac{\varepsilon}{g(x_n)} \rightarrow +\infty.$$

There is thus a contradiction with the fact that (x_n) is a minimising sequence. Therefore $g(x) < 0$. Since J_ε is continuous in $\text{Int}(K)$, we have

$$\inf_{x \in \text{Int}(K)} J_\varepsilon(x) = \lim_{n \rightarrow +\infty} J_\varepsilon(x_n) = J_\varepsilon(x),$$

and therefore x is a minimiser of J_ε in $\text{Int}(K)$. This minimiser is unique since J_ε is strictly convex : indeed, this is the sum of the strictly convex function f and of the function $x \rightarrow -\frac{\varepsilon}{g(x)}$ which is also strictly convex since, if $x, y \in \text{Int}(K)$ with $x \neq y$ and if $t \in]0, 1[$, then

$$g((1-t)x + ty) < (1-t)g(x) + tg(y),$$

hence, since $s \rightarrow -1/(\varepsilon s)$ is increasing and strictly convex on $] -\infty, 0[$,

$$-\frac{\varepsilon}{g((1-t)x + ty)} < -\frac{\varepsilon}{((1-t)g(x) + tg(y))} < -\frac{(1-t)\varepsilon}{g(x)} - \frac{t\varepsilon}{g(y)}.$$

therefore J_ε has a unique minimiser x_ε in $\text{Int}(K)$.

- (3) *Let $x \in \text{Int}(K)$. We have $J_\varepsilon(x) \geq J_\varepsilon(x_\varepsilon)$ since x_ε is the minimiser of J_ε . Moreover, we also have*

$$J_\varepsilon(x_\varepsilon) = f(x_\varepsilon) - \frac{\varepsilon}{g(x_\varepsilon)} \geq f(x_\varepsilon)$$

since $g(x_\varepsilon) < 0$. Therefore $J_\varepsilon(x) \geq J_\varepsilon(x_\varepsilon) \geq f(x_\varepsilon)$.

Let $\tilde{x} \in K$ be an accumulation point of (x_ε) when $\varepsilon \rightarrow 0^+$. Then there exists a subsequence (x_{ε_n}) which converges to $\tilde{x} \in K$ with $\varepsilon_n \rightarrow 0^+$. On a

$$\lim_{n \rightarrow +\infty} J_{\varepsilon_n}(x) = f(x) \geq \lim_{n \rightarrow +\infty} f(x_{\varepsilon_n}) = f(\tilde{x}).$$

therefore $f(x) \geq f(\tilde{x})$.

- (4) *We proved at the previous question that if a subsequence (x_{ε_n}) converges towards a given $\tilde{x} \in K$, then $f(x) \geq f(\tilde{x})$ for all x such that $g(x) < 0$. Let us show that this inequality still holds if $g(x) = 0$. Indeed, for all $t \in]0, 1[$,*

$$g((1-t)x + tx_0) \leq (1-t)g(x) + tg(x_0) \leq tg(x_0) < 0.$$

so the result of the previous question implies that $f((1-t)x + tx_0) \geq f(\tilde{x})$ for all $t \in]0, 1[$. Letting t go to 0^+ , we obtain the inequality $f(x) \geq f(\tilde{x})$. Therefore $\tilde{x}$ is a minimiser of f on K . This shows that $\tilde{x} = \bar{x}$.

Let us finally show that (x_ε) goes to $\bar{x}$ when $\varepsilon \rightarrow 0$ (the following argument is standard). Assume that this is not the case : then there exists a subsequence (x_{ε_n}) and a real number $\theta > 0$ such that $\|x_{\varepsilon_n} - \bar{x}\| \geq \theta$. Since

K is compact, we can again extract a subsequence $(x_{\varepsilon'_n})$ of (x_{ε_n}) which converges to a given $\bar{x}$. But then $\bar{x} = \bar{x}$ and we have a contradiction. Therefore (x_ε) goes to $\bar{x}$ when $\varepsilon \rightarrow 0$.

- (5) We assume that f and g are of class C^1 on $\mathbb{R}^d$. Since J_ε has for minimiser x_ε on the open set $\text{Int}(K)$, we have:

$$(*) \quad \nabla f(x_\varepsilon) + \frac{\varepsilon \nabla g(x_\varepsilon)}{(g(x_\varepsilon))^2} = 0.$$

If $g(\bar{x}) < 0$, the result is obvious since $\frac{\varepsilon \nabla g(x_\varepsilon)}{(g(x_\varepsilon))^2} \rightarrow 0$. Otherwise $g(\bar{x}) = 0$ and then, since $\nabla g(\bar{x}) \neq 0$, we have for $\varepsilon > 0$ small enough, taking the norm in expression (*)

$$\frac{\varepsilon}{(g(x_\varepsilon))^2} = \frac{\|\nabla f(x_\varepsilon)\|}{\|\nabla g(x_\varepsilon)\|} \rightarrow \frac{\|\nabla f(\bar{x})\|}{\|\nabla g(\bar{x})\|}.$$

Let $\lambda = \frac{\|\nabla f(\bar{x})\|}{\|\nabla g(\bar{x})\|}$. Going to the limit in (*), we thus have:

$$\nabla f(\bar{x}) + \lambda \nabla g(\bar{x}) = 0, \quad \text{with } \lambda \geq 0.$$

This proves the existence of a multiplier $\lambda \geq 0$ for $\bar{x}$.

- (6) Since $x_\varepsilon \rightarrow \bar{x}$ when $\varepsilon \rightarrow 0^+$, it suffices to approach x_ε . The latter is the minimiser of an unconstrained problem (more precisely: the constraint is open). We may thus use unconstrained optimisation methods (fixed-step gradient method or Newton method, for instance).

Solution 3.16. (1) Function J_ε is continuous and is the sum of function f , which is strictly convex and coercive, and of a positive convex function $(x \rightarrow \varepsilon^{-1} \max\{0, g(x)\})$, since g is convex. Therefore J_ε is strictly convex and coercive and has a unique minimiser x_ε in $\mathbb{R}^d$.

- (2) By the duality theorem, we know that function $x \rightarrow f(x) + \lambda g(x)$ has a global minimiser at $\bar{x}$ on $\mathbb{R}^d$. If $\lambda = 0$, then $\bar{x}$ is the global minimiser of f on $\mathbb{R}^d$. Since moreover $\bar{x} \in K$, for all x in $\mathbb{R}^d$:

$$J_\varepsilon(\bar{x}) = f(\bar{x}) \leq f(x) \leq J_\varepsilon(x)$$

therefore $x_\varepsilon = \bar{x}$. Assume now that $\lambda > 0$. We then have $g(\bar{x}) = 0$ by the exclusion condition. Therefore

$$J_\varepsilon(\bar{x}) = f(\bar{x}) + \frac{1}{\varepsilon} (\max\{0, g(\bar{x})\}) = f(\bar{x}) + \lambda g(\bar{x}) \leq f(x_\varepsilon) + \lambda g(x_\varepsilon).$$

If $g(x_\varepsilon) \geq 0$, then $\lambda g(x_\varepsilon) \leq \varepsilon^{-1} \max\{0, g(x_\varepsilon)\}$ since $\varepsilon \in]0, 1/\lambda]$. If $g(x_\varepsilon) \leq 0$, this inequality still holds. Therefore $J_\varepsilon(\bar{x}) \leq f(x_\varepsilon) + \lambda g(x_\varepsilon) \leq J_\varepsilon(x_\varepsilon)$ and therefore $\bar{x}$ is a minimiser of J_ε on $\mathbb{R}^d$. Since this minimiser is unique, we deduce that $x_\varepsilon = \bar{x}$.

- (3) Given the previous question, it suffices to show that $\lambda \leq M$, since then $\varepsilon \in]0, M^{-1}]$ implies that $\varepsilon \in]0, \lambda^{-1}]$. But the optimality conditions say that

$$\nabla f(\bar{x}) + \lambda \nabla g(\bar{x}) = 0,$$

and therefore

$$\lambda = \frac{\|\nabla f(\bar{x})\|}{\|\nabla g(\bar{x})\|} \leq M.$$

therefore $x_\varepsilon = \bar{x}$ if $\varepsilon \in]0, M^{-1}[$.

8. DUALITY: THEORY AND PRACTICE

In this section we will work (as this is now quite important) with N possibly non-linear constraints

$$\forall i \in \{1, \dots, N\}, g_i(x) \leq 0.$$

Nevertheless, so as not to bother with the qualification of constraints, the running assumptions are

$$(8.1) \quad \text{For any } i, g_i \text{ is convex and coercive}$$

and

$$(8.2) \quad \exists x_0, \forall i \in \{1, \dots, N\}, g_i(x_0) < 0.$$

We let

$$K := \cap_{i=1}^N \{g_i \leq 0\}$$

and we consider

$$(8.3) \quad \min_{x \in K} f(x)$$

where f is a $\mathcal{C}^1$, convex function.

8.1. The augmented Lagrangian method: a primer on duality.

8.1.1. *Basics.* The Karush-Kuhn-Tucker conditions suggest looking at the so-called Lagrangian

$$\mathcal{L}(x, \lambda) := f(x) + \sum_{i=1}^N \lambda_i g_i(x).$$

A key observation is the following one, that relates the saddle points of the Lagrangian to the solution of the constrained optimisation problem:

Theorem 8.1. *Let x^* be a solution of (8.3) and $\lambda(x^*) := (\lambda_1(x^*), \dots, \lambda_N(x^*))$ be the Lagrange multiplier given by Theorem 6.2. Then $(x^*, \lambda(x^*))$ is a saddle-point of $\mathcal{L}$ in the sense that*

$$\min_{x \in \mathbb{R}^d} \mathcal{L}(x, \lambda^*) = \mathcal{L}(x^*, \lambda(x^*)) = \max_{\lambda \geq 0} \mathcal{L}(x^*, \lambda)$$

where we use the notation $\lambda \geq 0$ to signify: for any i , $\lambda_i \geq 0$ and $\lambda(x^*) = (\lambda_1(x^*), \dots, \lambda_N(x^*))$.

Conversely, let (x, λ) be a saddle point of $\mathcal{L}$. Then $x \in K$, x solves (8.3) and λ are the associated the Lagrange multipliers.

Proof of Theorem 8.1. As all $\lambda_i(x^*)$ are non-negative, $x \mapsto \mathcal{L}(x, \lambda(x^*))$ is convex, so that the condition $\nabla_x \mathcal{L}(x^*, \lambda(x^*)) = 0$ implies the minimality of x^* . Furthermore, fix x^* and let $\mu \geq 0$. Assume, up to a relabelling,

$$g_1(x^*), \dots, g_n(x^*) = 0, g_i(x^*) < 0, i \geq n+1.$$

To check the optimality of $\lambda(x^*)$, it suffices to prove that

$$0 \leq \sum_{i=1}^N (\lambda_i(x^*) - \mu_i) g_i(x^*) = - \sum_{i=n+1}^N \mu_i g_i(x^*),$$

which is obvious.

Now, assume that (x, λ) is a saddle point of $\mathcal{L}$. The optimality in λ implies

$$\forall \mu \geq 0, \sum_{i=1}^N (\lambda_i - \mu_i) g_i(x) \geq 0.$$

This immediately implies (letting $\mu_i \rightarrow \infty$) that

$$\forall i \in \{1, \dots, N\}, g_i(x) \leq 0$$

or, in other words, that $x \in K$. Additionally, this gives

$$\sum_{i=1}^N \lambda_i g_i(x) \geq 0$$

and thus

$$(8.4) \quad \forall i \in \{1, \dots, N\}, \lambda_i g_i(x) = 0.$$

Furthermore, for any $x' \in K$, we have

$$f(x) = f(x) + \sum_{i=1}^N \lambda_i g_i(x) \leq f(x') + \sum_{i=1}^N \lambda_i g_i(x') \leq f(x'),$$

which gives the conclusion. $\square$

The main result of this part is the following:

Theorem 8.2. *Let x^* be a solution of (8.3) and $\lambda(x^*)$ be the associated Lagrange multiplier. Then*

$$\min_{x \in \mathbb{R}^d} \max_{\lambda \geq 0} \mathcal{L}(x, \lambda) = \mathcal{L}(x^*, \lambda(x^*)) = \max_{\lambda \geq 0} \min_{x \in \mathbb{R}^d} \mathcal{L}(x, \lambda).$$

Proof of Theorem 8.2. On the one hand, we have, for any (x, λ) ,

$$\inf_{y \in \mathbb{R}^d} \mathcal{L}(y, \lambda) \leq \mathcal{L}(x, \lambda) \leq \sup_{\mu \geq 0} \mathcal{L}(x, \mu)$$

so that

$$\sup_{\lambda \geq 0} \inf_{y \in \mathbb{R}^d} \mathcal{L}(y, \lambda) \leq \inf_{y \in \mathbb{R}^d} \sup_{\mu \geq 0} \mathcal{L}(x, \mu).$$

Furthermore,

$$\mathcal{L}(x^*, \lambda(x^*)) = \sup_{\lambda \geq 0} \mathcal{L}(x^*, \lambda)$$

so that

$$\inf_{y \in \mathbb{R}^d} \sup_{\lambda \geq 0} \mathcal{L}(y, \lambda) \leq \mathcal{L}(x^*, \lambda(x^*)).$$

Thus,

$$\mathcal{L}(x^*, \lambda(x^*)) = \inf_{x \in \mathbb{R}^d} \mathcal{L}(x, \lambda(x^*)) \leq \sup_{\lambda \geq 0} \inf_{x \in \mathbb{R}^d} \mathcal{L}(x, \lambda),$$

which provides the reverse inequality. $\square$

Using the terminology of convex optimisation, we say that convex problems have no duality gaps. At any rate, what matters most now is that we have replaced a potentially nasty optimisation problem with a (continuous) family of constrained optimisation problems, but where the constraints $\lambda \geq 0$ are particularly easy to handle (in particular, the projection operator is explicit). This is the basic idea behind the Uzawa algorithm, which will take up the rest of the chapter.

8.1.2. *An example.* Let us solve the following problem using a duality method:

$$\begin{aligned} \min_{\substack{x^2 + y^2 \leq 1 \\ y + z \leq 0}} \quad & \frac{1}{2} [(x-2)^2 + y^2 + z^2] \end{aligned}$$

Let $X = (x, y, z)$,

$$f(X) = \frac{1}{2} [(x-2)^2 + y^2 + z^2],$$

$g_1(X) = x^2 + y^2 - 1$, $g_2(X) = y + z$, and $K = \{X \in \mathbb{R}^3, g_i(X) \leq 0, i = 1, 2\}$.

- We are in the convex case (f and the g_i convex)
- f is coercive, K closed, therefore there exists a solution (unique, since f is strictly convex).
- the constraint is everywhere qualified since there exists $X = (0, 0, -1)$ such that $g_1(X) < 0$ and $g_2(X) < 0$.
- the dual problem is : $\sup_{\lambda \geq 0} \mathcal{G}(\lambda)$ where

$$\mathcal{G}(\lambda) = \inf_{X \in \mathbb{R}^3} L(X, \lambda) \text{ with } L(X, \lambda) = \frac{1}{2} [(x-2)^2 + y^2 + z^2] + \lambda_1(x^2 + y^2 - 1) + \lambda_2(y + z)$$

- We must first compute $\mathcal{G}(\lambda)$, by solving the corresponding minimisation problem. This is an unconstrained convex optimisation problem. The fact that the gradient of the objective cancels (that is, the gradient of $L(X, \lambda)$ with respect to X) is therefore a necessary and sufficient optimality condition. We get :

$$x(\lambda) = 2/(1 + 2\lambda_1); \quad y(\lambda) = -\lambda_2/(1 + 2\lambda_1); \quad z(\lambda) = -\lambda_2$$

hence

$$\mathcal{G}(\lambda) = L(X(\lambda), \lambda) = \frac{4\lambda_1}{1 + 2\lambda_1} - \frac{\lambda_2^2}{2(1 + 2\lambda_1)} - \frac{\lambda_2^2}{2} - \lambda_1$$

We now try to maximise $\mathcal{G}(\lambda)$ for $\lambda \geq 0$. We first get $\lambda_2 = 0$. It remains to maximise

$$\psi(\lambda_1) := \frac{4\lambda_1}{1 + 2\lambda_1} - \lambda_1$$

It may be checked that ψ is concave, and that its derivative cancels at $\lambda_1 = 1/2$. The global maximiser ψ is therefore 1/2, and therefore a fortiori the maximiser for the $\lambda_1 \geq 0$.

- **Conclusion.** The dual problem has a unique solution: $\hat{\lambda} = (1/2, 0)$, and value $G(\hat{\lambda}) = 1/2$. The primal problem has the same value. Its solution is : $X(\hat{\lambda}) = (1, 0, 0)$.

8.2. **The Uzawa algorithm.** The idea of the Uzawa algorithm is to generate two sequences $\{x_k\}_{k \in \mathbb{N}}$, $\{\lambda_k\}_{k \in \mathbb{N}}$ that should converge to x^* and $\lambda(x^*)$. We introduce the following notations:

$$\mathbb{R}_+^N := \{\lambda \in \mathbb{R}^N, \forall i \in \{1, \dots, N\}, \lambda_i \geq 0\}, g(x) = (g_1(x), \dots, g_N(x)).$$

An important observation is the following one: for any $\tau > 0$,

$$(8.5) \quad \lambda(x^*) = \Pi_{\mathbb{R}_+^N}(\lambda(x^*) + \tau g(x^*)).$$

The proof of (8.5) is straightforward: first, observe that for any $y \in \mathbb{R}^N$

$$\Pi_{\mathbb{R}_+^N}(y) = (\max(y_i, 0))_{i=1, \dots, N}.$$

Second, on each coordinate, either $g_i(x^*) < 0$ and $\lambda_i(x^*) = 0$, or $g_i(x^*) = 0$ and $\lambda_i(x^*) \geq 0$.

Back to the Uzawa algorithm, which goes as follows:

- (1) Start from any $\lambda_0 \in \mathbb{R}_+^N$ and fix a step size $\tau > 0$.
- (2) For any $k \in \mathbb{N}$, set

$$x_k := \operatorname{argmin}_{\mathbb{R}^d} (f(x) + \langle \lambda_k, g(x) \rangle)$$

and

$$\lambda_{k+1} := \Pi_{\mathbb{R}_+^N}(\lambda_k + \tau g(x_k)).$$

Now, can we expect to have convergence for this algorithm and, if so, at which rate? To do so, let us first remain at a formal level. Assume that for any $\lambda \in \mathbb{R}_+^N$ there exists a unique x_λ that solves

$$\ell(\lambda) := \min_{x \in \mathbb{R}^d} (f(x) + \langle \lambda, g(x) \rangle) = \mathcal{L}(x, \lambda).$$

Assume furthermore that the map $\lambda \mapsto x_\lambda$ is differentiable. Then, it appears that

$$\begin{aligned} \nabla \ell(\lambda) &= \nabla_x \mathcal{L}(x_\lambda, \lambda) \nabla_\lambda x_\lambda + \nabla_\lambda \mathcal{L}(x_\lambda, \lambda) \\ &= \nabla_\lambda \mathcal{L}(x_\lambda, \lambda) \text{ because of the optimality of } x_\lambda \\ &= g(x_\lambda), \end{aligned}$$

so that the Uzawa algorithm appears as a real projected gradient ascent. In particular, this means that we should expect convergence, and maybe even linear convergence. Unfortunately, while the convergence indeed holds under suitable assumption, the rate is not always known. We summarise this in the following theorem:

Theorem 8.3. *Assume that f is α -strongly convex and that g is convex and μ -Lipschitz. Then for any $\tau \in (0, \frac{2\alpha}{\mu^2})$, the sequence $\{x_k\}_{k \in \mathbb{N}}$ converges to the solution x^* of (8.3).*

Proof of Theorem 8.3. We proceed in several steps: first of all, observe that for any $\lambda \in \mathbb{R}_+^N$, letting x_λ be the unique solution of

$$\min_{x \in \mathbb{R}^d} (f(x) + \langle \lambda, g(x) \rangle)$$

there holds, for any $y \in \mathbb{R}^d$,

$$(8.6) \quad \langle \nabla f(x_\lambda), y - x_\lambda \rangle + \langle \lambda, g(y) - g(x_\lambda) \rangle \geq 0.$$

Proof of (8.6). This is a consequence of convexity: let $y \in \mathbb{R}^d$. Then for any $t \in [0, 1]$ there holds

$$f((1-t)x_\lambda + ty) - f(x_\lambda) + \langle \lambda, g((1-t)x_\lambda + ty) - g(x_\lambda) \rangle \geq 0.$$

As g is convex and $\lambda \geq 0$ this implies

$$f((1-t)x_\lambda + ty) - f(x_\lambda) + \langle \lambda, -tg(x_\lambda) + tg(y) \rangle \geq 0.$$

Dividing by t and letting $t \rightarrow 0$ provides the required inequality. $\square$

An important consequence of (8.6) is the following:

$$\begin{cases} \langle \nabla f(x_k), x^* - x_k \rangle + \langle \lambda_k, g(x^*) - g(x_k) \rangle \geq 0, \\ \langle \nabla f(x^*), x_k - x^* \rangle + \langle \lambda^*, g(x_k) - g(x^*) \rangle \geq 0. \end{cases}$$

Combining these two inequalities we deduce

$$(8.7) \quad \langle \nabla f(x^*) - \nabla f(x_k), x_k - x^* \rangle + \langle \lambda^* - \lambda_k, g(x_k) - g(x^*) \rangle \geq 0.$$

Using the α -strong convexity of f , (8.7) in turn implies

$$(8.8) \quad 0 \leq \alpha \|x_k - x^*\|^2 \leq \langle \lambda^* - \lambda_k, g(x_k) - g(x^*) \rangle.$$

Now, observe that from (8.5) and the fact that projections are 1-Lipschitz we also have the estimate

$$\begin{aligned} \|\lambda_{k+1} - \lambda^*\|^2 &\leq \|\lambda_k - \lambda^* + \tau(g(x_k) - g(x^*))\|^2 \\ &= \|\lambda_k - \lambda^*\|^2 + \tau^2 \mu^2 \|x_k - x^*\|^2 + 2\tau \langle \lambda_k - \lambda^*, g(x_k) - g(x^*) \rangle \\ &\leq \|\lambda_k - \lambda^*\|^2 + \|x_k - x^*\|^2 (\tau^2 \mu^2 - 2\alpha\tau). \end{aligned}$$

Thus, if τ is chosen so that

$$\tau < \frac{2\alpha}{\mu^2}$$

we obtain

$$\|\lambda_{k+1} - \lambda^*\|^2 \leq \|\lambda_k - \lambda^*\|^2.$$

The sequence $\{\|\lambda_k - \lambda^*\|^2\}_{k \in \mathbb{N}}$ is consequently non-increasing, and so it converges.

Passing to the limit in

$$\|\lambda_{k+1} - \lambda^*\|^2 \leq \|\lambda_k - \lambda^*\|^2 + \|x_k - x^*\|^2 (\tau^2 \mu^2 - 2\alpha\tau)$$

we obtain

$$\|x_k - x^*\|^2 \xrightarrow[k \rightarrow \infty]{} 0.$$

□

Observe that this proof does not give convergence to the solution of the dual problem; nevertheless, in certain specific cases, it is possible to show it. We refer to Exercise 3.23.

EXERCISES FOR THE FOURTH PART OF PART 3

Exercise 3.17. Solve the following problem using a duality method:

$$\begin{aligned} \min_{\substack{x^2 + y^2 \leq 1 \\ y + z \leq 0}} \quad & \frac{1}{2} [(x-2)^2 + y^2 + z^2] \end{aligned}$$

Exercise 3.18. Solve the following problem using a duality method:

$$\min_{\frac{1}{2}x^T A x \leq 1} c^T x$$

where A is a positive definite symmetric matrix of size $n \times n$, and c a vector of $\mathbb{R}^n$.

Exercise 3.19. Consider the problem

$$(P) \quad \min_{Cx \leq d} \frac{1}{2} x^T A x + b^T x$$

where A is a $n \times n$ positive definite symmetric matrix, b a vector of $\mathbb{R}^n$, C an $l \times n$ matrix, and d a vector of $\mathbb{R}^l$. The expression $Cx \leq d$ means that each coordinate of Cx is non-larger than the corresponding coordinate of d . Show that the dual problem of Problem (P) is the following problem:

$$\max_{\lambda \in \mathbb{R}_+^l} -\frac{1}{2} b^T A^{-1} b - \frac{1}{2} \lambda^T C A^{-1} C^T \lambda - (b^T A^{-1} C^T + d^T) \lambda$$

Exercise 3.20. Solve the following problem using a duality method:

$$\min_{\langle s, x \rangle \leq 0} \frac{1}{2} \|x\|^2 - \langle c, x \rangle$$

where s and c are non-zero vectors of $\mathbb{R}^n$.

Exercise 3.21. Let a_i ($i = 1, \dots, n$) be positive real numbers. Let x satisfy $\sum_{i=1}^n x_i^2 / a_i^2 > 1$. That is, x does not belong to the ellipsoid

$$E = \{u \in \mathbb{R}^n \mid \sum_{i=1}^n x_i^2 / a_i^2 \leq 1\}$$

Compute the dual problem $d(\lambda)$ of the problem

$$\min_{u \in E} \|u - x\|^2$$

Show that the maximum of $d(\lambda)$ satisfies

$$\sum_{i=1}^n \frac{a_i^2 x_i^2}{(a_i^2 + \lambda)^2} = 1.$$

Exercise 3.22 (The Danskin theorem). Let $f : \mathbb{R}^n \times \mathbb{R}^d \rightarrow \mathbb{R}$ be such that ∇f is locally Lipschitz. We assume that there exists $\alpha > 0$ such that

$$\forall (x, z) \in \mathbb{R}^n \times \mathbb{R}^d, \nabla_z^2 f(x, z) \geq \alpha \text{Id}.$$

We define

$$\ell(x) := \min_{z \in \mathbb{R}^d} f(x, z).$$

- (1) Show that ℓ is well-defined, and that the optimisation problem has a unique solution z_x .
- (2) Show that the map $x \mapsto z_x$ is Lipschitz continuous.

(3) Show that ℓ is differentiable, and that

$$\nabla \ell(x) = \nabla_x f(x, z_x).$$

Exercise 3.23. We want to investigate certain situations where the Uzawa algorithm provides not only convergence of the approximations of the primal problem, but also of the dual problem. To this end, we consider two functions f, g that satisfy the following assumptions:

- (1) f and g are convex, M_0 -Lipschitz with M_1 -Lipschitz gradients.
- (2) f is α -strongly convex.
- (3) There exists $x_0 \in \mathbb{R}^d$ such that $g(x_0) < 0$.

We consider the constrained optimisation problem

$$\min_{x, g(x) \leq 0} f(x),$$

and we let $\{x_k, \lambda_k\}_{k \in \mathbb{N}}$ be the sequence generated by the Uzawa algorithm. Similarly, we let

$$\mathcal{L}(x, \lambda) := f(x) + \lambda g(x)$$

be the associated Lagrangian.

- (1) Recall why, for any $\tau > 0$ small enough, the sequence generated by the Uzawa algorithm with step-size $\tau > 0$ converges to the solution of the primal problem.
- (2) Let $\lambda, \mu \geq 0$ be fixed, and let x_λ, x_μ minimise $\mathcal{L}(\cdot, \lambda)$ and $\mathcal{L}(\cdot, \mu)$ respectively. Show that there exists a constant M_2 (independent of λ and μ) such that

$$\|x_\lambda - x_\mu\| \leq M_2 |\lambda - \mu|.$$

- (3) Show that $\lambda \mapsto \ell(\lambda) := \min_{x \in \mathbb{R}^d} \mathcal{L}(x, \lambda)$ is either strictly concave or linear, and has a Lipschitz continuous gradient.
- (4) Conclude; you can use the following-admitted-fact: if f is convex, coercive and with a μ -Lipschitz gradient, if $K \subset \mathbb{R}^d$ is a compact, convex subset of $\mathbb{R}^d$ then for $\tau > 0$ we set

$$x_{k+1} := \Pi_K(x_k - \tau \nabla f(x_k)),$$

Π_K being the projection on K . Then

$$(8.9) \quad f(x_k) - \min_K f \xrightarrow{k \rightarrow \infty} 0.$$

This was proved in the midterm.

SOLUTIONS FOR THE FOURTH PART OF PART 3

Solution 3.17. Let $X = (x, y, z)$,

$$f(X) = \frac{1}{2}[(x-2)^2 + y^2 + z^2],$$

$g_1(X) = x^2 + y^2 - 1$, $g_2(X) = y + z$, and $K = \{X \in \mathbb{R}^3, g_i(X) \leq 0, i = 1, 2\}$.

- We are in the convex case (f and the g_i convex)
- f is coercive, K closed, therefore there exists a solution (unique, since f is strictly convex).
- the constraint is everywhere qualified since there exists $X = (0, 0, -1)$ such that $g_1(X) < 0$ and $g_2(X) < 0$.
- the dual problem is : $\sup_{\lambda \geq 0} \mathcal{G}(\lambda)$ where

$$\mathcal{G}(\lambda) = \inf_{X \in \mathbb{R}^3} L(X, \lambda) \text{ with } L(X, \lambda) = \frac{1}{2}[(x-2)^2 + y^2 + z^2] + \lambda_1(x^2 + y^2 - 1) + \lambda_2(y + z)$$

- We must first compute $\mathcal{G}(\lambda)$, by solving the corresponding minimization problem. This is an unconstrained convex optimisation problem. The fact that the gradient of the objective cancels (that is, the gradient of $L(X, \lambda)$ with respect to X) is therefore a necessary and sufficient optimality condition. We get :

$$x(\lambda) = 2/(1 + 2\lambda_1); \quad y(\lambda) = -\lambda_2/(1 + 2\lambda_1); \quad z(\lambda) = -\lambda_2$$

hence

$$\mathcal{G}(\lambda) = L(X(\lambda), \lambda) = \frac{4\lambda_1}{1 + 2\lambda_1} - \frac{\lambda_2^2}{2(1 + 2\lambda_1)} - \frac{\lambda_2^2}{2} - \lambda_1$$

We now try to maximize $\mathcal{G}(\lambda)$ for $\lambda \geq 0$. We first get $\lambda_2 = 0$. It remains to maximize

$$\psi(\lambda_1) := \frac{4\lambda_1}{1 + 2\lambda_1} - \lambda_1$$

It may be checked that ψ is concave, and that its derivative cancels at $\lambda_1 = 1/2$. The global maximizer ψ is therefore $1/2$, and therefore a fortiori the maximizer for the $\lambda_1 \geq 0$.

- **Conclusion.** The dual problem has a unique solution: $\hat{\lambda} = (1/2, 0)$, and value $G(\hat{\lambda}) = 1/2$. The primal problem primal has the same value. Its solution is : $X(\hat{\lambda}) = (1, 0, 0)$.

Solution 3.18. We may assume that $c \neq 0$: otherwise the problem is obvious. We are in the convex case ; K is compact (since $x \rightarrow x^T A x$ is coercive, therefore for all constant M , the set of vectors x such that $x^T A x \leq M$ is bounded, and in particular for $M = 1$), therefore there is a solution. Moreover, if we take $x = 0$, then $\frac{1}{2}x^T A x = 0 < 1$: therefore the constraint is qualified.

Let us now define and solve the dual problem. Let

$$\mathcal{L}(x, \lambda) = c^T x + \lambda(\frac{1}{2}x^T A x - 1)$$

Let us compute:

$$\mathcal{G}(\lambda) = \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \lambda) = \inf_{x \in \mathbb{R}^n} c^T x + \lambda(\frac{1}{2}x^T A x - 1) \quad \forall \lambda \geq 0.$$

Note that, if $\lambda = 0$, then $\mathcal{G}(0) = -\infty$ since $c \neq 0$ (take $x = -qc$ and let q go to $+\infty$). For $\lambda > 0$, the function $\mathcal{L}(\cdot, \lambda)$ is coercive and strictly convex and, due to the optimality conditions, the minimizer is the point $x^*(\lambda) \in \mathbb{R}^n$ satisfying

$$c + \lambda Ax^*(\lambda) = 0, \quad \text{that is } x^*(\lambda) = -\frac{A^{-1}c}{\lambda}.$$

therefore

$$\mathcal{G}(\lambda) = \mathcal{L}(x^*(\lambda), \lambda) = -\frac{1}{2\lambda} c^T A^{-1} c - \lambda \quad \text{si } \lambda > 0.$$

the dual problem is then :

$$\sup_{\lambda \geq 0} \mathcal{G}(\lambda) = \sup_{\lambda > 0} -\frac{1}{2\lambda} c^T A^{-1} c - \lambda = -\sqrt{2c^T A^{-1} c},$$

obtained for $\lambda = \sqrt{\frac{c^T A^{-1} c}{2}}$. The solution of the primal problem is therefore

$$x = -\sqrt{2} \frac{A^{-1}c}{\sqrt{c^T A^{-1} c}}.$$

Solution 3.19. The problem has a unique solution since the objective is strongly convex and that the constraint is convex, closed and nonempty. the constraint is qualified since affine. The Lagrangian of the problem is

$$\mathcal{L}(x, \lambda) = \frac{1}{2} x^T A x + b^T x + \lambda^T (Cx - d) \quad \forall (x, \lambda) \in \mathbb{R}^n \times (\mathbb{R}_+)^l.$$

We first compute $\mathcal{G}(\lambda) = \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \lambda)$. By strong convexity of $\mathcal{L}(\cdot, \lambda)$, there is a unique maximizer $x^*(\lambda)$ satisfying

$$Ax^*(\lambda) + b + C^T \lambda = 0, \quad \text{that is } x^*(\lambda) = -A^{-1}(b + C^T \lambda).$$

therefore

$$\mathcal{G}(\lambda) = -\frac{1}{2} b^T A^{-1} b - \frac{1}{2} \lambda^T C A^{-1} C^T \lambda - (b^T A^{-1} C^T + d^T) \lambda$$

Solution 3.20. The constraint is nonempty, convex, closed and qualified (affine). The objective is strongly convex therefore the problem has a unique solution. Let

$$\mathcal{L}(x, \lambda) = \frac{1}{2} \|x\|^2 - \langle c, x \rangle + \lambda \langle s, x \rangle \quad \forall (x, \lambda) \in \mathbb{R}^n \times \mathbb{R}_+.$$

Let us compute $\mathcal{G}(\lambda) = \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \lambda)$ for $\lambda \geq 0$. The minimizer is the point $x^*(\lambda)$ satisfying the optimality condition

$$x^*(\lambda) - c + \lambda s = 0, \quad \text{that is } x^*(\lambda) = c - \lambda s.$$

therefore

$$\mathcal{G}(\lambda) = -\frac{1}{2} \|c - \lambda s\|^2 = -\frac{1}{2} \|c\|^2 + \lambda \langle c, s \rangle - \frac{\lambda^2}{2} \|s\|^2, \quad \forall \lambda \geq 0.$$

This is a concave function with derivative $\mathcal{G}'(\lambda) = \langle c, s \rangle - \lambda \|s\|^2$, therefore $\mathcal{G}'(0) = \langle c, s \rangle$. The maximizer of $\mathcal{G}$ on $\mathbb{R}_+$ is therefore $\lambda^* = 0$ if $\langle c, s \rangle \leq 0$ and $\lambda^* = \langle c, s \rangle / \|s\|^2$ otherwise. Therefore the solution of the initial problem is

$$x^*(\lambda^*) = \begin{cases} c & \text{si } \langle c, s \rangle \leq 0, \\ c - \frac{\langle c, s \rangle c}{\|s\|^2} & \text{otherwise.} \end{cases}$$

Solution 3.21. The objective $f(u) = \|u - x\|^2$ is strictly convex and coercive and the constraint $\mathcal{E}$ is nonempty, convex and closed. Therefore the problem has a unique solution. Moreover the constraint is qualified since 0 satisfies the strict inequality in the definition of the constraint. Let

$$\mathcal{L}(u, \lambda) = \|u - x\|^2 + \lambda \left(\sum_{i=1}^n u_i^2 / a_i^2 - 1 \right) \quad \forall (u, \lambda) \in \mathbb{R}^n \times \mathbb{R}_+.$$

Let us compute $d(\lambda) = \inf_{u \in \mathbb{R}^n} \mathcal{L}(u, \lambda)$. The infimum is obtained at $u^*(\lambda)$ given by $u_i^*(\lambda) = a_i^2 x_i / (a_i^2 + \lambda)$ and its value is

$$d(\lambda) = \sum_{i=1}^n \frac{\lambda x_i^2}{a_i^2 + \lambda} - \lambda.$$

This is a concave function (see the course, or check it by computing its second order derivative), and

$$d'(\lambda) = \sum_{i=1}^n \frac{a_i^2 x_i^2}{(a_i^2 + \lambda)^2}$$

therefore $d'(0) = \sum_{i=1}^n \frac{x_i^2}{a_i^2} > 0$ since $x \notin \mathcal{E}$. Therefore at the maximizer of $d(\lambda)$, we have $d'(\lambda) = 0$, which gives the required equality.

Solution 3.22. (1) By strong convexity of $z \mapsto f(x, z)$, the existence and uniqueness of z_x are guaranteed.

(2) Let $x_1, x_2 \in \mathbb{R}^n$ and z_1, z_2 be such that $\ell(x_i) = f(x_i, z_i)$ ($i = 1, 2$). Then

$$\begin{aligned} \frac{\alpha}{2} \|z_1 - z_2\|^2 + \langle \nabla_z f(x_1, z_1), z_2 - z_1 \rangle &\leq f(x_1, z_2) - f(x_1, z_1) \\ &= f(x_2, z_1) - f(x_2, z_1) + f(x_1, z_2) - f(x_1, z_1) \\ &\leq f(x_2, z_1) - f(x_2, z_2) + f(x_1, z_2) - f(x_1, z_1). \end{aligned}$$

Introducing the function $F : x \mapsto f(x, z_1) - f(x, z_2)$, the right-hand side rewrites $F(x_2) - F(x_1)$, so that there exists $\xi \in [x_1; x_2]$ satisfying

$$F(x_2) - F(x_1) = \langle \nabla F(\xi), x_2 - x_1 \rangle.$$

As $\nabla F(\xi) = \nabla_x f(x, z_1) - \nabla_x f(x, z_2)$ we have

$$\nabla F(\xi) \leq L \|z_1 - z_2\|$$

and the conclusion follows by the Cauchy-Schwarz inequality.

(3) We let h be a vector. Then

$$\begin{aligned} \ell(x + h) - \ell(x) &\geq f(x, z_{x+h}) - f(x, z_{x+h}) \\ &= \langle \nabla_x f(x, z_{x+h}), h \rangle + o(\|h\|). \end{aligned}$$

Observe that $z_{x+h} \xrightarrow{h \rightarrow 0} z_x$. Similarly,

$$\begin{aligned} \ell(x + h) - \ell(x) &\leq f(x + h, z_x) - f(x, z_x) \\ &= \langle \nabla_x f(x, z_x), h \rangle + o(\|h\|). \end{aligned}$$

Passing to the limit provides the required result.

Solution 3.23. (1) We refer to the lecture notes.

(2) Observe that by strong convexity of f and convexity of g we have

$$\frac{\alpha}{2} \|x_\lambda - x_\mu\|^2 + \langle \nabla f(x_\lambda), x_\mu - x_\lambda \rangle + \lambda \langle \nabla g(x_\lambda), x_\mu - x_\lambda \rangle \leq \mathcal{L}(\lambda, x_\mu) - \mathcal{L}(\lambda, x_\lambda).$$

By optimality of x_λ , $\nabla f(x_\lambda) + \lambda \nabla g(x_\lambda) = 0$ whence

$$\frac{\alpha}{2} \|x_\lambda - x_\mu\|^2 \leq \mathcal{L}(\lambda, x_\mu) - \mathcal{L}(\lambda, x_\lambda).$$

However,

$$\begin{aligned} \mathcal{L}(\lambda, x_\mu) &= f(x_\mu) + \lambda g(x_\mu) \\ &= f(x_\mu) + \mu g(x_\mu) + (\lambda - \mu)g(x_\mu) \\ &\leq f(x_\lambda) + \mu g(x_\lambda) + (\lambda - \mu)g(x_\mu) \\ &= \mathcal{L}(\lambda, x_\lambda) + (\lambda - \mu)(g(x_\mu) - g(x_\lambda)) \end{aligned}$$

so that

$$\frac{\alpha}{2} \|x_\lambda - x_\mu\|^2 \leq |\lambda - \mu| \cdot |g(x_\mu) - g(x_\lambda)|.$$

The Lipschitzianity of g allows to conclude.

(3) The concavity of ℓ follows from the following observation: let $\lambda_1, \lambda_2 \geq 0$, let x_i be the minimiser associated with λ_i ($i = 1, 2$), define $\lambda_t := (1-t)\lambda_1 + t\lambda_2$ and let $x_t \in \mathbb{R}^d$ be the unique minimiser of $x \mapsto f(x) + \lambda_t g(x)$. We have

$$\begin{aligned} \ell(\lambda_t) &= f(x_t) + \lambda_t g(x_t) \\ &= (1-t)f(x_t) + tf(x_t) + (1-t)\lambda_1 g(x_t) + t\lambda_2 g(x_t) \\ &= (1-t)(f(x_t) + \lambda_1 g(x_t)) + t(f(x_t) + \lambda_2 g(x_t)) \\ &\geq (1-t)(f(x_1) + \lambda_1 g(x_1)) + t(f(x_2) + \lambda_2 g(x_2)). \end{aligned}$$

In particular, we deduce that:

- ℓ is concave,
- And that, if ℓ is not strictly concave, we can find $\lambda_1, \lambda_2 \geq 0$, $\lambda_1 \neq \lambda_2$ and $t \in (0; 1)$ such that x_1 minimises $f + \lambda_t g$. In particular, we obtain, for this t , two sets of optimality conditions: on the one-hand,

$$\nabla f(x_1) + \lambda_1 \nabla g(x_1) = 0$$

and, on the other,

$$\nabla f(x_1) + \lambda_t \nabla g(x_1) = 0.$$

In particular, subtracting these two equations, we obtain

$$(\lambda_1 - \lambda_t) \nabla g(x_1) = 0.$$

This implies that $\nabla f(x_1) = 0$ and, in turn, that for any $\mu \geq 0$,

$$\nabla f(x_1) + \mu \nabla g(x_1) = 0.$$

Thus, x_1 minimises $f + \mu g$ for any $\mu \geq 0$ and so

$$\ell(\lambda) = f(x_1) + \lambda g(x_1)$$

and is in particular linear.

Furthermore, by the Danskin theorem,

$$\ell'(\lambda) = g(x_\lambda),$$

and as $\lambda \mapsto x_\lambda$ is Lipschitz, so is ℓ' .

- (4) *It suffices to apply the result of the midterm in the case of a strictly concave function. If ℓ is, on the other hand, linear, it follows from a direct computation.*

Part 4. Calculus of variations & Optimal control

9. GENERAL INTRODUCTION

This section is a short introduction to optimal control which, roughly speaking, can be considered as a natural setting for infinite dimensional optimisation. In general, optimal control is concerned with differential systems depending on a parameter, often dubbed the “control” (or, in french, the “commande”), through which one might try to influence the outcome of the evolution. Situations vary wildly, from the control of the trajectory of a vehicle through a steering wheel, to the optimal administration of a drug within the context of medical therapy or to portfolio management. The goal is to find the control that optimises the outcome (*e.g.* the best return of investment, the fastest route from one point to another etc). However, there is a major difficulty: if we formalise what we just wrote a bit, it means that a control system typically writes

$$\dot{x}(t) = f(t, x(t)) + g(x(t), u(t))$$

where f is a non-linearity, and g a coupling between the state x and the control u . Thus, we are led to finding the best *function* u to optimise a certain criterion, whence the necessity to consider infinite dimensional settings. In this part of the lecture, we will proceed in steps of increasing complexity:

- (1) We will first present the Bellman dynamic programming principle for time-discrete systems: this principle explains that it is not necessary to explore all possibilities for the control u , but only a certain subset.
- (2) Second, to give a primer on infinite dimensional optimisation, we will consider the classical setting for calculus of variations, which will culminate with the derivation and analysis of the Euler-Lagrange equations.
- (3) Finally, we will discuss optimal control theory, where we will end with a formal derivation of the Pontryagin maximum principle.

10. THE DYNAMIC PROGRAMMING PRINCIPLE

10.1. Setting and main objective. In this first part, we consider a time-discrete dynamical system. Namely, we consider a parameter $N \in \mathbb{N}$, and a process $\{x_k\}_{k=0,\dots,N} \in (\mathbb{R}^d)^N$ starting at $x_0 = \bar{x}$ and whose state x_n at time $n \in \{0, \dots, N\}$ is determined by the system

$$(10.1) \quad x_{k+1} = f_k(x_k, u_k), \quad k \in \{1, \dots, N-1\}.$$

Here, f_k is a given non-linearity, and $u_k \in \mathbb{R}^m$ is the control. We will assume that the state of the system belongs to a fixed set $X \subset \mathbb{R}^d$, i.e. that $x_k \in X$ for all $k \in \{0, 1, \dots, N\}$. The control u_k belongs to a set U_k independent of x_k (this is to simplify notations and some statements, but most results remain true if the set U_k of available controls depends on the current state of the system x_k).

The typical optimal control problem is the following: consider a cost function depending on the initial state $\bar{x}$ and on the sequence of control $\{u_k\}_{k=0,\dots,N-1} := \mathbf{u}$ that writes

$$C(\bar{x}; \mathbf{u}) := g(x_N) + \sum_{k=0}^{N-1} \ell_k(x_k; u_k)$$

where ℓ_k are given functions dubbed “running cost at time k ” and g is the so-called “terminal cost”. The goal of optimal control is to study the optimisation problem

$$(10.2) \quad \min_{\mathbf{u} \in \mathbf{U} := \prod_{k=0}^{N-1} U_k} C(\bar{x}; \mathbf{u}) \text{ where } \mathbf{x} = \{x_k\}_{k=0,\dots,N} \text{ solves (10.1).}$$

At any rate, as always in optimisation, there are two main questions, that are always dealt with separately:

- (1) The existence of optimisers,
- (2) Their characterisation.

While the first question, that of existence, will often be treated through the usual tools of compactness and its multiple avatars, the second one is more delicate and will be our main focus here. One thing that is very important to keep in mind is the recursive structure of this system; this is what will allow us to simplify the search for optimisers.

A (formally easy) generalisation is to consider infinite time horizon problems of the type

$$(10.3) \quad \min_{\mathbf{u} \in \prod_{k=0}^{\infty} U_k} \sum_{k=0}^{+\infty} \beta^k \ell_k(x_k, u_k).$$

There, $\beta \in (0; 1)$ is a discount factor.

10.2. A basic example: the shortest path problem. This is a finite-horizon problem, the solution of which will lead us to a first understanding of the dynamic programming principle. The setting is the following: a traveler needs to get from city A to city E following certain roads that take him through several intermediate cities. Each road connecting two cities requires takes a certain amount of time. The possible paths are modelled by a graph, with A and E as initial and final vertices (see Figure 3). The other vertices are intermediate cities, and the labels on the edges correspond to the time one must take to travel down that road. Let us denote by $\Gamma(M)$ the successors of city M and, for $N \in \Gamma(M)$ we will denote MN as the time taken to complete the MN route. Figure 3 describes the values of $\Gamma(M)$

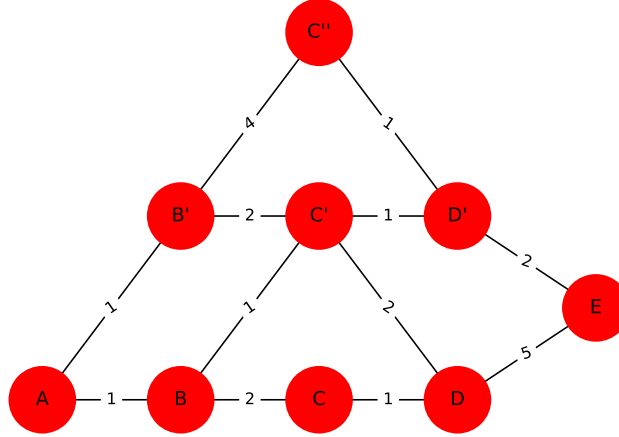


FIGURE 3. Graph of cities potentially crossed by the traveler

for each city and the journey times between stages. For example, $\Gamma(A) = \{B, B'\}$ and $AB = AB' = 1$.

Our traveler wants to take the shortest possible path. In other words, the control problem is: given that I am at a city M , what is the next city I am going to go through? The cost function is $\ell(M, N) = MN$ where M is the current city and $N \in \Gamma(M)$ is the next step city which constitutes our control.

To determine the shortest route or routes, we could try them all, but it will be more efficient to use the following fact: *if an optimal path from A to E goes through a certain city M , then this path is still optimal to travel from M to E* . This simple, almost naive, observation, is the core of the dynamic programming principle.

Let us introduce the value function $V(M)$ equal to the minimum amount of time required to travel from M to E . We wish to calculate $V(A)$. Let us progress backwards: starting from E , what are the possible ways to reach E in one step? First of all, clearly,

$$V(D) = 5, \quad V(D') = 2.$$

We then go back to the previous cities. The dynamic programming principle gives

$$V(C) = 6, \quad V(C') = \min(1 + V(D'), 2 + V(D)) = 1 + V(D') = 3, \quad V(C'') = 3.$$

Reiterating the argument, we get:

$$V(B) = \min(2 + V(C), 1 + V(C')) = 1 + V(C') = 4$$

$$V(B') = \min(2 + V(C'), 4 + V(C'')) = 5$$

and finally

$$V(A) = \min(1 + V(B), 1 + V(B')) = 1 + V(B) = 5.$$

The minimum travel time is therefore 5 and corresponds to the $ABC'D'E$ route alone.

This elementary example is instructive in several respects:

- (1) It is easy to see how the previous strategy can be generalised to more general problems of the form (12.2): we introduce the value functions at each step, and we compute it backwards, starting from the end.
- (2) In the example, we have not explored all possible paths, but only the optimal ones starting from M . As a result, previous reasoning shows, for example, that if the traveler gets lost in B' (which is not on the optimal path), then it will be optimal to go via $C'D'E$.
- (3) It may seem odd, in order to solve a problem from point A , to try to solve all problems from intermediate points. Let us give a few arguments to overcome this objection: firstly, the solution strategy is robust: if a mistake is made at a given point and leads to a city M that is not on the optimal path, we can subsequently make up for it by following the optimal path from M . Second, this strategy is natural (choosing the next city according to the city where we are now, rather than following a plan drawn up in advance) and allows to reduce the problem to a succession of static problems. Above all, considering only the optimal paths from M reduces the number of paths from the origin to the final destination and going through M that are relevant to consider. A naive approach, consisting of determining all paths from origin to destination, calculating the cost of each of these paths, and choosing the best one, comes up against the fact that, for problems with many steps, there can quickly be more paths from origin to destination than atoms in the universe!

10.3. The dynamic programming principle for finite time horizon problems.

10.3.1. *Statement of the dynamic programming principle.* We consider the problem (12.2) and we introduce the *value function* of the problem as

$$V(x; p) := \inf_{u \in \prod_{k=p}^N U_k} \sum_{k=p}^{N-1} \ell_k(x_k, u_k) + g(x_N) \text{ subject to } \begin{cases} x_{k+1} = f_k(x_k, u_k), k = p, \dots, N-1, \\ x_p = x. \end{cases}$$

With this function, solving (12.2) amounts to computing $V(\bar{x}; 0)$. The main result of this section is the following:

Theorem 10.1. *For any x (the set of admissible states) there holds*

$$\begin{cases} V(x, N) = g(x), \\ V(x; p) = \inf_{u \in U_p} (\ell_p(x_p; u) + V(f_p(x_p, u); p+1)). \end{cases}$$

The conclusion of this theorem is valid whether an optimal control exists or not; nevertheless, in order to keep the proof short and to the point, we will assume existence of an optimal control (at every step).

Proof of Theorem 10.1. We introduce the auxiliary function

$$W(x; p) := \inf_{u \in U_p} (\ell_p(x_p; u) + V(f_p(x_p, u); p+1)),$$

and we seek to show that $V = W$. To this end, we proceed by double inequality:

- (1) Let us first prove that $V \geq W$. To this end, we fix x and p . First, let $u_p^*, \dots, u_{N-1}^*$ be an optimal strategy for V . Thus,

$$V(x; p) = \sum_{k=p}^{N-1} \ell_k(x_k; u_k^*) + g(x_N).$$

Now, $u_{p+1}^*, \dots, u_{N-1}^*$ is an admissible strategy starting from $x' = f_p(x_p; u_p^*)$, and so

$$V(x; p) \geq \ell_p(x_p; u_p^*) + V(f_p(x_p; u_p^*); p+1) \geq W(x; p).$$

- (2) We now show that $W \geq V$. To this end, let u_p^* be an optimal strategy for the optimisation problem defining W . Then

$$W(x; p) = \ell_p(x_p, u_p^*) + V(f_p(x_p; u_p^*); p+1).$$

Let $u_{p+1}^*, \dots, u_{N-1}^*$ denote the optimal strategy for $V(f_p(x_p; u_p^*); p+1)$. Then, the strategy $u_p^*, \dots, u_{N-1}^*$ is an admissible strategy starting from x , and it follows that

$$W(x; p) \geq V(x; p),$$

so that the conclusion follows. $\square$

One often encounters, in applications, the terminology “optimal feedback” to describe the optimal controls. Let us briefly justify this terminology, which essentially states that the optimal control at time k only depends on the state x_k , and not on any of the previous steps $x_0, \dots, x_{k-1}$.

10.3.2. *An example of application of the dynamic programming principle.* Let us for instance consider the problem

$$\min_u \sum_{t=1}^4 -2u_t - 3(x_t - u_t)$$

with x_t satisfying the dynamics $x_{t+1} = 0.8u_t + 0.5(x_t - u_t)$, under the constraints $0 \leq u_t$ and $0 \leq (x_t - u_t)$

Let $V_5(x) = 0$ and, for $n = 1, \dots, 4$,

$$V_n(x) = \min_u \sum_{t=n}^4 -2u_t - 3(x_t - u_t)$$

with x_t satisfying the dynamics $x_{t+1} = 0.8u_t + 0.5(x_t - u_t)$ under the constraint $x_n = x$, $0 \leq u_t$ and $0 \leq (x_t - u_t)$. By Theorem 10.1, for $n = 1, \dots, 4$,

$$V_n(x) = \min_{0 \leq u \leq x} -2u - 3(x - u) + V_{n+1}(4u/5 + (x - u)/2).$$

therefore

$$V_4(x) = \min_{0 \leq u \leq x} -2u - 3(x - u) = -3x,$$

with minimizer $\hat{u}_4(x) = 0$. Then:

$$V_3(x) = \min_{0 \leq u \leq x} -2u - 3(x - u) - 3(4u/5 + (x - u)/2) = -9x/2,$$

with minimizer $\hat{u}_3(x) = 0$.

$$V_2(x) = \min_{0 \leq u \leq x} -2u - 3(x - u) - 9(4u/5 + (x - u)/2)/2 = -\frac{28}{5}x,$$

with minimizer $\hat{u}_2(x) = x$.

$$V_1(x) = \min_{0 \leq u \leq x} -2u - 3(x - u) - 28(4u/5 + (x - u)/2)/5 = -\frac{162}{25}x,$$

with minimizer $\hat{u}_1(x) = x$. We thus find $V_1(x) = -\frac{162}{25}x$, with the optimal control and trajectory given by $x_1 = x$, $u_1 = x_1 = x$, $x_2 = 4u_1/5 = 4x/5$, $u_2 = x_2 = 4x/5$, $x_3 = 4u_2/5 = 16x/25$, $u_3 = 0$, $x_4 = x_3/2 = 8x/25$, $u_4 = 0$, $x_5 = x_4/2 = 4x/25$. Note that for the set of possible trajectories to be non empty we need to assume $x \geq 0$.

The following result justifies the terminology “optimal feedback” that you might encounter, and which simply means that the sequence of optimal controls at a given step k only depends on the state x_k , and not on any of the previous steps; the proof is immediate, and we leave it as an exercise.

Proposition 10.1. *Let $\bar{x} \in X$ be a fixed initial condition. If we define by induction the sequences $\{u_k\}_{k \in \mathbb{N}}$ and $\{x_k\}_{k \in \mathbb{N}}$ as*

$$\bar{x}_0 = \bar{x}, \quad \bar{u}_k = u^*(x_k), \quad \bar{x}_{k+1} = f_k(\bar{x}_k, \bar{u}_k),$$

then the sequence $\{\bar{u}_k\}_{k \in \mathbb{N}}$ is optimal for the discrete control problem:

$$V(0, \bar{x}) = \sum_{k=0}^{N-1} \ell_k(\bar{x}_k, \bar{u}_k) + g(\bar{x}_N).$$

10.4. The dynamic programming principle for infinite time horizon problems.

10.4.1. *The Bellman dynamic programming principle.* We now state the dynamic programming principle for infinite time horizon problems; as the proof is similar to that of Theorem 10.1, we will skip it and rather focus on the specificities of the infinite time horizon setting. The setting is the following: the set of controls U is independent of the time step, as is the non-linearity f . The current cost $\ell : X \times U \rightarrow \mathbb{R}$ is bounded and independent of the time step, and the discount factor is a given $\beta \in (0, 1)$. For any $\bar{x} \in X$, we let

$$V(\bar{x}) = \inf_{\mathbf{u} \in U^{\mathbb{N}}} \sum_{k=0}^{+\infty} \beta^k \ell(x_k; u_k)$$

where $\{x_k\}_{k \in \mathbb{N}}$ is defined by induction by

$$\begin{cases} x_0 = \bar{x} \\ x_{k+1} = f(x_k, u_k), k \in \mathbb{N} \end{cases}$$

Note that the sum $\sum_{n=0}^{+\infty} \beta^n \ell(x_n, u_n)$ is indeed convergent because ℓ is bounded and $\beta \in (0, 1)$.

Theorem 10.2 (Dynamic programming). *For any $x \in X$, we have*

$$(10.4) \quad V(x) = \inf_{u \in U} \{ \ell(x, u) + \beta V(f(x, u)) \}.$$

The equation (10.4) is referred to as the “Bellman equation”.

Proof of Theorem 10.2. Likewise, we assume that there exists an optimal control; the proof when the infimum is not a minimum is similar. We introduce

$$W(x) := \inf_{u \in U} (\ell(x, u) + \beta V(f(x, u))),$$

and our goal is to show that $V = W$. Let x be a given state and $\{u_n\}_{n \in \mathbb{N}}$ be a sequence of optimal controls for x . It follows that

$$\begin{aligned} V(x) &= \sum_{k=0}^{\infty} \beta^k \ell(x_k, u_k) \\ &= \ell(x, u_0) + \beta \sum_{k=0}^{\infty} \ell(x_{k+1}, u_{k+1}) \\ &\geq \ell(x, u_0) + \beta V(f(x, u_0)) \\ &\geq W(x). \end{aligned}$$

To prove the converse inequality, we let $u_W^* \in U$ be an optimal control for W , and we let $\{u_{V,k}^*\}_{k \in \mathbb{N}}$ be an optimal sequence for $V(f(x, u))$, and $\{x_k\}_{k \in \mathbb{N}}$ be the associated sequence of states. If we then define the control $\{u_k^{**}\}_{k \in \mathbb{N}}$ as

$$u_0^{**} = u_W^*, \quad u_{k+1}^{**} = u_{V,k}^* \quad (k \geq 0)$$

we deduce that

$$W(x) = \ell(x, u_W^*) + \beta \sum_{k=0}^{\infty} \beta^k \ell(x_k, u_k) \geq V(x).$$

This concludes the proof. $\square$

As is clear from the statement of the theorem, the main difficulty in infinite time-horizon problems is that the function V appears in both the right and left hand sides of the Bellman equation (10.4), so that one can not simply compute it at final time and work backwards. Besides, this implies that, in a certain framework, V is the unique solution of an explicit equation. Of course, as is usually the case with implicit equations, studying the existence of a solution V and providing a numerical approximation scheme of the solution are often done in one single step, through the study of iterative algorithm.

10.4.2. *Approximation scheme for the value function V .* We consider the norm

$$\|h\|_{\infty} := \sup_{(x,u) \in X \times U} |h(x, u)|$$

and define $B(X)$ as the set of bounded mappings of X in $\mathbb{R}$, endowed with the $\|\cdot\|_{\infty}$ norm. Let us define the (non-linear) operator $T : B(X) \rightarrow B(X)$ by

$$T(h)(x) := \inf_{u \in U} \{\ell(x, u) + \beta h(f(x, u))\} \quad \forall h \in B(X).$$

Note that $T(h) \in B(X)$ since, for all $x \in X$,

$$|T(h)(x)| \leq \inf_{u \in U} \{|\ell(x, u)| + \beta |h(f(x, u))|\} \leq \|L\|_{\infty} + r \|h\|_{\infty}.$$

In particular, we deduce that $\|T(h)\|_{\infty} \leq \|L\|_{\infty} + r \|h\|_{\infty}$, and that $T(h) \in B(X)$.

Theorem 10.3. *The operator T is contracting in $B(X)$:*

$$\|T(h) - T(h')\|_{\infty} \leq \beta \|h' - h\|_{\infty} \quad \forall h, h' \in B(X),$$

and the value function V is its unique fixed point in $B(X)$.

Proof. First note that as ℓ is bounded, so is V and

$$\|V\|_\infty \leq \sum_{k=0}^{\infty} \beta^k \|L\|_\infty \leq \frac{\|L\|_\infty}{1-\beta}.$$

Theorem 10.2 implies that V is a fixed point of T . To check the uniqueness of this fixed point, it suffices to establish that T is a contraction. To this end, let $h, h' \in B(X)$ and $x \in X$. For any $u \in U$ we have

$$\ell(x, u) + \beta h(f(x, u)) \leq \ell(x, u) + \beta h'(f(x, u)) + \beta \|h' - h\|_\infty.$$

Taking the inf with respect to u on both sides, we obtain:

$$\begin{aligned} T(h)(x) &= \inf_{u \in U} \{L(x, u) + \beta h(f(x, u))\} \\ &\leq \inf_{u \in U} \{L(x, u) + \beta h'(f(x, u))\} + \beta \|h' - h\|_\infty = T(h')(x) + \beta \|h' - h\|_\infty. \end{aligned}$$

We deduce that

$$T(h)(x) - T(h')(x) \leq r \|h' - h\|_\infty,$$

and, by symmetry, we deduce that for any $x \in X$ there holds

$$|T(h)(x) - T(h')(x)| \leq r \|h' - h\|_\infty.$$

This gives

$$\|T(h) - T(h')\|_\infty \leq r \|h' - h\|_\infty,$$

which proves that T is a contraction since $\beta \in (0, 1)$. $\square$

The previous result yields an approximation scheme for the value function V . Namely, for any given $h_0 \in B(X)$, we define $h_{k+1} = T(h_k)$. The fixed-point theorem provides the (uniform) convergence of $\{h_k\}_{k \in \mathbb{N}}$ to the value function V , in the following quantified way:

$$\|V - h_k\|_\infty \leq \beta^k \|V - h_0\|_\infty.$$

The value function can also be used to describe optimal feedbacks. Let us assume that, for any $x \in X$, there exists an "optimal feedback" $u^*(x) \in U$, i.e. such that

$$\ell(x, u^*(x)) + \beta V(f(x, u^*(x))) = \inf_{u \in U} \{\ell(x, u) + \beta V(f(x, u))\}.$$

We can show that, if U and X are metric sets, if U is compact and if the functions $f : X \times U \rightarrow X$ and $\ell : X \times U \rightarrow \mathbb{R}$ are continuous, then such feedback exists; this follows from the continuity of V , which we omit here.

The following proposition is intuitively obvious; its proof is left as an exercise,

Proposition 10.2. *Let $\bar{x} \in X$ be an initial condition. If we define by induction the sequences $\{u_k\}_{k \in \mathbb{N}}$ and $\{x_k\}_{k \in \mathbb{N}}$ as*

$$\bar{x}_0 = \bar{x}, \quad \bar{u}_k = u^*(x_k), \quad \bar{x}_{k+1} = f(\bar{x}_k, \bar{u}_k),$$

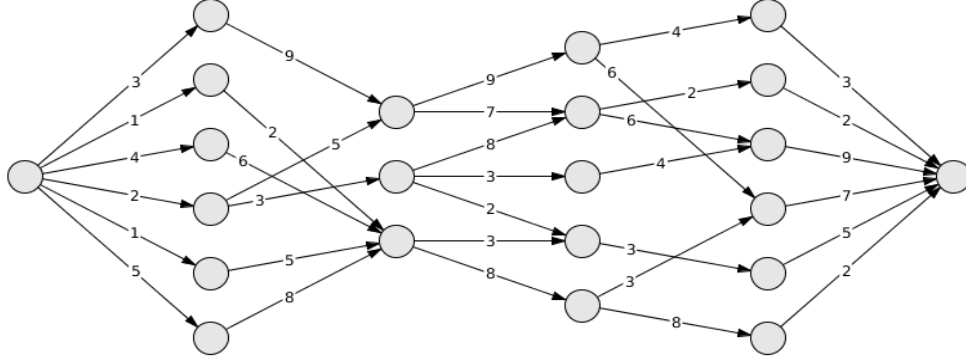
then the sequence $\{\bar{u}_k\}_{k \in \mathbb{N}}$ is optimal for the discrete control problem:

$$V(\bar{x}) = \sum_{k=0}^{+\infty} \beta^k \ell(\bar{x}_k, \bar{u}_k).$$

In some specific case, the optimal policy can be computed, but this is usually a very delicate endeavour and relies on finer tools of optimisation; we refer to the exercises of the chapter.

EXERCISES FOR THE FIRST PART OF PART 4

Exercise 4.1. Find the shortest path for the following problem:



Exercise 4.2. Solve the optimal control problem

$$\min_{u_t} \sum_{t=0}^2 (x_t^2 + tu_t^2) + x_3^2$$

with x_t satisfying the dynamics $x_{t+1} = x_t - u_t$, $x_0 = 1$.

Exercise 4.3. Find the extrema of

$$\sum_{t=0}^3 (te^{u_t} + x_t) - 2x_4$$

with x_t satisfying the dynamic $x_{t+1} = x_t + u_t$ and $x_0 = x$ under the constraint $0 \leq u_t \leq 1$.

Exercise 4.4. Solve the optimal control problem

$$\min_u \sum_{t=0}^3 \frac{1}{2} (x_t^2 + u_t^2) + \frac{1}{2} x_4^2$$

with x_t satisfying the dynamics $x_{t+1} = x_t - u_t$ with $x_0 = 1$.

Exercise 4.5. For $x \geq 0$ and $N \in \mathbb{N}^*$, consider problem

$$W(x) := \inf \left\{ \sum_{i=0}^{N-1} u_i^2, \text{ where } u_i \geq 0, \sum_{i=0}^{N-1} u_i = x \right\}$$

This exercise compares two methods to compute $W(x)$:

- (1) Compute $W(x)$ using the Karush-Kuhn-Tucker theorem.
- (2) Formulate the problem as a discrete time control problem with final time N and use the dynamic programming principle. There are various ways to do so. Here is one:
 - (a) Rewrite the problem, letting $U_n(x_n) = [0, x_n]$ for $n \leq N-2$, $U_{N-1}(x) = x$, $f_n(x, u) = x - u$, $g = 0$, and $\ell_n(x, u) = u^2$.
 - (b) Write the dynamic programming principle for value functions V_n .
 - (c) Deduce that $V_n(x) = x^2/(N - n)$.
 - (d) Compute $W(x)$.

- (3) Compare both methods. Suggest a third, simpler method, based on the convexity of $x \rightarrow x^2$.

Exercise 4.6 (Difficult). We study here the optimal growth problem described by

$$\sup_{(k_t)} \sum_{t=0}^{\infty} \beta^t \ln(k_t^\alpha - k_{t+1})$$

under constraints : $k_0 = k$ (where $k > 0$ is given), $k_{t+1} \in [0, k_t^\alpha]$ for all $t \in \mathbb{N}$. The discount rate $\beta \in]0, 1[$ and the power $\alpha \in]0, 1[$ are given.

For $k > 0$, let $W(k)$ denote the value of this problem. The economic interpretation is that (k_t) represents the capital at time t , the difference $k_t^\alpha - k_{t+1}$ being the consumption (the difference between the production k_t^α and the investment k_{t+1} at time t).

- (1) For $k > 0$, let $v(k)$ be the value function of the problem

$$v(k) := \sup_{(k_t)} \sum_{t=0}^{\infty} \beta^t \ln(k_t^\alpha)$$

under the same constraints as for W . Show that $v(k) = \frac{\alpha \ln(k)}{1-\alpha\beta}$ and that $W \leq v$.

- (2) Show that W is solution of the fixed point equation $f = Tf$, where T is the operator defined by

$$Tf(x) := \sup_{y \in [0, x^\alpha]} \{ \ln(x^\alpha - y) + \beta f(y) \}.$$

for all $x > 0$.

- (3) Why can't we say directly that W is the unique solution of the Bellman equation?
- (4) Show that $Tv = v + c$ where c is a nonpositive constant to be determined.
- (5) Compute the iterated maps $T^n v$ for $n \in \mathbb{N}$, show that this sequence converges towards a limit v_∞ to be made explicit. Finally, show that $Tv_\infty = v_\infty$.
- (6) Show that $W \leq v_\infty$.
- (7) Show that $W \geq v_\infty$ (difficult) and conclude.
- (8) Show that the initial problem has a unique solution and compute it. Denote by (k_t^*) the optimal policy.
- (9) Study this optimal policy (or optimal dynamic) (k_t^*) (monotony, convergence).

SOLUTIONS FOR THE FIRST PART OF PART 4

Solution 4.1. *To be written...*

Solution 4.2. *This is an optimal control problem with a time horizon $T = 3$. Let $V_3(x) = x^2$, and, for all $x \in \mathbb{R}$ and $n \in \{0, \dots, 2\}$,*

$$V_n(x) = \min_{x_t} \sum_{t=n}^2 (x_t^2 + tu_t^2) + x_3^2,$$

with x_t satisfying the dynamics $x_{t+1} = x_t - u_t$ (for $t = n, \dots, 2$) and $x_n = x$. We want to compute $V_0(1)$. By the dynamic programming principle, for $n \in \{0, \dots, 2\}$,

$$V_n(x) = \min_u x^2 + nu^2 + V_{n+1}(x - u).$$

therefore

$$V_2(x) = \min_u x^2 + 2u^2 + (x - u)^2 = \frac{5x^2}{3},$$

where the minimum is obtained at $\hat{u}_2(x) = x/3$. Then

$$V_1(x) = \min_u x^2 + u^2 + \frac{5}{3}(x - u)^2 = \frac{13x^2}{8},$$

where the minimum is obtained at $\hat{u}_1(x) = 5x/8$. Finally,

$$V_0(x) = \min_u x^2 + \frac{13}{8}(x - u)^2 = x^2,$$

where the minimum is obtained at $\hat{u}_0(x) = x$.

The problem's solution is therefore $x_0 = 1$, $u_0 = x$, $x_1 = 0$, $u_1 = 0$, $x_2 = 0$, $u_2 = 0$, $x_3 = 0$.

Solution 4.3. *Let us begin with the minimisation problem. Let $V_4(x) = -2x$ et, for $x \in \mathbb{R}$ and $n \in \{0, \dots, 3\}$,*

$$V_n(x) = \inf \sum_{t=n}^3 (te^{u_t} + x_t) - 2x_4$$

with x_t satisfying the dynamics $x_{t+1} = x_t + u_t$ and $x_n = x$, under the constraint $0 \leq u_t \leq 1$. By the dynamic programming principle, for $n = 0, \dots, 3$,

$$V_n(x) = \inf_{0 \leq u \leq 1} ne^u + x + V_{n+1}(x + u).$$

therefore

$$V_3(x) = \inf_{0 \leq u \leq 1} 3e^u + x - 2(x + u) = 3 - x.$$

where the minimum is obtained for $\hat{u}_3(x) = 0$. Then

$$V_2(x) = \inf_{0 \leq u \leq 1} 2e^u + x + 3 - (x + u) = 5.$$

where the minimum is obtained for $\hat{u}_2(x) = 0$. Then

$$V_1(x) = \inf_{0 \leq u \leq 1} e^u + x + 5 = 6 + x.$$

where the minimum is obtained for $\hat{u}_1(x) = 0$. Finally

$$V_0(x) = \inf_{0 \leq u \leq 1} x + 6 + (x + u) = 6 + 2x.$$

where the minimum is obtained for $\hat{u}_3(x) = 0$. The solution of the minimisation problem is therefore $x_t = x$ and $u_t = 0$ for all t and $V_0(x) = 6 + 2x$.

Let us now study the maximisation problem. Let $V_4(x) = -2x$ and, for $x \in \mathbb{R}$ and $n \in \{0, \dots, 3\}$,

$$V_n(x) = \sup_{t=n}^3 (te^{u_t} + x_t) - 2x_4$$

with x_t satisfying the dynamics $x_{t+1} = x_t + u_t$ and $x_n = x$, under the constraint $0 \leq u_t \leq 1$. By the dynamic programming principle, for $n = 0, \dots, 3$,

$$V_n(x) = \sup_{0 \leq u \leq 1} ne^u + x + V_{n+1}(x + u).$$

therefore

$$V_3(x) = \sup_{0 \leq u \leq 1} 3e^u + x - 2(x + u) = 3e - 2 - x.$$

where the maximum is obtained for $\hat{u}_3(x) = 1$. Then

$$V_2(x) = \sup_{0 \leq u \leq 1} 2e^u + x + 3e - 2 - (x + u) = 5e - 3.$$

where the maximum is obtained for $\hat{u}_2(x) = 1$. Then

$$V_1(x) = \sup_{0 \leq u \leq 1} e^u + x + 5e - 3 = 6e - 3 + x.$$

where the maximum is obtained for $\hat{u}_1(x) = 1$. Finally

$$V_0(x) = \sup_{0 \leq u \leq 1} x + 6e - 3 + (x + u) = 6e - 2 + 2x.$$

where the maximum is obtained for $\hat{u}_3(x) = 1$. The solution to the maximisation problem is therefore $x_t = x + t$ and $u_t = 1$ for all t , and $V_0(x) = 6e - 2 + 2x$.

Solution 4.4. Let $V_4(x) = x^2/2$ and, for $n = 0, \dots, 3$ and $x \in \mathbb{R}$,

$$V_n(x) = \min_u \sum_{t=n}^3 \frac{1}{2}(x_t^2 + u_t^2) + \frac{1}{2}x_4^2$$

with x_t satisfying the dynamics $x_{t+1} = x_t - u_t$ with $x_n = x$.

Let us show by backward induction that $V_n(x)$ is of the form $V_n(x) = \frac{\alpha_n}{2}x^2$. Indeed, V_4 is of this form. Assume that V_n takes this form (for $n \in \{1, \dots, 4\}$), and let us show that this is also the case of V_{n-1} . By the dynamic programming principle, we have

$$V_{n-1}(x) = \min_{u \in \mathbb{R}} \left(\frac{1}{2}(x^2 + u^2) + V_n(x - u) \right) = \frac{1}{2} \min_{u \in \mathbb{R}} (x^2 + u^2 + \alpha_n(x - u)^2) = \frac{\alpha_{n-1}}{2}x^2,$$

where $\alpha_{n-1} = \frac{1+2\alpha_n}{1+\alpha_n}$, the minimiser being $\bar{u} = \frac{\alpha_n}{1+\alpha_n}x$.

We then find, using this induction relation, and the fact that $\alpha_4 = 1$,

$$\alpha_3 = \frac{1+2\alpha_4}{1+\alpha_4} = \frac{3}{2}, \quad \alpha_2 = \frac{8}{5}, \quad \alpha_1 = \frac{21}{13}, \quad \alpha_0 = \frac{55}{34}.$$

The minimiser of the initial problem is therefore $V_0(1) = \frac{\alpha_0}{2}1^2 = \frac{55}{68}$.

Solution 4.5. let us first interpret the problem. Think of a hiker who has n days to walk x kilometers in total, and may stop anywhere each night. Assume that after n days, she has still x_n kilometers to go. If $n = N - 1$, this is the last day, and she has no choice: she has to walk the remaining x_{N-1} kilometers on that day. If $n \leq N - 2$, then she chooses the number u_n of kilometers she will be walking on that day, and then she will have $x_{n+1} = x_n - u_n$ kilometers to go until her final destination. When solving via Kuhn and Tucker, we directly look for the vector $(u_1, \dots, u_{N-1})$ specifying the number of kilometers to walk on each day. When using the dynamic programming principle, we first solve the problem when there is a single step, then when there are two steps, etc. (of course, an induction is needed to solve the problem for an arbitrary number of steps). Now, to the mathematics.

- (1) If $x = 0$, the constraint boils down to $u_i = 0$ for all i and the problem's solution is therefore obvious. Assume that $x > 0$. The problem is convex since the objective and the constraints are convex; the problem has a solution since the constraint is compact and nonempty and the objective continuous; finally the constraint is qualified everywhere as affine. We know that the necessary optimality conditions are sufficient. Therefore, it suffices to find (u_i) in the constraint for which there exists multipliers λ_i ($i = 0, \dots, N - 1$) and $\mu \in \mathbb{R}$ such that

$$2u_i - \lambda_i + \mu = 0, \quad \lambda_i u_i = 0, \quad \text{pour } i = 0, \dots, N - 1.$$

Given the problem's symmetry, it is intuitive to try $u_i = x/N$, $\lambda_i = 0$ and so $\mu = -2x/N$. It may be checked that this is indeed solution. We thus find:

$$W(x) = \sum_{i=0}^{N-1} (x/N)^2 = \frac{x^2}{N} \quad \forall x \geq 0.$$

Note that other arguments may be used: a) we may solve Kuhn and Tucker conditions without guessing the solution; b) since the constraint is convex and the objective strictly convex, the solution is unique, therefore due to the problem's symmetry, all x_i should be equal (if a solution is such that $x_i \neq x_j$, then exchanging the values of x_i and x_j would lead to another solution, contradicting the fact that the solution is unique).

- (2) (a) The problem may be written as a discrete-time optimal control problem $W(x) = V_0(x)$ with for $n \in \{0, \dots, N - 1\}$,

$$V_n(x_n) = \inf_{(u_i)} \sum_{i=n}^{N-1} \ell_i(x_i, u_i)$$

where $x_0 = x$, (u_i) and (x_i) satisfy $u_i \in U_i(x_i)$ and $x_{i+1} = x_i - u_i$ for all $i = n, \dots, N - 1$, $\ell(x_i, u_i) = u_i^2$ and $g = 0$.

Indeed, if (u_i) satisfies $u_i \geq 0$ for all i and $\sum_{i=0}^{N-1} u_i = x$, then, letting $x_i = x - \sum_{j=0}^{i-1} u_j$, we have that $x_0 = x$, $x_i = x_{i-1} - u_{i-1}$, $u_i \in U_i(x_i)$ (the condition $u_{N-1} \in U_{N-1}(x_{N-1})$ is due to the constraint $\sum_{i=0}^{N-1} u_i = x$) and

$$\sum_{i=0}^{N-1} \ell(x_i, u_i) = \sum_{i=0}^{N-1} u_i^2.$$

Conversely, if (x_i) and (u_i) are as in the above control problem, then we have

$$\sum_{i=0}^{N-1} u_i = \sum_{i=0}^{N-1} (x_i - x_{i+1}) = x_0 - x_N = x$$

since $x_N = x_{N-1} - u_{N-1}$ with $u_{N-1} \in U_{N-1}(x_{N-1}) = \{x_{N-1}\}$. Therefore the (u_i) are in the constraint of the optimisation problem and the criteria are the same as we saw above.

(b) Given the the dynamic programming principle principle, we have:

$$V_n(x_n) = \inf_{u_n \in U_n(x_n)} \{u_n^2 + V_{n+1}(x - u_n)\}.$$

(c) Let us show by backward induction that $V_n(x_n) = x_n^2/(N - n)$. For $n = N - 1$, the result is obvious since, by definition of U_{N-1} ,

$$V_{N-1}(x_{N-1}) = \inf_{u_{N-1} \in U_{N-1}(x_{N-1})} u_{N-1}^2 = x_{N-1}^2.$$

Assume the result true for $n + 1 \leq N - 1$ and let us show that it is true for n (note that $n < N_1$). Given the previous question and the induction hypothesis,

$$V_n(x_n) = \inf_{u_n \in U_n(x_n)} \{u_n^2 + V_{n+1}(x_n - u_n)\} = \inf_{u_n \in [0, x_n]} \left\{ u_n^2 + \frac{(x_n - u_n)^2}{N - n - 1} \right\}.$$

Solving for the point where the derivative cancels, we find that the minimiser is $u_n = x_n/(N - n) \in [0, x_n]$. It follows that $V_n(x_n) = x_n^2/(N - n)$.

By induction, we conclude that the equality $V_n(x_n) = x_n^2/(N - n)$ is true for all $n \in \{0, \dots, N - 1\}$.

(d) In particular, $W(x) = x^2/N$.

(3) The second method is longer but straightforward. The reason why the first method was quicker is also that we guessed the solution, which is not always easy. A third method uses that the function $f : y \rightarrow y^2$ is convex. Therefore, $\frac{1}{n} \sum_{i=0}^{N-1} f(u_i) \geq f\left(\frac{\sum_{i=0}^{N-1} u_i}{n}\right) = f(x/n)$. It follows that $\sum_{i=0}^{N-1} f(u_i) \geq nf(x/n) = \sum_{i=0}^{N-1} f(x/n)$. Therefore $u = (x/n, \dots, x/n)$ is solution, and using that f is strictly convex, it may be shown that this is the only one.

Solution 4.6. (1) Let $k > 0$ and $\bar{k}_t = k^{\alpha^t}$. Then $\bar{k}_{t+1} = \bar{k}_t^\alpha$. It is easy to show by induction that, if (k_t) satisfies $k_0 = k$ and $k_{t+1} \in [0, k_t^\alpha]$, then $k_t \leq \bar{k}_t$. Therefore

$$\sum_{t=0}^{\infty} \beta^t \ln(k_t^\alpha) \leq \sum_{t=0}^{\infty} \beta^t \ln(\bar{k}_t^\alpha) = \sum_{t=0}^{\infty} \alpha(\alpha\beta)^t \ln(k) = \frac{\alpha \ln(k)}{1 - \alpha\beta}.$$

Taking the sup on sequences (k_t) leads to the inequality $v(k) \leq \frac{\alpha \ln(k)}{1 - \alpha\beta}$ and therefore the equality $\frac{\alpha \ln(k)}{1 - \alpha\beta} = \sum_{t=0}^{\infty} \beta^t \ln(\bar{k}_t^\alpha)$.

For all sequences (k_t) such that $k_0 = k$, $k_{t+1} \in [0, k_t^\alpha]$, we have $\ln(k_t^\alpha) \geq \ln(k_t^\alpha - k_{t+1})$, and therefore

$$\sum_{t=0}^{\infty} \beta^t \ln(k_t^\alpha) \geq \sum_{t=0}^{\infty} \beta^t \ln(k_t^\alpha - k_{t+1}).$$

Taking the sup on sequences (k_t) on both side of the inequality leads to $v(k) \geq W(k)$.

- (2) The same argument as in the course shows that W is a fixed point of T .
- (3) By contrast, we cannot be sure as in the course that W is the unique fixed point of T , since T does not here on bounded functions (the log is not bounded) and T is not contracting in the space of continuous functions on $]0, +\infty[$.
- (4) We have:

$$Tv(k) = \sup_{y \in]0, k^\alpha[} \left\{ \ln(k^\alpha - y) + \beta \frac{\alpha \ln(y)}{1 - \alpha\beta} \right\}.$$

The function to be maximized is coercive (in the sense that it goes to $-\infty$ in 0^+ and in $(k^\alpha)^-$) and there exists therefore a maximum $\bar{y}$ in $]0, k^\alpha[$, which satisfies the necessary optimality condition

$$-\frac{1}{k^\alpha - \bar{y}} + \frac{\alpha\beta}{\bar{y}(1 - \alpha\beta)} = 0.$$

Therefore $\bar{y} = \alpha\beta k^\alpha$. It follows that

$$\begin{aligned} Tv(k) &= \ln(k^\alpha - \alpha\beta k^\alpha) + \beta \frac{\alpha \ln(\alpha\beta k^\alpha)}{1 - \alpha\beta} = \frac{\alpha \ln(k)}{1 - \alpha\beta} + \ln(1 - \alpha\beta) + \frac{\alpha\beta \ln(\alpha\beta)}{1 - \alpha\beta} \\ &= v(k) + c \end{aligned}$$

where $c = \ln(1 - \alpha\beta) + \frac{\alpha\beta \ln(\alpha\beta)}{1 - \alpha\beta} < 0$ $\alpha\beta \in (0, +\infty)$.

- (5) Let us show by induction that $T^n v = v + \frac{1 - \beta^n}{1 - \beta} c$ for $n \in \mathbb{N}^*$. It follows from the previous question that this holds for $n = 1$. If this relation is true at step n , then

$$\begin{aligned} T^{n+1}v(k) &= \sup_{y \in]0, k^\alpha[} \{ \ln(k^\alpha - y) + \beta v^n(y) \} = \sup_{y \in]0, k^\alpha[} \{ \ln(k^\alpha - y) + \beta v(y) \} + \beta \frac{1 - \beta^n}{1 - \beta} c \\ &= v(k) + c + \frac{\beta - \beta^{n+1}}{1 - \beta} c = v(k) + \frac{1 - \beta^{n+1}}{1 - \beta} c. \end{aligned}$$

We deduce that $T^n v$ converges towards $v + c_\infty$ where $c_\infty = \frac{c}{1 - \beta}$ when $n \rightarrow +\infty$.

Finally note that

$$T(v_\infty) = T(v + \frac{1}{1 - \beta} c) = T(v) + \beta \frac{1}{1 - \beta} c = v + c + \beta \frac{1}{1 - \beta} c = v_\infty.$$

therefore v_∞ is indeed a fixed point of T .

- (6) Recall that $W \leq v$ and that W is a fixed point of T . Moreover, the operator T is monotone. Therefore by induction, we have $W \leq T^n v = v + \frac{1 - \beta^{n+1}}{1 - \beta} c$. Going to the limit in n shows that $W \leq v_\infty$.
- (7) Let k_t^* be defined inductively by $k_0^* = k$ and $k_{t+1}^* = \alpha\beta(k_t^*)^\alpha$. Note that the sequence (k_t^*) satisfies $k_{t+1}^* \in [0, (k_t^*)^\alpha]$ and therefore

$$W(k) \geq \sum_{t=0}^{\infty} \beta^t \ln((k_t^*)^\alpha - k_{t+1}^*).$$

Given the computation of question 4, we know that

$$Tv_\infty(k_t^*) = \sup_{y \in]0, (k_t^*)^\alpha[} \{ \ln((k_t^*)^\alpha - y) + \beta v_\infty(y) \} = \{ \ln((k_t^*)^\alpha - k_{t+1}^*) + \beta v_\infty(k_{t+1}^*) \} = v_\infty(k_t^*).$$

Therefore $\ln((k_t^*)^\alpha - k_{t+1}^*) = v_\infty(k_t^*) - \beta v_\infty(k_{t+1}^*)$. This proves that for all $T > 0$,

$$\sum_{t=0}^T \beta^t \ln((k_t^*)^\alpha - k_{t+1}^*) = \sum_{t=0}^T \beta^t (v_\infty(k_t^*) - \beta v_\infty(k_{t+1}^*)) = v_\infty(k) - \beta^{T+1} v_\infty(k_{T+1}^*).$$

In order to be able to pass to the limit as $T \rightarrow +\infty$, we need to clarify the limit of the term on the right. By definition (and a simple induction), we have $k_t^* = (\alpha\beta)^{\frac{1-\alpha^t}{1-\alpha}} k^{\alpha^t}$. Therefore

$$\beta^T v_\infty(k_{T+1}^*) = \beta^T \left(\frac{\alpha \ln(k_{T+1}^*)}{1 - \alpha\beta} + c_\infty \right) = \frac{\alpha\beta^T}{1 - \alpha\beta} \left(\frac{1 - \alpha^{T+1}}{1 - \alpha} \ln(\alpha\beta) + \alpha^{T+1} \ln(k) \right) + \beta^T c_\infty,$$

which goes to 0 as $T \rightarrow +\infty$. We deduce that

$$W(k) \geq \sum_{t=0}^{\infty} \beta^t \ln((k_t^*)^\alpha - k_{t+1}^*) = v_\infty(k),$$

which is the desired result : $W = v_\infty$.

- (8) It follows from the previous question that the sequence $(k_t^* = (\alpha\beta)^{\frac{1-\alpha^t}{1-\alpha}} k^{\alpha^t})$ is optimal for $v_\infty = W$.
- (9) Note that $k_t^* = \exp\{\frac{1-\alpha^t}{1-\alpha} \ln(\alpha\beta) + \alpha^t \ln(k)\}$. This sequence is strictly increasing if $\ln(k) < \ln(\alpha\beta)/(1 - \alpha)$, that is, $k < (\alpha\beta)^{1/(1-\alpha)}$, strictly decreasing if $k > (\alpha\beta)^{1/(1-\alpha)}$, and constant otherwise. It goes to $k_\infty^* = (\alpha\beta)^{1/(1-\alpha)}$.

11. A BIT OF CALCULUS OF VARIATIONS

As we said, our end-goal in this part is to study optimal control problems; to get acquainted with infinite dimensional optimisation, we begin with the simple setting of calculus of variations. Here, the optimisation variable is a whole function, typically denoted by y . This time, one has to be extremely careful with the topology used on the set of functions, as it strongly influences the existence of solutions and the derivation of the optimality conditions. The goal of this chapter is to help you derive, at least formally, the so-called Euler-Lagrange equation. In order to keep things as light as possible, we will only be working in the one-dimensional case.

11.1. General introduction. In this section and more generally in this chapter, we will be considering problems of the following type: an interval $I = [a; b]$ being given, we investigate the problem

$$(11.1) \quad \min_{y \in \mathcal{Y}} \int_a^b \mathcal{L}(x, y(x), y'(x)) dx$$

where $\mathcal{Y}$ is a set of functions defined in such a way that the quantity $\int \mathcal{L}$ makes sense for any $y \in \mathcal{Y}$. A typical (but usually not so good) choice is to take

$$\mathcal{Y} := \{y \in \mathcal{C}^1([a; b]; \mathbb{R}^d), y(a) = y_0, y(b) = y_1\},$$

where y_0, y_1 correspond to two given points. Typical examples of problems of the type (11.1) are the following:

- (1) Shortest path problems: in that setting, one seeks a path $y : [0; 1] \rightarrow \mathbb{R}^d$ connecting an initial point y_0 to another point y_1 . The underlying functional is thus $\mathcal{L}(x, p, q) = \|q\|$, and the set of constraints is

$$\mathcal{Y} = \{y \in \mathcal{C}^1([0; 1]; \mathbb{R}^d), y(0) = y_0, y(1) = y_1\}.$$

- (2) Isoperimetric type problems (the queen Dido problem): the idea is to find a function $y : [-1; 1] \rightarrow \mathbb{R}$ in such a way that the area of the graph enclosed by the function is maximal, under the constraint that the graph has length L . In other words, we aim at solving

$$\min_{y \in \mathcal{Y}} \int_{-1}^1 y,$$

with

$$\mathcal{Y} := \left\{ y \in \mathcal{C}^1([-1; 1]; \mathbb{R}), y(\pm 1) = 0, \int_{-1}^1 \sqrt{1 + (y')^2} = L \right\}.$$

As always, one should be extremely careful: in general, optimisers might fail to exist. A typical example of functional to keep in mind is the following one, due to Weierstraß: consider the minimisation problem

$$\min_{y \in \mathcal{Y}} \int_{-1}^1 x^2 (y'(x))^2 dx,$$

under the constraint that $y(-1) = 0$, $y(1) = 1$ and y is $\mathcal{C}^1$. Then by considering a smoothed version of $y_k := \mathbb{1}_{\{x \geq \frac{1}{k}\}} + \frac{x + \frac{1}{k}}{\frac{2}{k}} \mathbb{1}_{\{-\frac{1}{k} \leq x \leq \frac{1}{k}\}}$ we see that the infimum is zero, but can not be reached by any $\mathcal{C}^1$ function.

11.2. The Euler-Lagrange equation.

11.2.1. *From the optimisation problem to the Euler-Lagrange equation.* In this section, we explain how to derive the optimality conditions for (11.1) under the assumption that the set $\mathcal{Y}$ consists of $\mathcal{C}^1$ functions with fixed end-points conditions, that is,

$$\mathcal{Y} := \{y \in \mathcal{C}^1([a; b]; \mathbb{R}^d), y(a) = y_0, y(b) = y_1\}.$$

It is possible to handle other types of constraints, but, as always, other sets of constraints are reflected in the choices of admissible perturbation.

We start with an instructive example, that of the least energy path between two points. In other words, we consider

$$\min_{y \in \mathcal{C}^1([0; 1]; \mathbb{R}^d), y(0)=y_0, y(1)=y_1} \mathcal{F}(y) := \frac{1}{2} \int_0^1 \|y'(t)\|^2 dt.$$

If y^* is an optimiser, it follows that for any perturbation h such that $y^* + \delta h$ is still admissible⁴ for $|\delta|$ small enough, we have

$$\mathcal{F}(y^*) \leq \mathcal{F}(y^* + \delta h).$$

However,

$$\mathcal{F}(y^* + \delta h) = \frac{1}{2} \int_0^1 \|(y^*)'\|^2 + \delta \int_0^1 \langle (y^*)', h' \rangle + O(\delta^2).$$

In particular, we deduce that for any h such that $y^* + \delta h$ remains admissible for small enough $|\delta|$ there holds

$$\int_0^1 \langle (y^*)', h' \rangle = 0.$$

But now, the conditions ensuring that $y^* + \delta h$ remains admissible are, on the one hand, a regularity one, which imposes $h \in \mathcal{C}^1$, and, on the other hand, a fixed end points condition that imposes $h(0) = h(1) = 0$. We thus obtain that for any h of class $\mathcal{C}^1$ with $h(0) = h(1) = 0$ we have

$$\int_0^1 \langle (y^*)', h' \rangle = 0.$$

Now, this is still a very implicit formulation; the goal is now to rewrite this condition as a linear form in h , and not in h' : the orthogonality to a set of derivatives of a given ensemble of functions is much more complicated than the orthogonality to a set of functions. If we have faith, and assume that y^* is, in fact, $\mathcal{C}^2$, then we might do an integration by parts and deduce that

$$\begin{aligned} 0 &= \int_0^1 \langle (y^*)', h' \rangle \\ &= [\langle (y^*)', h \rangle]_0^1 - \int_0^1 \langle (y^*)'', h \rangle \\ &= - \int_0^1 \langle (y^*)'', h \rangle. \end{aligned}$$

⁴Theoretically, we would need to work in the cone of admissible perturbations, but we do not bother with these technical aspects here.

We thus obtain the following fact: for any $h \in \mathcal{C}^1$ satisfying $h(0) = h(1) = 0$ there holds

$$\int_0^1 \langle (y^*)'', h \rangle = 0.$$

We recall the following simple fact (exercise: prove it!):

Lemma 11.1. *[Du Bois-Reymond lemma] Let $f \in \mathcal{C}^0([a; b]; \mathbb{R})$ be such that, for any $g \in \mathcal{C}^0([a; b]; \mathbb{R})$ satisfying $g(a) = g(b) = 0$ there holds $\int_a^b fg = 0$. Then $f \equiv 0$ in $[a; b]$.*

Applying this lemma component wise, we deduce that

$$(y^*)'' = 0$$

or, in other words, that $y^* = ty_1 + (1 - t)y_0$. Thus, the path of least energy is the straight line.

Let us now move back to the general case and see what we can obtain for minimisers y^* of

$$\min_{y \in \mathcal{C}^1([a; b]; \mathbb{R}), y(a)=y_0, y(b)=y_1} \mathcal{F}(y) := \int_a^b \mathcal{L}(t, y(t), y'(t)) dt.$$

To simplify notations, we assume that y is scalar valued, but the conclusion would hold for vector-valued functions. Rather than giving a complete set of hypothesis on $\mathcal{L}$ from the get-go, we will, as we go along, see what type of regularity we need on $\mathcal{L}$, and in which topology. Likewise, for any $h \in \mathcal{C}^1([a; b]; \mathbb{R})$ such that $h(a) = h(b) = 0$ and any $|\delta|$ small enough, there holds

$$\mathcal{F}(y^*) \leq \mathcal{F}(y^* + \delta h)$$

or, equivalently,

$$\int_a^b (\mathcal{L}(t, y^*(t) + \delta h(t), (y^*)'(t) + \delta h'(t)) - \mathcal{L}(t, y^*(t), (y^*)'(t))) dt \geq 0.$$

We can thus (provided $\mathcal{L}$ is regular enough) do a Taylor expansion of $\mathcal{L} = \mathcal{L}(x, p, q)$, which gives that for any admissible h we have

$$\int_a^b \frac{\partial \mathcal{L}}{\partial p}(t, y^*, (y^*)') h(t) dt + \int_a^b \frac{\partial \mathcal{L}}{\partial q}(t, y^*, (y^*)') h'(t) dt = 0.$$

Now, we would like to use Lemma 11.1 again, which requires rewriting this condition as an orthogonality condition to the set of all h 's while here it still involves h' . To handle this, we here again need a bit of faith, and to assume that $\frac{\partial \mathcal{L}}{\partial q}(t, y^*, (y^*)')$ is a $\mathcal{C}^1$ function. If this is the case, then we can do an integration by parts, and obtain that for any $h \in \mathcal{C}^1$ such that $h(a) = h(b) = 0$ there holds

$$(11.2) \quad \int_a^b h \left\{ \frac{\partial \mathcal{L}}{\partial p}(t, y^*, (y^*)') - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial q}(t, y^*, (y^*)') \right\} dt = 0$$

or, in compact form (from Lemma 11.1),

$$(11.3) \quad \left\{ \frac{\partial \mathcal{L}}{\partial p}(t, y^*, (y^*)') - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial q}(t, y^*, (y^*)') \right\} = 0 \text{ in } [a; b].$$

Equation (11.3) is called the Euler-Lagrange equation associated with the problem. Now, be careful: the operator $\frac{d}{dt}$ is *not* a partial derivative of the Lagrangian $\mathcal{L}$ with respect to t . In other words, this is not the quantity

$$\frac{\partial^2 \mathcal{L}}{\partial q \partial t}(t, y^*, (y^*)').$$

Here, this is the derivative of the map $t \mapsto g(t) := \mathcal{L}(t, y^*, (y^*)')$.

As practice makes perfect, a portion of the chapter will be devoted to the investigation of classical examples in the calculus of variations. Nevertheless, before we do so, let us make several important comments.

11.2.2. *Some crucial comments.* Here are, without any specific order, some key aspects to keep in mind:

- (1) Unless you are working in degenerate cases, the Euler-Lagrange equation yields a differential equation that is of second order in y^* . This was already the case when computing the path of least energy.
- (2) As usual, the Euler-Lagrange equation is a purely local equation: you can not, in general, hope to derive global optimality from the fact that the Euler-Lagrange equation is satisfied. We will see, in a forthcoming sections, some cases where you can actually go back.
- (3) There are two aspects we swept under the rug: first, the existence of an optimiser. This goes beyond the scope of this course, and rather hinges on fine tools in functional analysis. Second, the *a priori* justification of the validity of the integration by parts that we do. This is also a very delicate matter: take the case of the least energy path joining two points. Then, although we worked in $\mathcal{C}^1$ (which is the natural space given by the problem), we assumed at some point that the optimiser was, in fact, $\mathcal{C}^2$. Proving that this is valid is possible, but very difficult, and is a core question in calculus of variations, that of regularity theory. This is also far beyond the scope of this class to investigate such properties (but is a fascinating question!!).

11.2.3. *Some examples of Euler-Lagrange equations.* In this paragraph, we show how to derive the Euler-Lagrange equation on some basic examples—the exercise sheet contains many more. It should be noted that, as is the case with the dynamic programming principle, although writing the equation down is simple, solving it can prove extremely challenging. Indeed, it will not always be the case that the resulting differential equation can be integrated in a straightforward manner. We will give some examples of situations where the form of the equation can be simplified. Simplification when $\mathcal{L}$ does not depend on y . We begin with a simple case, when $\mathcal{L}(t, p, q) = \mathcal{L}(t, q)$. We have the following identity:

Proposition 11.1. *Let y^* be a $\mathcal{C}^2$ solution of the optimisation problem*

$$\min_{y \in \mathcal{C}^1, y(a)=y_0, y(b)=y_1} \int_a^b \mathcal{L}(t, y'(t)) dt.$$

Then there holds

$$\frac{\partial \mathcal{L}}{\partial q}(t, y'(t)) \equiv \alpha$$

for some constant α .

This is obvious from the Euler-Lagrange equation, but nevertheless has nice implications; in particular, the ODE that this condition provides is a first-order ODE. A situation we encountered where this is what was used was the path of least energy.

Simplification when $\mathcal{L}$ does not depend on t . We begin with a situation where $\mathcal{L}$ does not depend on the variable t . To illustrate our point, consider one of the oldest problems in physics, that of the catenary. Namely, we consider a chain hanging from two points in the plane, say (a_0, b_0) and (a_1, b_1) with $a_0 < a_1$. What shape does this chain have? Here, we have to rely on a principle of physics: to each possible configuration of the chain is associated an energy, that has to be minimal in the configuration we see in nature. Here, a configuration of the chain is a function $y : [a_0; a_1] \rightarrow \mathbb{R}$ satisfying $y(a_i) = b_i$. The only thing we take into account in order to describe the energy of y is the potential energy due to the gravitational force and a simple reasoning show that the energy is given (up to a multiplicative constant) by

$$\mathcal{E}(y) = \int_{a_0}^{a_1} y \sqrt{1 + (y')^2}.$$

Thus, the problem of determining the shape of the chain is the following calculus of variations problem: solve

$$\min_{y \in \mathcal{C}^1, y(a_i) = b_i} \mathcal{E}(y).$$

The Lagrangian is

$$\mathcal{L}(x, p, q) = p \sqrt{1 + q^2}.$$

In particular,

$$\frac{\partial \mathcal{L}}{\partial p} = \sqrt{1 + q^2}, \quad \frac{\partial \mathcal{L}}{\partial q} = \frac{pq}{\sqrt{1 + q^2}}$$

so that the Euler-Lagrange equation at an optimiser y^* writes

$$\sqrt{1 + ((y^*)')^2} - \frac{d}{dt} \frac{y^*(y^*)'}{\sqrt{1 + ((y^*)')^2}} = 0.$$

From this point one, we might proceed to tediously develop the differential, and obtain an atrocious second order differential equation. However, here, we can use that $\mathcal{L}$ does not depend on t , which we summarise in the following proposition, sometimes referred to as the Beltrami identity:

Proposition 11.2. *Let y^* be a $\mathcal{C}^2$ solution of the optimisation problem*

$$\min_{y \in \mathcal{C}^1, y(a)=y_0, y(b)=y_1} \int_a^b \mathcal{L}(y, y').$$

Then there holds

$$\mathcal{H}(t) := \mathcal{L}(y^*, (y^*)') - (y^*)' \frac{\partial \mathcal{L}}{\partial q}(y^*, (y^*)') \equiv cst.$$

Proof of Proposition 11.2. We compute in a straightforward way:

$$\begin{aligned} \frac{d}{dt} \mathcal{H}(t) &= (y^*)' \frac{\partial \mathcal{L}}{\partial p} + (y^*)'' \frac{\partial \mathcal{L}}{\partial q} - (y^*)'' \frac{\partial \mathcal{L}}{\partial q} \\ &\quad - (y^*)' \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial q}(y^*, (y^*)') \end{aligned}$$

$$\begin{aligned}
 &= (y^*)' \left(\frac{\partial \mathcal{L}}{\partial p} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial q} \right) \\
 &= 0 \text{ by the Euler-Lagrange equation.}
 \end{aligned}$$

□

Now, if we apply the Beltrami identity to the catenary problem, we deduce that

$$y^* \sqrt{1 + ((y^*)')^2} - \frac{y^* (y^*)' (y^*)'}{\sqrt{1 + ((y^*)')^2}} \equiv \alpha$$

for some constant α , which implies

$$\frac{(y^*)^2}{1 + ((y^*)')^2} \equiv \alpha^2.$$

Integrating this equation is still not trivial (and will not be done here), but we point out that this gives a first-order ode, to be compared with the second-order ode we should have gotten. At this stage, you might be wondering: but a first order ODE only has a one-dimensional family of solutions; that is correct, but the constant α provides a second degree of freedom, so that this is not contradictory.

11.2.4. From the Euler-Lagrange equation to the optimality. The Euler-Lagrange equation (11.3) is akin to the first-order condition $\nabla f = 0$, and this begs the question: when can we say that satisfying (11.3) implies optimality? As always, the key word here is convexity; we summarise this fact in the following theorem:

Theorem 11.1. *Consider*

$$(11.4) \quad \min_{y \in \mathcal{Y}} \mathcal{F}(y) := \int_a^b \mathcal{L}(t, y(t), y'(t)) dt$$

where

$$\mathcal{Y} = \{y \in \mathcal{C}^1([a; b]; \mathbb{R}), y(a) = y_0, y(b) = y_1\}.$$

Assume that $\mathcal{L}$ is of class $\mathcal{C}^1$ and that, for all $t \in [0, 1]$, the map $(p, q) \rightarrow \mathcal{L}(t, p, q)$ is convex. If $y^* \in \mathcal{Y}$ solves (11.3) then y^* solves (11.4).

Proof of Theorem 11.1. We let $y \in \mathcal{Y}$ and we consider the map

$$\varphi : t \mapsto \mathcal{F}((1-t)y^* + ty).$$

By convexity of $\mathcal{L}$, we deduce that φ is convex. But, as already computed,

$$\varphi'(0) = \int_a^b \left\{ \frac{\partial \mathcal{L}}{\partial p} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial q} \right\} (y - y^*) = 0,$$

so that 0 is a minimiser. This concludes the proof. □

11.3. What happens with other types of constraints?

11.3.1. *The Lagrange multiplier rule.* To conclude this introduction to calculus of variations, we come back to the isoperimetric problem we alluded to at the (very) beginning of the chapter; without getting into specifics, what should be noted is that the difference with what we did up until that point is that we have an additional equality constraints. Namely, we consider a functional of the type

$$J(y) := \int_a^b \mathcal{J}(x, y, y')$$

and the admissible set

$$\mathcal{Z} := \{y \in \mathcal{C}^1([a; b]; \mathbb{R}^d), y(a) = y_0, y(b) = 1, J(y) = 0\}$$

and we investigate the constrained optimisation problem

$$(11.5) \quad \min_{y \in \mathcal{Z}} \mathcal{F}(y) := \int_a^b \mathcal{L}(t, y, y').$$

Now, just as in the case of constrained finite-dimensional optimisation, the optimality conditions for this problem involve a Lagrange multiplier. Recall that, in the finite dimensional case, for the optimisation problem $\min_{g=0} f$, we deduced the existence of a Lagrange multiplier λ such that $\nabla f = \lambda \nabla g$ at an optimiser. The situation here is completely analogous, albeit more delicate to prove. We simply state the following result, and illustrate it on some simple examples:

Theorem 11.2. *If $\mathcal{L}, \mathcal{J}$ are $\mathcal{C}^1$ functions, if $y^* \in \mathcal{Z} \cap \mathcal{C}^2$ solves (11.5), there exists $\lambda \in \mathbb{R}$ such that*

$$\frac{\partial \mathcal{L}}{\partial p}(t, y^*, (y^*)') - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial q}(t, y^*, (y^*)') = \lambda \left(\frac{\partial \mathcal{J}}{\partial p}(t, y^*, (y^*)') - \frac{d}{dt} \frac{\partial \mathcal{J}}{\partial q}(t, y^*, (y^*)') \right).$$

11.3.2. *Two basic examples.* We give two examples of constrained calculus of variations problems.

Eigenvalue problems: We consider the minimisation problem

$$\min_{u \in \mathcal{C}^1([-1; 1]; \mathbb{R}), u(\pm 1) = 0, \int_{-1}^1 u^2 = 1} \int_{-1}^1 (u')^2.$$

Let $\mathcal{L}(t, p, q) = q^2$, $\mathcal{J}(t, p, q) = p^2$. At a minimiser v (we assume that it exists), it follows from Theorem 11.2 that there exists λ such that

$$-u'' = \lambda u, u(\pm 1) = 0, \int_{-1}^1 u^2 = 1.$$

In other words, we obtain an eigenvalue for the differential operator $-\frac{d^2}{dt^2}$.

Isoperimetric type problems: we aim at solving

$$\min_{y \in \mathcal{Y}} \int_{-1}^1 y,$$

with

$$\mathcal{Y} := \left\{ y \in \mathcal{C}^1([-1; 1]; \mathbb{R}), y(\pm 1) = 0, \int_{-1}^1 \sqrt{1 + (y')^2} = L \right\}.$$

Likewise, introduce $\mathcal{L}(t, p, q) := p$, $\mathcal{J}(t, p, q) := \sqrt{1 + q^2}$. By the Lagrange multiplier rule, at an optimiser z (we assume that it exists) there exists $\lambda \in \mathbb{R}$ such that

$$1 = \lambda \frac{d}{dt} \frac{z'}{\sqrt{1 + (z')^2}}.$$

Expanding the derivative, this yields the equation (with $\mu = \frac{1}{\lambda}$)

$$\frac{z''}{(1 + (z')^2)^{\frac{3}{2}}} = \mu.$$

Observe that, simply from this equation, we see that the solution y should be either convex or concave. Now, if z were convex, it would be negative in $(-1; 1)$ and would obviously not be an optimiser, so that $\mu < 0$. We thus deduce that z' changes sign at most once (and, for the sake of simplicity, we assume that it changes sign at 0). Then, on $[0; 1]$, z solves

$$z' = \mu' t (1 + z^2)^{\frac{1}{2}}$$

for some $\mu' < 0$...and it remains to solve this ODE explicitly: this would give that z describes a portion of a circle! The computations, once again, are tedious.

EXERCISES FOR THE SECOND PART OF PART 4

Exercise 4.7. Write the Euler-Lagrange equation for the following problems and compute their solutions.

$$J_1(x) = \int_0^1 2tx(t) - x^2(t) + 3x^2(t)x'(t) dt$$

and

$$J_2(x) = \int_0^1 t\sqrt{1 + (x'(t))^2} dt$$

Exercise 4.8. Same questions for the following problems, but now taking into account the initial and terminal conditions.

$$J_1(x) = \int_0^1 (x'(t))^2 + 12tx(t) dt \quad \text{with } x(0) = 2, x(1) = 3;$$

$$J_2(x) = \int_0^1 x'(t)(1 + (1+t)^2 x'(t)) dt \quad \text{with } x(0) = 3, x(1) = 2.$$

Exercise 4.9. Consider the problem

$$\inf_{x \in C^1([0,1]), x(0)=1, x(1)=0} \int_0^1 t\sqrt{1 + (x'(t))^2} dt.$$

- (i) Check that for all $x \in C^1([0,1])$, $\int_0^1 t\sqrt{1 + (x'(t))^2} dt \geq 1/2$.
- (ii) Show that the infimum is equal to $1/2$ (hint: compute the criterion for functions of the form $x_n(t) = (1-t)^n$, or functions of the form $y_n(t) = 1-nt$ if $t \leq 1/n$ and 0 otherwise).
- (iii) Show that the problem has no solution.

Exercise 4.10. Consider the calculus of variation problem (with no terminal condition) :

$$(P) \quad \inf_{x \in X, x(0)=A} \int_0^1 L(t, x(t), x'(t)) dt + g(x(1)),$$

where X is the set of C^1 functions from $[0,1]$ to $\mathbb{R}$. Functions $L = L(t, x, p)$ and $g = g(x)$ are assumed of class $\mathcal{C}^1$ on $[0,1] \times \mathbb{R} \times \mathbb{R}$ and $\mathbb{R}$, respectively. Assume that x is a solution (a minimizer) of the problem.

- (i) Show that for all functions $v : [0,1] \rightarrow \mathbb{R}$ of class C^1 such that $v(0) = 0$:

$$\int_0^1 \frac{\partial L}{\partial x}(t, x(t), x'(t))v(t) + \frac{\partial L}{\partial p}(t, x(t), x'(t))v'(t) dt + g'(x(1))v(1) = 0.$$

- (ii) Use Dubois-Raymond Lemma to prove that x satisfies the Euler equation : function $t \rightarrow \frac{\partial L}{\partial p}(t, x(t), x'(t))$ is of class $\mathcal{C}^1$ on $[0,1]$ and

$$(11.6) \quad \frac{d}{dt} \frac{\partial L}{\partial p}(t, x(t), x'(t)) = \frac{\partial L}{\partial x}(t, x(t), x'(t)) \quad \forall t \in [0,1].$$

- (iii) Deduce that for all functions $v : [0,1] \rightarrow \mathbb{R}$ of class C^1 with $v(0) = 0$:

$$\left[\frac{\partial L}{\partial p}(1, x(1), x'(1)) + g'(x(1)) \right] v(1) = 0.$$

- (iv) Show that x satisfies the “transversality condition” : $\frac{\partial L}{\partial p}(1, x(1), x'(1)) + g'(x(1)) = 0$.

(v) *Let us admit that the problem*

$$(P) \quad \inf_{x \in X, x(0)=1} \int_0^1 (x'(t))^2 dt + (x(1))^2$$

has a minimum. Find it.

SOLUTIONS FOR THE SECOND PART OF PART 4

Solution 4.7. Recall that the Euler-Lagrange equation is given by:

$$\frac{d}{dt} \frac{\partial L}{\partial p}(t, x(t), x'(t)) = \frac{\partial L}{\partial x}(t, x(t), x'(t)).$$

For J_1 , $L(t, x, p) = 2tx - x^2 + 3x^2p$, so that $\frac{\partial L}{\partial p}(t, x, p) = 3x^2$ and $\frac{\partial L}{\partial x}(t, x, p) = 2t - 2x + 6xp$. Therefore the Euler-Lagrange equation is

$$\frac{d}{dt} (3(x(t))^2) = 2t - 2x(t) + 6xx'(t),$$

that is,

$$6x(t)x'(t) = 2t - 2x(t) + 6xx'(t).$$

We thus find $x(t) = t$ for all $t \in [0, 1]$.

For J_2 , we have $L(t, x, p) = t\sqrt{1+p^2}$ and therefore $\frac{\partial L}{\partial p}(t, x, p) = \frac{tp}{\sqrt{1+p^2}}$ and $\frac{\partial L}{\partial x}(t, x, p) = 0$. The Euler-Lagrange equation thus reduces to

$$0 = \frac{d}{dt} \left(\frac{tx'(t)}{\sqrt{1+(x'(t))^2}} \right).$$

This shows that there exists a constant $c \in \mathbb{R}$ such that $\frac{tx'(t)}{\sqrt{1+(x'(t))^2}} = c$ for all $t \in [0, 1]$. But for $t = 0$, implies that $c = 0$. It follows that $x'(t) = 0$ for all $t \in [0, 1]$: only constant functions satisfy the Euler-Lagrange equation.

Solution 4.8. For J_1 , $L(t, x, p) = p^2 + 12tx$, and therefore $\frac{\partial L}{\partial p}(t, x, p) = 2p$ and $\frac{\partial L}{\partial x}(t, x, p) = 12t$. Euler equation thus takes the form: $2x''(t) = 12t$, so $x'(t) = 3t^2 + c_1$ (where $c_1 \in \mathbb{R}$) and $x(t) = t^3 + c_1t + c_2$ (where $c_2 \in \mathbb{R}$). Since $x(0) = 2$ and $x(1) = 3$, we must have $c_2 = 2$ and $c_1 = 0$. Finally, we find $x(t) = t^3 + 2$.

For J_2 , $L(t, x, p) = p(1 + (1+t)^2p)$, and therefore $\frac{\partial L}{\partial p}(t, x, p) = 1 + 2p(1+t)^2$ and $\frac{\partial L}{\partial x}(t, x, p) = 0$. Euler equation thus takes the form:

$$\frac{d}{dt} (1 + 2(1+t)^2x'(t)) = 0,$$

and there exists a constant $c_1 \in \mathbb{R}$ such that $1 + 2(1+t)^2x'(t) = c_1$. Therefore $x'(t) = (c_1 - 1)/(1+t)^2$, so $x(t) = -(c_1 - 1)/(1+t) + c_2$ (where $c_2 \in \mathbb{R}$). Since $x(0) = 3$ and $x(1) = 2$, we must have $c_1 = -1$ and $c_2 = 1$. Finally, we find $x(t) = 2/(1+t) + 1 = \frac{t+3}{t+1}$ for $t \in [0, 1]$.

Solution 4.9. (i) For all x in $C^1([0, 1])$, $t\sqrt{1+(x'(t))^2}dt \geq t$, hence

$$\int_0^1 t\sqrt{1+(x'(t))^2}dt \geq \int_0^1 tdt = 1/2.$$

(ii) For $n \in \mathbb{N}$ and $t \in [0, 1]$, let $x_n(t) = (1-t)^n$. Using the fact that $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ for all $a, b \geq 0$ and that $x'_n \leq 0$, we get

$$\begin{aligned} \int_0^1 t\sqrt{1+(x'_n(t))^2}dt &\leq \int_0^1 t(1+|x'_n(t)|)dt \leq \frac{1}{2} - \int_0^1 tx'_n(t)dt \\ &= \frac{1}{2} - [tx_n(t)]_0^1 + \int_0^1 x_n(t)dt \rightarrow \frac{1}{2} \end{aligned}$$

since $x_n(1) = 0$ and $\int_0^1 x_n(t)dt = 1/(n+1) \rightarrow 0$ as $n \rightarrow +\infty$ (the latter also follows from a dominated convergence argument, since $x_n(t) \rightarrow 0$ for $t \in]0, 1[$ and $|x_n(t)| \leq 1$ for all $t \in [0, 1]$). Moreover, the infimum of the problem is greater or equal to $1/2$ as seen in (i). Therefore, the infimum is equal to $1/2$.

Alternatively, the value of the objective for functions y_n is easy to compute explicitly and goes to $1/2$. The problem is that functions y_n are not of class C^1 . This can be arranged by arguing (we will not do it precisely here, but there are standard techniques to do so) that y_n could be slightly modified into a C^1 function $\tilde{y}_n$ while changing the value of the objective by less than $1/n$, so that the value of the objective for $\tilde{y}_n$ goes to $1/2$ as well.

- (iii) If the problem has a solution $t \rightarrow x(t)$ (with x of class C^1 , $x(0) = 0$, $x(1) = 1$), then it follows from question (i) and the fact that the infimum is $1/2$ that $x'(t) = 0$ for almost all t . Since x is of class C^1 , this implies that $x'(t) = 0$ for all t . But $x(0) = 0$, which implies that $x(t) = 0$ for all t , and therefore $x(1) = 0$. This contradicts the assumption $x(1) = 1$. Therefore, the problem has no solution.

Solution 4.10. (i) For $y \in X$, let $J_1(y) = \int_0^1 L(t, y(t), y'(t))dt$, $J_2(y) = g(y(1))$ and $J = J_1 + J_2$. We know from the course that J_1 is C^1 on X with

$$J'_1(y)(v) = \int_0^1 \frac{\partial L}{\partial x}(t, y(t), y'(t))v(t) + \frac{\partial L}{\partial p}(t, y(t), y'(t))v'(t) dt$$

for all $y, v \in X$. Moreover,

$$J_2(y+v) = g(y(1)+v(1)) = g(y(1))+g'(y(1))v(1)+|v(1)|\varepsilon(v(1)) = g(y(1))+g'(y(1))v(1)+\|v\|_X\tilde{\varepsilon}(v)$$

where $\tilde{\varepsilon}(v) \rightarrow 0$ as $v \rightarrow 0$ in X . This shows that $J'_2(y)(v) = g'(y(1))v(1)$. Therefore

$$J'(y)(v) = \int_0^1 \frac{\partial L}{\partial x}(t, y(t), y'(t))v(t) + \frac{\partial L}{\partial p}(t, y(t), y'(t))v'(t) dt + g'(y(1))v(1)$$

for all $y, v \in X$. For $h \in \mathbb{R}$, let $x_h(t) = x(t) + hv(t)$. Then x_h is of class C^1 and satisfies $x_h(0) = A$ since $x(0) = A$ and $v(0) = 0$. Since x is a minimizer of the problem, $J(x_h) \leq J(x)$ for all $h \in \mathbb{R}$. We deduce that $\frac{d}{dh}J(x_h) = 0$ at $h = 0$, with $\frac{d}{dh}J(x_h) = J'(x_h)(v)$ since $x_h = x + hv$. Therefore

$$\int_0^1 \frac{\partial L}{\partial x}(t, x(t), x'(t))v(t) + \frac{\partial L}{\partial p}(t, x(t), x'(t))v'(t) dt + g'(x(1))v(1) = 0.$$

- (ii) In particular, for any function $v \in X$ with $v(0) = v(1) = 0$,

$$\int_0^1 \frac{\partial L}{\partial x}(t, x(t), x'(t))v(t) + \frac{\partial L}{\partial p}(t, x(t), x'(t))v'(t) dt = 0.$$

Let $M(t) = \int_0^t \frac{\partial L}{\partial x}(s, x(s), x'(s))ds$. An integration by parts leads to:

$$\int_0^1 \frac{\partial L}{\partial x}(t, x(t), x'(t))v(t)dt = [M(t)v(t)]_0^1 - \int_0^1 M(t)v'(t)dt = - \int_0^1 M(t)v'(t)dt,$$

since $v(0) = v(1) = 0$. Therefore, for any function $v \in X$ with $v(0) = v(1) = 0$, we have

$$\int_0^1 \left(-M(t) + \frac{\partial L}{\partial p}(t, x(t), x'(t)) \right) v'(t) dt = 0.$$

We deduce from Dubois-Raymond lemma that the function $t \rightarrow -M(t) + \frac{\partial L}{\partial p}(t, x(t), x'(t))$ is constant. Since M is of class C^1 (this is a primitive of a continuous function), the function $t \rightarrow \frac{\partial L}{\partial p}(t, x(t), x'(t))$ is of class $\mathcal{C}^1$ on $[0, 1]$, and

$$\frac{d}{dt} \frac{\partial L}{\partial p}(t, x(t), x'(t)) = M'(t) = \frac{\partial L}{\partial x}(t, x(t), x'(t)) \quad \forall t \in [0, 1].$$

(iii) Let $v : [0, 1] \rightarrow \mathbb{R}^N$ be of class C^1 with $v(0) = 0$. Using the notation of the previous question, we have

$$\int_0^1 \frac{\partial L}{\partial x}(t, x(t), x'(t)) v(t) dt = [M(t)v(t)]_0^1 - \int_0^1 M(t)v'(t) dt = M(1)v(1) - \int_0^1 M(t)v'(t) dt,$$

since $v(0) = 0$. We then deduce from question (i) that

$$\begin{aligned} 0 &= \int_0^1 \frac{\partial L}{\partial x}(t, x(t), x'(t)) v(t) dt + \frac{\partial L}{\partial p}(t, x(t), x'(t)) v'(t) dt + g'(x(1))v(1) \\ &= \int_0^1 \left(-M(t) + \frac{\partial L}{\partial p}(t, x(t), x'(t)) \right) v'(t) dt + (M(1) + g'(x(1)))v(1). \end{aligned}$$

Since the function $t \rightarrow -M(t) + \frac{\partial L}{\partial p}(t, x(t), x'(t))$ is constant and $v(0) = 0$,

$$\int_0^1 \left(-M(t) + \frac{\partial L}{\partial p}(t, x(t), x'(t)) \right) v'(t) dt = \left(-M(1) + \frac{\partial L}{\partial p}(1, x(1), x'(1)) \right) v(1)$$

and therefore

$$0 = \left(\frac{\partial L}{\partial p}(1, x(1), x'(1)) + g'(x(1)) \right) v(1).$$

(iv) We choose $v(t) = t$. Then since $v \in X$ and $v(0) = 1$, the previous question implies that

$$0 = \left(\frac{\partial L}{\partial p}(1, x(1), x'(1)) + g'(x(1)) \right) v(1) = \frac{\partial L}{\partial p}(1, x(1), x'(1)) + g'(x(1)),$$

which is the required “transversality condition”.

(v) Let x be a minimizer of problem

$$(P) \quad \inf_{x \in X, x(0)=1} \int_0^1 (x'(t))^2 dt + (x(1))^2$$

It follows from previous questions (with $L(t, x, p) = p^2$, $g(x) = x^2$) that

$$\frac{d}{dt} (2x'(t)) = 0, \quad 2x'(1) + 2x(1) = 0,$$

Therefore $x(t) = At + B$ (since $x''(t) = 0$), with $B = 1$ (since $x(0) = 1$) and $2A + 2(1 + A) \times 1 = 0$ (since $2x'(1) + 2x(1) = 0$). We find $A = -1/2$ and therefore $x(t) = 1 - t/2$ for all $t \in [0, 1]$.

12. OPTIMAL CONTROL OF DIFFERENTIAL EQUATIONS: HAMILTON-JACOBI EQUATIONS & THE PONTRYAGIN MAXIMUM PRINCIPLE

In this lecture, we will investigate the optimal control of differential equations. In other words, we will be considering the following problem: let, for any $u : [0; T] \rightarrow \mathbb{R}^d$ satisfying regularity assumptions to be made precise later on, the solution y_u to

$$(12.1) \quad \begin{cases} y'_u = f(t, y_u(t), u(t)), \\ y_u(0) = 0. \end{cases}$$

The function u is called the control. The optimal control problem under consideration is the following: for a given time horizon T , solve

$$(12.2) \quad \min_{u \text{ admissible control}} \int_0^T \ell(t, y_u(t), u(t)) + g(y_u(T)).$$

We will be discussing the two main tools (that are actually closely related) used in dealing with this type of problems. From the get go, observe that (12.2) blends problems coming from dynamic programming (as we have an evolutionary process) and from calculus of variations, as we are dealing with optimisation variables that are functions. Just as in the case of calculus of variations, the existence of optimisers will not be discussed here.

Finally, we will always pretend that the underlying dynamics (12.1) is sufficiently well-posed; the proper setting is provided by the Picard-Lindelöf theorem (or Cauchy-Lipschitz theorem), which we recall here for the sake of completeness.

We assume that all controls u take value in a compact subset U of $\mathbb{R}^d$ and that $f : [0, T] \times \mathbb{R}^N \times U \rightarrow \mathbb{R}^N$ is globally continuous and uniformly Lipschitz with respect to the space variable: there exists $K > 0$ such that

$$\|f(t, x, u) - f(t, y, u)\| \leq K\|x - y\| \quad \forall (t, x, y, u) \in [0, T] \times \mathbb{R}^d \times \mathbb{R}^d \times U.$$

Theorem 12.1 (Picard-Lindelöf). *For any initial position $x_0 \in \mathbb{R}^d$ and for any measurable $u : [0; T] \rightarrow U$, there exists a unique solution of the ODE (12.1).*

12.1. The dynamic programming principle & Hamilton-Jacobi equations.

12.1.1. *The dynamic programming principle.* We begin with the first of these tools, the dynamic programming principle of Bellman. Just as, for time-discrete optimal control problems, the relevant function to study was the value function, which provided the optimal strategy in feedback form, we need to introduce the value function associated with (12.2). Namely, we define the function V as follows: for any $t \in [0; T]$ and any $x \in \mathbb{R}^d$, set

$$V(t, x) := \inf_{u \text{ admissible}} \int_t^T \ell(s, y_u(s), u(s)) + g(y_u(T)) \text{ subject to } \begin{cases} y'_u(s) = f(s, y_u(s), u(s)) \\ y_u(t) = x. \end{cases} \quad \text{in } [t; T],$$

Theorem 12.2. [Bellman dynamic programming principle] *We have, for all $0 \leq t_0 < t_1 \leq T$ and any $x_0 \in \mathbb{R}^d$*

$$V(t_0, x_0) = \inf_{u \text{ admissible}} \int_{t_0}^{t_1} \ell(t, y_u(t), u(t)) dt + V(t_1, y_u(t_1))$$

under the constraint that the pair $(y_u(\cdot), u(\cdot))$ solves

$$\begin{cases} y'_u(s) = f(s, y_u(s), u(s)) & \text{in } [t; T], \\ y_u(t_0) = x_0. \end{cases}$$

As was the case for the time discrete dynamic programming principle, the proof of this result is not very instructive; what matters is its interpretation, which is exactly the same in this time-continuous setting. But still, this does not give us a way to compute the value function, or the optimal strategy. This is where Hamilton-Jacobi equations come into play.

12.1.2. Link with Hamilton-Jacobi equations. How to use Theorem 12.2? Indeed, it only gives an implicit relation on the value function V . However, assume that the function V is $\mathcal{C}^1$ (in t and x) and fix a given (t_0, x_0) . Consider a small time perturbation h around t_0 . It follows from Theorem 12.2 that

$$V(t_0, x_0) = \inf_u \int_{t_0}^{t_0+h} \ell(t, y_u, u) + V(t_0 + h, y_u(t_0 + h)).$$

Now, up to some technicalities on the uniformity of Taylor expansion, do the Taylor expansions

$$\int_{t_0}^{t_0+h} \ell(t, y_u, u) = h\ell(t_0, y_u(t_0), u(t_0)) + o(h)$$

and

$$y_u(t_0 + h) = y_u(t_0) + y'_u(t_0)h + o(h) = y_u(t_0) + f(t_0, y_u(t_0), u(t_0))h + o(h).$$

It follows that

$$V(t_0+h, y_u(t_0+h)) = V(y_u(t_0)) + \partial_t V(t_0, x_0)h + h\langle \nabla V(y_u(t_0)), f(t_0, x_0, u(t_0)) \rangle + o(h),$$

still under some regularity assumption on V . We thus “deduce” from the dynamic programming principle that

$$V(t_0, x_0) = \inf_u h \{ \ell(t_0, x_0, u(t_0)) + \langle \nabla_x V(t_0, x_0), f(t_0, x_0, u_0) \rangle \} + h\partial_t V(t_0, x_0) + o(h).$$

If we divide by h and take the limit h to 0 we thus obtain

$$-\partial_t V - \inf_u \{ \ell(t_0, x_0, u(t_0)) + \langle \nabla_x V(t_0, x_0), f(t_0, x_0, u_0) \rangle \} = 0,$$

supplemented with the terminal condition $V(T, x) = g(x)$. If we introduce the Hamiltonian of the system as

$$(12.3) \quad H(t, x, \xi) := \sup_u \{ -\ell(t, x, u) - \langle \xi, f(t, x, u) \rangle \}$$

this rewrites as the backwards Hamilton-Jacobi equation

$$(12.4) \quad \begin{cases} -\partial_t V + H(t, x, \nabla_x V) = 0 & \text{in } [0; T], \\ V(T, x) = g(x), \end{cases}$$

and the optimal control is given by the u^* solving the optimisation problem (12.3).

We summarise the (correct and rigorous) result below:

Theorem 12.3. *Assume that the value function V is of class $\mathcal{C}^1$ in $(0, T) \times \mathbb{R}^d$ and $\mathcal{C}^0$ in $[0, T] \times \mathbb{R}^N$. Then V satisfies the Hamilton-Jacobi-Bellman (HJB) equation*

$$(12.5) \quad \begin{cases} -\partial_t V + H(t, x, \nabla_x V) = 0 & \text{in } [0; T], \\ V(T, x) = g(x), \end{cases}$$

and an optimal control is given by u^* maximising the function

$$u \mapsto -\ell(t, x, u) - \langle \nabla_x V(t, x), u \rangle.$$

This theorem is limited in scope, particularly given that we require V to be $\mathcal{C}^1$. Nevertheless, it remains true even when V is merely continuous (in which case one has to understand what “solving (12.5)” actually means). This is the field of viscosity solutions, which goes far beyond the scope of this class.

Now, a natural question is: if a function W solves (12.5), is it the value function associated with the optimal control problem (12.2)? If so, can we deduce the optimal control from W ? The answer is yes, provided the function W is smooth enough. This is the content of the following, so-called “verification” theorem:

Theorem 12.4. *Assume that*

- (1) $W : [0, T] \times \mathbb{R}^N \rightarrow \mathbb{R}$ is of class $\mathcal{C}^1$ in $(0, T) \times \mathbb{R}^d$ and $\mathcal{C}^0$ in $[0, T] \times \mathbb{R}^d$,
- (2) W satisfies (12.5),
- (3) There exists a continuous mapping $\tilde{u}^* : (0, T) \times \mathbb{R}^d \rightarrow U$ such that, for all $(t, x) \in (0, T) \times \mathbb{R}^d$,

$$-\langle \nabla_x W, f(t, x, \tilde{u}^*(t, x)) \rangle - \ell(t, x, \tilde{u}^*(t, x)) = H(t, x, \nabla_x W).$$

That is, $\tilde{u}^*(t, x)$ is a solution of the maximization problem defining $H(t, x, \nabla_x W)$.

Then $W = V$ and an optimal feedback is given by $\tilde{u}^*$.

12.1.3. An example of application. Let us give a simple example of application of the Hamilton-Jacobi-Bellman equation. We consider the problem (in one dimension)

$$(12.6) \quad \min_u \int_0^T (y_u^2 + u^2) + 2y_u^2(T) \text{ subject to } \begin{cases} y'_u(t) = u(t), \\ y_u(0) = x. \end{cases}$$

To solve this problem, we first compute the associated Hamiltonian of the system, which here writes

$$H(t, x, \xi) = \sup_{u \in \mathbb{R}} \{-u^2 - x^2 - \xi u\}.$$

The maximum is reached at $u^* = -\frac{\xi}{2}$ and thus

$$H(t, x, \xi) = -x^2 + \frac{\xi^2}{4}, \quad u^*(\xi) = -\frac{\xi}{2}.$$

If we assume that the value function V associated with (12.6) is of class $\mathcal{C}^1$ we deduce that V solves

$$(12.7) \quad \begin{cases} -\partial_t V + \frac{|\nabla_x V|^2}{4} = x^2 & \text{in } [0; T] \times \mathbb{R}, \\ V(T, x) = 2x^2. \end{cases}$$

It is not unreasonable to expect that the solution V of (12.7) writes $V(t, x) = \alpha(t)x^2$, where α solves

$$-\alpha'(t) + \alpha(t)^2 = 1, \quad \alpha(T) = 1.$$

If we can find a smooth solution α to this differential equation, the verification theorem will ensure that we have found the correct value function, and also provide us with the optimal control. Now, this ODE rewrites

$$-\alpha'(t) = 1 - \alpha^2, \quad \alpha(T) = 2$$

which can be solved explicitly as

$$\alpha(t) = \frac{1 + \beta e^{2t}}{1 - \beta e^{2t}}$$

with $\beta = \frac{1}{3}e^{-2T}$. In particular, α is smooth, and we thus deduce by the verification theorem that

$$u^*(t, x) = -\frac{1 + \beta e^{2t}}{1 - \beta e^{2t}}x.$$

12.2. The Pontryagin maximum principle.

12.2.1. *Derivation of the maximum principle.* We conclude these lectures with the Pontryagin maximum principle. This is a subtle result (arguably the subtlest of this class) that is foundational and central to optimal control theory. Rather than discussing a rigorous proof of it, we will mostly explain the Pontryagin maximum principle, and why it is to some extent reasonable. The basic idea is the following: when considering an optimal control problem of the form (12.2), one should see the dynamics

$$y'_u = f(t, y_u, u)$$

as an *infinity* of constraints. In other words, if we introduce, for any t , the function

$$\Psi_t(y, u) = -y' + f(t, y, u)$$

the problem (12.2) becomes:

$$(12.8) \quad \min_{y, u \in \mathcal{C}^1, \forall t > 0, \Psi_t(y, u) = 0, y(0) = x_0} \int_0^T \ell(t, y, u) + g(y(T)).$$

Seeing (12.8) as a constrained optimisation problem associated with an infinity of constraints, and reasoning by analogy with Theorem 5.1, it makes sense to introduce a family $\{\lambda(t)\}_{t \in (0; T]}$, each associated with the constraint Ψ_t , and to consider the Lagrangian

$$\mathcal{L}(y, u; \lambda) := \int_0^T \ell(t, y, u) + \langle \lambda(t), f(t, y, u) - y' \rangle + g(y(T)).$$

Assuming-formally-that λ is $\mathcal{C}^1$, this rewrites

$$\mathcal{L}(y, u; \lambda) = \int_0^T (\ell(t, y, u) + \langle \lambda'(t), y(t) \rangle + \langle \lambda(t), f(t, y, u) \rangle) \langle \lambda(T), y(T) \rangle + \langle \lambda(0), y(0) \rangle + g(y(T)).$$

Let us pretend that everything works as in constrained optimisation, so that $((y_u^*, u^*), \lambda^*)$ is a saddle point (for a certain Lagrange multiplier λ^*). The first order optimality conditions write

$$\frac{\partial \mathcal{L}}{\partial y} = 0, \quad \frac{\partial \mathcal{L}}{\partial u} = 0, \quad \frac{\partial \mathcal{L}}{\partial \lambda} = 0.$$

If we compute these partial derivatives, the last one gives

$$f(t, y, u) - y' = 0$$

so that $y = y_u$. The first one gives

$$\frac{\partial \ell}{\partial y}(t, y_u^*, u^*) + \left\langle \lambda^*(t), \frac{\partial f}{\partial y}(t, y_u^*, u^*) \right\rangle + (\lambda^*)'(t) = 0, \quad \lambda^*(T) = g'(y_u^*(T)).$$

The second one yields

$$\frac{\partial \ell}{\partial u}(t, y_u^*, u^*) + \left\langle \lambda^*(t), \frac{\partial f}{\partial u}(t, y_u^*, u^*) \right\rangle = 0.$$

This leads to introducing the Hamiltonian of the system

$$\mathcal{H}(t, y, u, \lambda) := \ell(t, y, u) + \langle \lambda(t), f(t, y, u) \rangle,$$

which allows to rewrite the previous equations in a more compact form as follows:

$$\frac{\partial \mathcal{H}}{\partial u}(t, y_u^*, u^*, \lambda^*) = 0, (\lambda^*)'(t) + \frac{\partial \mathcal{H}}{\partial y}(t, y_u^*, u^*, \lambda^*) = 0, \lambda^*(T) = g'(y_u^*(T)).$$

We finally observe that, as a consequence (then again really believing that y_u^*, u^*, λ^* is a saddle point) that

$$\mathcal{H}(t, y_u^*, u^*, \lambda^*) = \min_u \mathcal{H}(t, y_u^*, u, \lambda^*).$$

Now, for one last simplification of the problem: as $y = \frac{\partial \mathcal{H}}{\partial \lambda}(t, y, u)$, we obtain that

$$\lambda'(t) + \frac{\partial \mathcal{H}}{\partial \lambda}(t, y_u^*, u^*) = 0,$$

so that we have a nice, elegant (Hamiltonian) structure in the following form

$$(y_u^*)' = \frac{\partial \mathcal{H}}{\partial \lambda}, (\lambda^*)' = -\frac{\partial \mathcal{H}}{\partial y}.$$

All of these information are summarised in a rigorous manner in the following version of the Pontryagin maximum principle:

Theorem 12.5 (Pontryagin maximum principle). *Let*

$$H(t, x, \lambda) := \min_u \mathcal{H}(t, x, u, \lambda).$$

If (y_u^, u^*) is optimal in (12.8), there exists a function $\lambda^* \in \mathcal{C}^1([0; T]; \mathbb{R}^d)$ such that the pair (y_u^*, λ^*) solves*

$$\begin{cases} (y_u^*)' = \frac{\partial \mathcal{H}}{\partial \lambda}, \\ y_u^*(0) = x_0, \\ (\lambda^*)' = -\frac{\partial \mathcal{H}}{\partial y}, \\ \lambda^*(T) = g'(y_u^*(T)). \end{cases}$$

In addition

$$\mathcal{H}(t, y_u^*, u^*, \lambda^*) = \min_u \mathcal{H}(t, y_u^*, u, \lambda^*).$$

12.2.2. *An example of application.* For a given $x \in \mathbb{R}$, consider the optimal control problem:

$$\inf \left\{ \int_0^T u^2(s) ds + x(T), \text{ where } x(0) = x, \dot{x}(t) = x(t) + u(t), u(t) \in \mathbb{R} \right\}.$$

With our notations $f(t, x, u) = x + u$, $\ell(t, x, u) = u^2$, and $g(x) = x$. Therefore, by a straightforward computation

$$H(t, x, \lambda) = \lambda x - \frac{\lambda^2}{4}.$$

with maximizer $u^{opt}(t, x, \lambda) = -\frac{\lambda}{2}$. If u is an optimal control for an initial position x_0 at time 0, and y is the associated trajectory, we deduce that

$$\begin{cases} y'(t) = y(t) - \frac{\lambda(t)}{2} \\ \lambda'(t) = -\lambda(t) \\ y(0) = x_0, \lambda(T) = 1. \end{cases}$$

Thus, $\lambda(t) = e^{T-t}$ and upon integration we obtain

$$y(t) = x_0 e^t - \frac{1}{4} e^T (e^t - e^{-t}), \quad t \in [0, T].$$

EXERCISES FOR THE THIRD PART OF PART 4

Exercise 4.11. Consider the optimal control problem in $\mathbb{R}$:

$$V(t_0, x_0) := \inf_{u(\cdot)} g(x(T))$$

under the constraint that $u : [t_0, T] \rightarrow [-1, 1]$ is measurable and that $x(\cdot)$ is the unique solution of this ODE

$$\begin{cases} \dot{x}(t) = u(t), & t \in [t_0, T] \\ x(t_0) = x_0 \in \mathbb{R} \end{cases}$$

Function $g : \mathbb{R} \rightarrow \mathbb{R}$ is assumed continuous.

- (1) Show that the value function of the control problem is given by:

$$V(t, x) = \min_{y \in [x-(T-t), x+(T-t)]} g(y) \quad \forall (t, x) \in [0, T] \times \mathbb{R}.$$

- (2) Show that V is continuous, but not necessarily of class C^1 on $[0, T] \times \mathbb{R}$.
 (3) Assume that g is convex and of class C^1 . Show that V is of class C^1 on $[0, T] \times \mathbb{R}$ and satisfies the Hamilton-Jacobi equation:

$$\begin{cases} -\partial_t V(t, x) + \left| \frac{\partial V}{\partial x}(t, x) \right| = 0 & \text{in }]0, T[\times \mathbb{R} \\ V(T, x) = g(x) & \text{in } \mathbb{R}. \end{cases}$$

Exercise 4.12. Assuming that the problem has a solution, find it.

- (1) Let $T > 0$ and $x_0 \in \mathbb{R}$. Consider the following problem:

$$(P1) \quad \inf_{u(\cdot)} \int_0^T u^2(s) ds + \frac{x^2(T)}{2}, \quad x(0) = x_0, \quad \dot{x}(t) = u(t),$$

where the infimum is taken on all bounded, measurable controls $u : [0, T] \rightarrow \mathbb{R}$.

- (2) For fixed $T > 0$ and $x \in \mathbb{R}$, consider the optimal control problem:

$$\inf \left\{ \int_0^T e^{-s} u^2(s) ds + x(T), \quad \text{where } x(0) = x, \quad x'(t) = x(t) + u(t), \quad u(t) \in \mathbb{R} \right\},$$

where the infimum is taken on all bounded, measurable controls $u : [0, T] \rightarrow \mathbb{R}$.

- (3) Let $T > 0$, $x = (x_1, x_2) \in \mathbb{R}^2$ and $c = (c_1, c_2) \neq (0, 0)$. Consider the optimal control problem:

$$\inf_{u \in \mathcal{U}} \{c_1 y_1(T) + c_2 y_2(T)\},$$

where (y_1, y_2) is solution of

$$y'_k(t) = u_k(t), \quad y_k(0) = x_k, \quad k = 1, 2, \quad t \in [0, T],$$

and $\mathcal{U}$ is the set of control processes with value in $U := \{(u_1, u_2) \in \mathbb{R}^2 : u_1^2 + u_2^2 = 1\}$.

Exercise 4.13. Compute the value functions associated to the optimal control problems of Exercise 4.12 and find their solutions thanks to a verification theorem.

Exercise 4.14. Let $\lambda > 0$. For a fixed initial condition $x_0 \in \mathbb{R}^N$, consider the infinite horizon optimal control problem:

$$V(x_0) := \inf_{(x,u)} \int_0^{+\infty} e^{-\lambda t} L(x(t), u(t)) dt$$

under the constraint that $u : [0, +\infty[\rightarrow U$ is measurable and that the couple $(x(\cdot), u(\cdot))$ satisfies the ODE

$$\begin{cases} \dot{x}(t) = f(x(t), u(t)), & t \in [0, +\infty[\\ x(0) = x_0 \end{cases}$$

Define the system's Hamiltonian $H : \mathbb{R}^N \times \mathbb{R}^N \rightarrow \mathbb{R}$ by

$$H(x, p) := \sup_{u \in U} \{-\langle p, f(x, u) \rangle - L(x, u)\}.$$

The mapping $L : \mathbb{R}^N \times U \rightarrow \mathbb{R}$ is assumed continuous and bounded, and f satisfies usual conditions.

- (1) Show that V satisfies the dynamic programming principle: for all $t \geq 0$,

$$V(x_0) = \inf_{(x,u)} \int_0^t e^{-\lambda s} L(x(s), u(s)) ds + e^{-\lambda t} V(x(t))$$

- (2) Assume that V is of class C^1 . Show that V is solution of the Hamilton-Jacobi equation

$$\lambda V(x) + H(x, \frac{\partial V}{\partial x}) = 0, \quad x \in \mathbb{R}^N.$$

- (3) Conversely, assume that there exists a bounded function $W : \mathbb{R}^N \rightarrow \mathbb{R}$, of class C^1 , satisfying

$$\lambda W(x) + H\left(x, \frac{\partial W}{\partial x}\right) = 0, \quad x \in \mathbb{R}^N.$$

Assume that there exists a continuous feedback $\tilde{u}^* : \mathbb{R}^N \rightarrow U$ such that

$$-\langle \frac{\partial W}{\partial x}, f(x, \tilde{u}^*(x)) \rangle - L(x, \tilde{u}^*(x)) = H(x, \frac{\partial W}{\partial x}), \quad x \in \mathbb{R}^N.$$

Show that $W = V$ and that $\tilde{u}^*$ is an optimal feedback.

SOLUTIONS FOR THE THIRD PART OF PART 4

Solution 4.11. (1) Let $u : [t_0, T] \rightarrow [-1, 1]$ be a measurable mapping and $t \rightarrow x(t)$ a solution of the differential equation

$$\begin{cases} \dot{x}(t) = u(t), & t \in [t_0, T] \\ x(t_0) = x_0 \in \mathbb{R} \end{cases}$$

Then

$$|x(T) - x_0| = \left| \int_0^T x'(t) dt \right| = \left| \int_0^T u(t) dt \right| \leq \int_0^T |x'(t)| dt \leq T - t_0.$$

therefore $x(T) \in [x_0 - (T - t_0), x_0 + (T - t_0)]$ and

$$V(t_0, x_0) = \inf_{u(\cdot)} g(x(T)) \geq \inf_{y \in [x_0 - (T - t_0), x_0 + (T - t_0)]} g(y).$$

Conversely, let $y \in [x_0 - (T - t_0), x_0 + (T - t_0)]$. Define the constant control $u(t) = \bar{u} = (y - x_0)/(T - t_0)$ for all $t \in [t_0, T]$. Then $u(t) \in [-1, 1]$ for all $t \in [0, T]$ since $y \in [x_0 - (T - t_0), x_0 + (T - t_0)]$ and the solution of

$$\begin{cases} \dot{x}(t) = u(t), & t \in [t_0, T] \\ x(t_0) = x_0 \in \mathbb{R} \end{cases}$$

is given by $x(t) = x_0 + (t - t_0)\bar{u}$ for $t \in [t_0, T]$. Therefore $x(T) = x_0 + (T - t_0)\bar{u} = y$ by the choice of $\bar{u}$ and we have $V(t_0, x_0) \leq g(x(T)) = g(y)$. Since $y \in [x_0 - (T - t_0), x_0 + (T - t_0)]$ is arbitrary, we deduce that

$$V(t_0, x_0) \leq \inf_{y \in [x_0 - (T - t_0), x_0 + (T - t_0)]} g(y).$$

hence the equality.

- (2) Assume by contradiction that V is not continuous at a point (t_0, x_0) . Then there exists $\varepsilon > 0$ and (t_n, x_n) which goes to (t_0, x_0) and such that $|V(t_n, x_n) - V(t_0, x_0)| \geq \varepsilon$ for all n . By the previous question and the continuity of g , there exists $y_n \in [x_n - (T - t_n), x_n + (T - t_n)]$ such that $V(t_n, x_n) = g(y_n)$. The sequence (y_n) being bounded, there exists a subsequence (y_{n_k}) which converges to an element y of $[x_0 - (T - t_0), x_0 + (T - t_0)]$. Thus, since $|V(t_n, x_n) - V(t_0, x_0)| \geq \varepsilon$, $|g(y) - V(t_0, x_0)| \geq \varepsilon$. Since moreover $V(t_0, x_0) \leq g(y)$, we actually have $V(t_0, x_0) \leq g(y) - \varepsilon$.

Let $\tilde{y} \in [x_0 - (T - t_0), x_0 + (T - t_0)]$ be such that $V(t_0, x_0) = g(\tilde{y})$. We claim that there exists $\tilde{y}_n$ in $[x_n - (T - t_n), x_n + (T - t_n)]$ which converges to $\tilde{y}$: indeed, if $\tilde{y}$ is in $(x_0 - (T - t_0), x_0 + (T - t_0))$, then $\tilde{y}$ is in $[x_n - (T - t_n), x_n + (T - t_n)]$ for n large enough. Otherwise, either $\tilde{y} = x_0 - (T - t_0)$, and the sequence $(\tilde{y}_n = x_n - (T - t_n))$ is suitable, or $\tilde{y} = x_0 + (T - t_0)$, and the sequence $(\tilde{y}_n = x_n + (T - t_n))$ is suitable. In all cases, we showed that there exists $\tilde{y}_n$ in $[x_n - (T - t_n), x_n + (T - t_n)]$ which converges to $\tilde{y}$. Then

$$V(t_{n_k}, x_{n_k}) \leq g(y_{n_k}),$$

and finally

$$g(y) = \lim_k V(t_{n_k}, x_{n_k}) \leq \lim_k V(t_{n_k}, x_{n_k})g(\tilde{y}_{n_k}) = g(\tilde{y}) = V(t_0, x_0) \leq g(y) - \varepsilon.$$

This is a contradiction, therefore V is continuous.

Let us show by an example that V need not be of class C^1 . Assume that g is an even function of class C^1 and that $g' < 0$ on $]0, +\infty[$. Then

$$V(t, x) = \min_{y \in [x-(T-t), x+(T-t)]} g(y) = \begin{cases} g(x + (T-t)) & \text{if } x \geq 0 \\ g(x - (T-t)) & \text{if } x \leq 0 \end{cases}$$

Indeed, if $x \geq 0$, either $x - (T-t) \geq 0$ and then for all $y \in [x - (T-t), x + (T-t)]$ we have $y \geq 0$ and $g(y) \geq g(x + (T-t))$ since g is decreasing on $\mathbb{R}_+$, or $x - (T-t) < 0$ and then, since g is increasing on $[x - (T-t), 0]$ and decreasing on $[0, x + (T-t)]$, we have for all $y \in [x - (T-t), x + (T-t)]$ that $g(y) \leq \min\{g(x - (T-t)), g(x + (T-t))\}$, with $g(x - (T-t)) = g(-x + T - t) > g(x + T - t)$ since g is even and decreasing on $\mathbb{R}_+$. Therefore, if $x \geq 0$, then $V(t, x) = g(x + T - t)$. Similarly, if $x \leq 0$, then $V(t, x) = g(x - (T-t))$.

Let us check that $x \rightarrow V(t, x)$ is not differentiable at 0 if $t < T$. We have

$$\lim_{h \rightarrow 0^+} \frac{V(t, h) - V(t, 0)}{h} = \lim_{h \rightarrow 0^+} \frac{g(h + T - t) - g(T - t)}{h} = g'(T - t) < 0,$$

while

$$\lim_{h \rightarrow 0^-} \frac{V(t, h) - V(t, 0)}{h} = \lim_{h \rightarrow 0^-} \frac{g(h - (T-t)) - g(-(T-t))}{h} = g'(-(T-t)) > 0,$$

which proves that the right derivative differs from the left derivative, and therefore that V is not differentiable at (t, x) .

- (3) Let us first show that V is convex. Let $(t_1, x_1), (t_2, x_2) \in [0, T] \times \mathbb{R}$ and $\lambda \in [0, 1]$. There exists $y_1 \in [x_1 - (T - t_1), x_1 + (T - t_1)]$ and $y_2 \in [x_2 - (T - t_2), x_2 + (T - t_2)]$ such that $V(t_1, x_1) = g(y_1)$ and $V(t_2, x_2) = g(y_2)$. Let $(t, x) = (1 - \lambda)(t_1, x_1) + \lambda(t_2, x_2)$ and $y = (1 - \lambda)y_1 + \lambda y_2$. Then $y \in [x - (T - t), x + (T - t)]$ and therefore

$$V(t, x) \leq g(y) \leq (1 - \lambda)g(y_1) + \lambda g(y_2) = (1 - \lambda)V(t_1, x_1) + \lambda V(t_2, x_2),$$

which proves the convexity of V .

To show that V is of class C^1 and satisfies the Hamilton-Jacobi equation, let us consider 3 cases : if g' is always positive, if g' is always negative, or if g' takes the value 0.

In the first case (where $g' > 0$), the function g is increasing and therefore

$$V(t, x) = \min_{y \in [x-(T-t), x+(T-t)]} g(y) = g(x - (T-t)) \text{ for all } (t, x) \in [0, T] \times \mathbb{R}.$$

Then

$$-\partial_t V(t, x) + \left| \frac{\partial V}{\partial x}(t, x) \right| = -g'(x - (T-t)) + |g'(x - (T-t))| = 0,$$

since $g' > 0$. The case where $g' < 0$ is similar, since then $V(t, x) = g(x + (T-t))$ for all (t, x) .

Let us now consider the case where g' vanishes at a point x_0 . Then, since g is convex, g' is non-decreasing. Therefore, $g' \leq 0$ on $(-\infty, x_0]$ and $g' \geq 0$ on $[x_0, +\infty)$. Therefore g is non-increasing on $(-\infty, x_0]$ and non-decreasing on $[x_0, +\infty)$ (in particular, g has a global minimum at x_0).

Therefore,

$$V(t, x) = \min_{y \in [x-(T-t), x+(T-t)]} g(y) = \begin{cases} g(x + (T-t)) & \text{if } x + (T-t) \leq x_0 \\ g(x - (T-t)) & \text{if } x - (T-t) \geq x_0 \\ g(x_0) & \text{if } x_0 \in [x - (T-t), x + (T-t)] \end{cases}$$

In particular

$$\frac{\partial V}{\partial t}(t, x) = \begin{cases} -g'(x + (T-t)) & \text{if } x + (T-t) < x_0 \\ g'(x - (T-t)) & \text{if } x - (T-t) > x_0 \\ 0 & \text{if } x_0 \in (x - (T-t), x + (T-t)) \end{cases}$$

and

$$\frac{\partial V}{\partial x}(t, x) = \begin{cases} g'(x + (T-t)) \leq 0 & \text{if } x + (T-t) < x_0 \\ g'(x - (T-t)) \geq 0 & \text{if } x - (T-t) > x_0 \\ 0 & \text{if } x_0 \in (x - (T-t), x + (T-t)) \end{cases}$$

Since $g'(x_0) = 0$, the partial derivatives of V can be continuously extended to $[0, T] \times \mathbb{R}$ and therefore V is of class C^1 . We then check as above that V satisfies the Hamilton-Jacobi equation everywhere.

Solution 4.12. (1) The system's Hamiltonian is:

$$H(t, x, p) = \sup_{u \in \mathbb{R}} \{-up - u^2\} = \frac{p^2}{4} \quad \forall (t, x, p) \in [0, T] \times \mathbb{R} \times \mathbb{R},$$

(with maximizer $u^{\text{opt}}(t, x, p) = -p/2$) while the terminal condition is $g(x) = x^2/2$. Therefore, it follows from the maximum principle that if (x, u) is optimal, then there exists p such that

$$\begin{cases} \dot{x}(t) = -\frac{\partial H}{\partial p}(t, x(t), p(t)) = -\frac{1}{2}p(t) \\ \dot{p}(t) = \frac{\partial H}{\partial x}(t, x(t), p(t)) = 0 \\ x(0) = x_0, p(T) = \frac{\partial g}{\partial x}(x(T)) = x(T) \end{cases}$$

We thus find $p(t) = p(T) = x(T)$ and $x(t) = x_0 - x(T)t/2$. To conclude, note that for $t = T$, we get $x(T) = x_0 - x(T)T/2$, hence $x(T) = 2x_0/(2+T)$. Therefore, if (x, u) is optimal, then $x(t) = x_0(1 - t/(2+T))$ and $u(t) = -p(t)/2 = -x(t)/2 = -x_0/(2+T)$.

(2) The Hamiltonian is:

$$H(t, x, p) = \sup_{u \in \mathbb{R}} \{-p(x+u) - e^{-t}u^2\} = -px + \frac{e^t}{4}p^2 \quad \forall (t, x, p) \in [0, T] \times \mathbb{R} \times \mathbb{R},$$

(with maximizer $u^{\text{opt}}(t, x, p) = -e^t p/2$) and the terminal cost is $g(x) = x$. According to the maximum principle, if (x, u) is solution, then there exists p such that

$$\begin{cases} \dot{x}(t) = -\frac{\partial H}{\partial p}(t, x(t), p(t)) = x(t) - \frac{e^t}{2}p(t) \\ \dot{p}(t) = \frac{\partial H}{\partial x}(t, x(t), p(t)) = -p(t) \\ x(0) = x, p(T) = \frac{\partial g}{\partial x}(x(T)) = 1 \end{cases}$$

We deduce that $p(t) = e^{(T-t)}$ and therefore

$$\frac{d}{dt} (e^{-t}x(t)) = -\frac{1}{2}e^{(T-t)},$$

hence

$$x(t) = x_0 e^t + \frac{1}{2}(e^T - e^{T+t}).$$

The optimal control is then given by $u(t) = u^{opt}(t, x(t), p(t)) = -e^t/2$.

(3) The Hamiltonian is

$$H(t, x, p) = \sup_{u \in U} \{-p_1 u_1 - p_2 u_2\} = \|p\| \quad \forall (t, x, p) \in [0, T] \times \mathbb{R}^2 \times \mathbb{R}^2,$$

(with maximizer $u^{opt}(t, x, p) = -p/\|p\|$ if p is non-zero, and u^{opt} arbitrary otherwise) and the terminal cost is $g(x) = c_1 x_1 + c_2 x_2$. It follows from the maximum principle that if (x, u) is solution then there exists p such that

$$\begin{cases} \dot{x}_1(t) = -\frac{\partial H}{\partial p_1}(t, x(t), p(t)) = -\frac{p_1(t)}{\|p(t)\|} \\ \dot{x}_2(t) = -\frac{\partial H}{\partial p_2}(t, x(t), p(t)) = -\frac{p_2(t)}{\|p(t)\|} \\ \dot{p}_1(t) = \frac{\partial H}{\partial x_1}(t, x(t), p(t)) = 0 \\ \dot{p}_2(t) = \frac{\partial H}{\partial x_2}(t, x(t), p(t)) = 0 \\ x(0) = x, p_1(T) = \frac{\partial g}{\partial x_1}(x(T)) = c_1, p_2(T) = \frac{\partial g}{\partial x_2}(x(T)) = c_2 \end{cases}$$

We deduce that $p(t) = c$ for all $t \in [0, T]$ and therefore $\dot{x}(t) = x - (c/\|c\|)t$. Moreover the optimal control is given by $u(t) = u^{opt}(t, x(t), p(t)) = -c/\|c\|$ for all $t \in [0, T]$.

Solution 4.13. (1) We make the same kind of computation as in Exercise 4.12 but for an arbitrary initial condition (s, x_0) . We find (for instance by replacing T and t by $T - s$ and $t - s$ respectively, which is valid since the Hamiltonian is independent of time) that if (x, u) is an optimal solution, then $x(t) = x_0(1 - (t - s)/(2 + T - s)) = x_0(2 + T - t)/(2 + T - s)$ and $u(t) = x_0/(2 + T - s)$. We expect that the value function equals

$$\begin{aligned} V(s, x_0) &= \int_s^T u^2(t) dt + x^2(T)/2 = \frac{x_0^2}{(2 + T - s)^2} (T - s) + \frac{2x_0^2}{(2 + T - s)^2} \\ &= \frac{x_0^2}{2 + T - s}. \end{aligned}$$

Function V is a C^∞ (hence C^1) solution of the Hamilton-Jacobi equation

$$-\frac{\partial V}{\partial t}(t, x) + \frac{1}{4} \left| \frac{\partial V}{\partial x}(t, x) \right|^2 = 0 \text{ if } (t, x) \in]0, T[\times \mathbb{R} \quad V(T, x) = \frac{x^2}{2} \text{ if } x \in \mathbb{R}.$$

From the verification theorem, we deduce that $\tilde{u}^*(t, x) = -\frac{\partial V}{\partial x}(t, x)/2 = \frac{x}{2 + T - t}$ (which is the u that maximizes the pre-Hamiltonian at $(t, x, \frac{\partial V}{\partial x}(t, x))$, see exercise 4.12) is an optimal feedback. Therefore, we find that the optimal trajectory starting from x_0 at time $t_0 = 0$ satisfies

$$\dot{x}(t) = \tilde{u}^*(t, x(t)) = \frac{x(t)}{2 + T - t}, \quad x(0) = x_0,$$

and is therefore equal to the solution $x(t) = x_0(2 + T - t)/(2 + T)$ given by the necessary optimality conditions.

(2) We make the same kind of computation as in Exercise 4.12 for an arbitrary initial condition. By the maximum principle, if an optimal trajectory x for the problem with initial condition (s, x_0) exists, then there exists p such that

$$\begin{cases} \dot{x}(t) = -\frac{\partial H}{\partial p}(t, x(t), p(t)) = x(t) - \frac{e^t}{2} p(t) \\ \dot{p}(t) = \frac{\partial H}{\partial x}(t, x(t), p(t)) = -p(t) \\ x(s) = x_0; p(T) = \frac{\partial g}{\partial x}(x(T)) = 1 \end{cases}$$

We deduce that $p(t) = e^{(T-t)}$ and therefore

$$\frac{d}{dt}(e^{-t}x(t)) = -\frac{1}{2}e^{(T-t)},$$

Let

$$x(t) = x_0 e^{t-s} + \frac{1}{2}(e^T - e^{T+t-s}).$$

The optimal control is then given by $u(t) = -e^T/2$. We expect that the value function be given by

$$\begin{aligned} V(s, x_0) &= \int_s^T e^{-t} u^2(t) dt + x(T) = \frac{1}{4} \int_s^T e^{-t} e^{2T} dt + x_0 e^{T-s} + \frac{1}{2}(e^T - e^{2T-s}) \\ &= x_0 e^{T-s} + \frac{1}{4}(e^T - e^{2T-s}). \end{aligned}$$

We then check easily that V is a C^∞ (hence C^1) solution of the Hamilton-Jacobi equation

$$-\frac{\partial V}{\partial t}(t, x) - x \frac{\partial V}{\partial x}(t, x) + \frac{e^t}{4} \left| \frac{\partial V}{\partial x}(t, x) \right|^2 = 0 \text{ si } (t, x) \in]0, T[\times \mathbb{R} \quad V(T, x) = x \text{ si } x \in \mathbb{R}.$$

By the verification theorem, we deduce that $\bar{u}^*(t, x) = -e^t \frac{\partial V}{\partial x}(t, x)/2 = -e^T/2$ (which is the u that maximizes the pre-Hamiltonian at $(t, x, \frac{\partial V}{\partial x}(t, x))$, see exercise 4.12) is an optimal feedback. We then find that the optimal trajectory with initial condition x_0 at time $t_0 = 0$ satisfies

$$\dot{x}(t) = x(t) + u^*(t, x(t)) = x(t) - e^T/2, \quad x(0) = x_0,$$

and is therefore equal to the solution $x(t) = x_0 e^t + \frac{1}{2}(e^T - e^{T+t})$ given by the necessary optimality conditions.

- (3) Again, we do the same kind of computation as in Exercice 4.12 but for a generic initial condition (s, x_0) . Since the Hamiltonian is time-independent, it actually suffices to replace everywhere t by $t-s$ and T by $T-s$. If (x, u) is an optimal solution for the initial condition $(s, x_0) \in [0, T] \times \mathbb{R}^2$, we have $x(t) = x_0 - (c/\|c\|)(t-s)$ and the optimal control is given by $u(t) = -c/\|c\|$ for all $t \in [0, T]$. Therefore, we expect that the value function be given by

$$V(s, x_0) = c_1 x_1(T) + c_2 x_2(T) = c_1 x_{0,1} + c_2 x_{0,2} - \|c\|(T-s),$$

which is a C^∞ (hence C^1) solution of the Hamilton-Jacobi equation

$$-\frac{\partial V}{\partial t}(t, x) + \left\| \frac{\partial V}{\partial x}(t, x) \right\| = 0 \text{ if } (t, x) \in]0, T[\times \mathbb{R} \quad V(T, x) = x \text{ if } x \in \mathbb{R}.$$

The verification theorem thus says that the optimal trajectory starting from $(0, x_0)$ satisfies

$$\dot{x}(t) = -\frac{\partial V}{\partial x}(t, x) / \left\| \frac{\partial V}{\partial x}(t, x) \right\| = -c/\|c\|, \quad t \in [0, T], \quad x(0) = x_0,$$

which is the solution $x(t) = x_0 - (c/\|c\|)t$ given by the necessary optimality conditions.

Solution 4.14. (1) Let us show that V satisfies the dynamic programming principle : for $t \geq 0$, let

$$W(x_0) = \inf_{(x,u)} \int_0^t e^{-\lambda s} L(x(s), u(s)) ds + e^{-\lambda t} V(x(t))$$

and let us show that $W(x_0) = V(x_0)$.

Let $\varepsilon > 0$ and let u be ε -optimal for $V(x_0)$:

$$(12.9) \quad \varepsilon + V(x_0) \geq \int_0^{+\infty} e^{-\lambda s} L(x(s), u(s)) \, ds$$

where $(x(\cdot), u(\cdot))$ satisfies the ODE

$$\begin{cases} \dot{x}(s) = f(x(s), u(s)), & s \in [0, +\infty[\\ x(0) = x_0 \end{cases}$$

Let $\tilde{x}(s) = x(t+s)$, $\tilde{u}(s) = u(t+s)$ for all $s \geq 0$. Then $(\tilde{x}(\cdot), \tilde{u}(\cdot))$ satisfies the ODE

$$\begin{cases} \dot{\tilde{x}}(s) = f(\tilde{x}(s), \tilde{u}(s)), & s \in [0, +\infty[\\ \tilde{x}(0) = x(t) \end{cases}$$

as f does not depend on time. In particular,

$$V(x(t)) \leq \int_0^{+\infty} e^{-\lambda s} L(\tilde{x}(s), \tilde{u}(s)) \, ds.$$

Then

$$\begin{aligned} \int_0^{+\infty} e^{-\lambda s} L(x(s), u(s)) \, ds &= \int_0^t e^{-\lambda s} L(x(s), u(s)) \, ds + \int_t^{+\infty} e^{-\lambda s} L(x(s), u(s)) \, ds \\ &= \int_0^t e^{-\lambda s} L(x(s), u(s)) \, ds + \int_0^{+\infty} e^{-\lambda(t+s)} L(x(t+s), u(t+s)) \, ds \\ &= \int_0^t e^{-\lambda s} L(x(s), u(s)) \, ds + e^{-\lambda t} \int_0^{+\infty} e^{-\lambda s} L(\tilde{x}(s), \tilde{u}(s)) \, ds \\ &\geq \int_0^t e^{-\lambda s} L(x(s), u(s)) \, ds + V(x(t)) \geq W(x_0), \end{aligned}$$

where the last inequality comes from the definition of $W(x_0)$. By Eq. (12.9), we thus get $\varepsilon + V(x_0) \geq W(x_0)$. Since ε is arbitrary, it follows that $V(x_0) \geq W(x_0)$.

Conversely, let $\varepsilon > 0$ and let u_1 be an ε -optimal control for $W(x_0)$:

$$\varepsilon + W(x_0) \geq \int_0^t e^{-\lambda s} L(x_1(s), u_1(s)) \, ds + e^{-\lambda t} V(x_1(t)),$$

where (x_1, u_1) satisfies the ODE with $x_1(0) = x_0$. Let u_2 be an ε -optimal control for $V(x_1(t))$:

$$\varepsilon + V(x_1(t)) \geq \int_0^{+\infty} e^{-\lambda s} L(x_2(s), u_2(s)) \, ds$$

where (x_2, u_2) satisfies the ODE with $x_2(0) = x_1(t)$. Let $(x(s), u(s)) = (x_1(s), u_1(s))$ if $s \in [0, t]$ and $(x(s), u(s)) = (x_2(s-t), u_2(s-t))$ if $s \geq t$. It is then easy to check that (x, u) is solution of the ODE with $x(0) = x_0$. Then

$$\begin{aligned} \varepsilon + W(x_0) &\geq \int_0^t e^{-\lambda s} L(x_1(s), u_1(s)) \, ds + e^{-\lambda t} V(x_1(t)) \\ &\geq \int_0^t e^{-\lambda s} L(x_1(s), u_1(s)) \, ds + e^{-\lambda t} \left(-\varepsilon + \int_0^{+\infty} e^{-\lambda s} L(x_2(s), u_2(s)) \, ds \right) \end{aligned}$$

$$\begin{aligned}
 &\geq \int_0^t e^{-\lambda s} L(x(s), u(s)) ds + \int_0^{+\infty} e^{-\lambda(t+s)} L(x(t+s), u_2(t+s)) ds - \varepsilon e^{-\lambda t} \\
 &\geq \int_0^{+\infty} e^{-\lambda s} L(x(s), u(s)) ds - \varepsilon e^{-\lambda t} \geq V(x_0) - \varepsilon e^{-\lambda t}.
 \end{aligned}$$

Since ε is arbitrary, we deduce that $W(x_0) \geq V(x_0)$ and therefore that $W(x_0) = V(x_0)$.

- (2) Fix $u \in U$ and choose as control $u(s) = u$ for all $s \geq 0$. Let (x, u) be the solution of the ODE with $x(0) = x_0$. By dynamic programming, we get

$$\begin{aligned}
 V(x_0) &\leq \int_0^t e^{-\lambda s} L(x(s), u) ds + e^{-\lambda t} V(x(t)) \\
 &= V(x_0) + \int_0^t e^{-\lambda s} \left(L(x(s), u) - \lambda V(x(s)) + \left\langle \frac{\partial V}{\partial x}(s, x(s)), \dot{x}(s) \right\rangle \right) ds.
 \end{aligned}$$

Simplifying by $V(x_0)$, then dividing by $t > 0$ and letting t go to 0^+ , we get

$$0 \leq L(x_0, u) - \lambda V(x_0) + \left\langle \frac{\partial V}{\partial x}(s, x_0), f(x_0, u) \right\rangle$$

since $\dot{x}(s) = f(x(s), u)$. Since $u \in U$ is arbitrary, we obtain as in the notes of the course:

$$\lambda V(x_0) + \sup_{u \in U} \{-L(x_0, u) - \left\langle \frac{\partial V}{\partial x}(s, x_0), f(x_0, u) \right\rangle\} \leq 0,$$

that is,

$$\lambda V(x_0) + H(x_0, \frac{\partial V}{\partial x}(x_0)) \leq 0.$$

We show the opposite inequality by contradiction. Assume that

$$\lambda V(x_0) + H(x_0, \frac{\partial V}{\partial x}(x_0)) < 0.$$

Then by continuity and definition of H , there exists $\delta > 0$ and $r > 0$ such that

$$(12.10) \quad \lambda V(x) - L(x, u) - \left\langle \frac{\partial V}{\partial x}(x), f(x, u) \right\rangle \leq -\delta$$

for all $x \in \mathbb{R}^N$ with $|x - x_0| < r$ and for all $u \in U$. Fix $h > 0$ and consider u_h an h^2 -optimal control for the dynamic programming principle :

$$V(x_0) + h^2 \geq \int_0^h e^{-\lambda s} L(x_h(s), u_h(s)) ds + e^{-\lambda h} V(x_h(h))$$

where (x_u, u_h) satisfies the ODE with $x_h(0) = x_0$. We then have

$$V(x_0) + h^2 \geq V(x_0) + \int_0^h e^{-\lambda s} \left(L(x_h(s), u_h(s)) ds - \lambda V(x_h(s)) + \left\langle \frac{\partial V}{\partial x}(x_h(s)), \dot{x}_h(s) \right\rangle \right) ds$$

Since (x_u, u_h) satisfies the ODE with $x_h(0) = x_0$, there exists $h_0 > 0$ such that, for $h \in]0, h_0[$, $|x_h(s) - x_0| < r$ for all $s \in [0, h]$. We may then use (12.10) to deduce that

$$\begin{aligned}
 h^2 &\geq \int_0^h e^{-\lambda s} \left(L(x_h(s), u_h(s)) ds - \lambda V(x_h(s)) + \left\langle \frac{\partial V}{\partial x}(x_h(s)), \dot{x}_h(s) \right\rangle \right) ds \\
 &\geq \int_0^h e^{-\lambda s} \delta ds.
 \end{aligned}$$

Dividing by $h > 0$ and letting h go to 0 leads to $0 \geq \delta$, a contradiction. We deduce that

$$\lambda V(x_0) + H(x_0, \frac{\partial V}{\partial x}(x_0)) = 0.$$

- (3) Let u be an arbitrary control and let (x, u) be the solution of the ODE with $x(0) = x_0$. We have

$$\begin{aligned} \frac{d}{dt} e^{-\lambda t} W(x(t)) &= e^{-\lambda t} \left(-\lambda W(x(t)) + \left\langle \frac{\partial W}{\partial x}(x(t)), \dot{x}(t) \right\rangle \right) \\ &= e^{-\lambda t} \left(-\lambda W(x(t)) + \left\langle \frac{\partial W}{\partial x}(x(t)), f(x(t), u(t)) \right\rangle \right) \end{aligned}$$

since W satisfies the Hamilton-Jacobi equation, we have for all $x \in \mathbb{R}^N$ and all $u \in U$,

$$\lambda W(x) - L(x, u) - \left\langle \frac{\partial W}{\partial x}(x), f(x, u) \right\rangle \leq 0.$$

Therefore

$$\frac{d}{dt} e^{-\lambda t} W(x(t)) \geq -e^{-\lambda t} L(x(t), u(t)).$$

We integrate this inequality between $t = 0$ and $+\infty$ and, using that W is bounded, we get that

$$-W(x_0) \geq - \int_0^{+\infty} e^{-\lambda t} L(x(t), u(t)) dt.$$

Changing the signs and using that u is an arbitrary control leads to :

$$W(x_0) \leq \inf_{(x, u)} \int_0^{+\infty} e^{-\lambda t} L(x(t), u(t)) dt. = V(x_0).$$

To get the reverse inequality, consider a solution $\bar{x}$ of

$$\begin{cases} \dot{\bar{x}}(s) = f(\bar{x}(s), \tilde{u}^*(\bar{x}(s))), & s \in [0, +\infty[\\ \bar{x}(0) = x_0 \end{cases}$$

and let $\bar{u}(s) = \tilde{u}^*(\bar{x}(s))$. Then $(\bar{x}, \bar{u})$ is solution of the ODE with $\bar{x}(0) = x_0$. We proceed as above, noting that for all t ,

$$\begin{aligned} \lambda W(\bar{x}(t)) - L(\bar{x}(t), \bar{u}(t)) - \left\langle \frac{\partial W}{\partial x}(\bar{x}(t)), f(\bar{x}(t), \bar{u}(t)) \right\rangle \\ = \lambda W(\bar{x}(t)) - L(\bar{x}(t), \tilde{u}^*(\bar{x}(t))) - \left\langle \frac{\partial W}{\partial x}(\bar{x}(t)), f(\bar{x}(t), \tilde{u}^*(\bar{x}(t))) \right\rangle = 0, \end{aligned}$$

since W satisfies the Hamilton-Jacobi equation. Therefore

$$\frac{d}{dt} e^{-\lambda t} W(\bar{x}(t)) = -e^{-\lambda t} L(\bar{x}(t), \bar{u}(t)),$$

which implies, after integrating, that

$$-W(x_0) = - \int_0^{+\infty} e^{-\lambda t} L(\bar{x}(t), \bar{u}(t)) dt.$$

It follows that

$$W(x_0) = \int_0^{+\infty} e^{-\lambda t} L(\bar{x}(t), \bar{u}(t)) dt \geq V(x_0) \geq W(x_0),$$

where the first inequality come from the definition of V and the second from the previous step. We conclude that there is equality everywhere, and therefore that $\bar{u}$ is an optimal control. It follows that $\tilde{u}^$ is an optimal feedback.*

MIDTERMS & EXAMS

Midterm (October 2025). Duration: 2 hours. No documents allowed. The presentation of your paper will be taken into account in the grading. You can write in French or in English, but be consistent (*i.e.* do not solve one exercise in English, and the following one in French). The number of points per question might be modified later on.

Notations: $\|\cdot\|$ stands for the euclidean norm on $\mathbb{R}^d$.

- Exercise 4.15** (Basics). (1) (0.5 points) Let $f \in \mathcal{C}^1(\mathbb{R}^d; \mathbb{R})$ be convex. Is it true that f has a unique critical point?
- (2) (1 point) Let $f \in \mathcal{C}^2(\mathbb{R}^d; \mathbb{R})$ be α -strongly convex for some $\alpha > 0$. Does f admit a minimiser? Show it, or provide a counter-example.
- (3) (2,5 points) Study the optimisation problem (*i.e.* show the existence of optimisers, compute and classify the critical points)

$$\min_{(x,y) \in \mathbb{R}^2} x^4 + y^4 - 2(x-y)^2.$$

- Solution 4.15** (Basics). (1) (0.5 points) No, just take a constant function f .
- (2) (1 point) A strongly convex function f is coercive, so that it admits a minimiser. Indeed, by a Taylor expansion, for some $\alpha > 0$ we have

$$f(x) \geq f(0) + \langle \nabla f(0), x \rangle + \frac{\alpha}{2} \|x\|^2$$

and the Cauchy-Schwarz inequality allows to conclude.

- (3) On the one hand, using the inequality

$$(x-y)^2 \leq C(x^2 + y^2)$$

for some constant C as well as the fact that for any $\alpha > 0$ there exist $\beta > 0$ such that

$$x^4 + y^4 \geq \alpha(x^2 + y^2) - \beta$$

we deduce that f is coercive. In particular, it admits a minimiser. Furthermore, the criticality condition $\nabla f(x, y) = 0$ rewrites

$$\begin{cases} 4x^3 - 4(x-y) = 0, \\ 4y^3 + 4(x-y) = 0. \end{cases}$$

We deduce that $x = -y$ and that

$$x^3 = x, \quad y^3 = y.$$

Thus, we obtain as possible critical points $(0, 0)$, $(1, -1)$, $(-1, 1)$. To classify them, we need to compute the Hessian at each of these points. As

$$\nabla^2 f(x, y) = \begin{pmatrix} 12x^2 - 2 & 2 \\ 2 & 12y^2 - 2 \end{pmatrix}$$

we obtain that $(0, 0)$ is an indeterminate critical point but it is easy, consider $\varepsilon \mapsto f(\varepsilon, -\varepsilon)$, to see that $(0, 0)$ is not a local minimiser. Observing that $f(x, -y) = f(-x, y)$ and the fact that f has a local minimiser, we deduce that $(-1, 1)$ and $(1, -1)$ are local (and thus, global) minimisers of f .

- Exercise 4.16** (A constrained optimisation problem). (2 points) We consider the set

$$K := \{(x, y) \in \mathbb{R}^2, x^2 + y^2 = 16\}$$

and the point

$$(x_0, y_0) := (2, 1).$$

Solve the optimisation problem

$$\inf_{(x,y) \in K} \|(x_0, y_0) - (x, y)\|.$$

Solution 4.16. As K is compact, the existence of a minimiser follows from the Weierstraßtheorem. Introduce (for the sake of convenience we square the objective function) the function

$$f : (x, y) \mapsto (x - 2)^2 + (y - 1)^2$$

as well as the constraint

$$\varphi : (x, y) \mapsto x^2 + y^2 - 16.$$

As $\nabla\varphi(x, y) = 2(x, y)^T \neq 0$ for any $(x, y) \neq (0, 0)$ we can apply the Lagrange multiplier theorem: there exists λ such that

$$\nabla f(x^*, y^*) + \lambda \nabla \varphi(x^*, y^*) = 0.$$

This rewrites

$$2(x^* - 2) + 2\lambda x^* = 0, \quad 2(y^* - 1) + 2\lambda y^* = 0.$$

If $\lambda = 0$ then $x^* = 2, y^* = 1$, which contradicts the constraint $\varphi(x^*, y^*) = 0$. Thus, $\lambda \neq 0$ and we deduce that $x^*(1 + \lambda) = 2, y^*(1 + \lambda) = 1$. It follows that $(1 + \lambda) \neq 0$, and that

$$x^* = \frac{2}{1 + \lambda}, \quad y^* = \frac{1}{1 + \lambda}.$$

From the constraint

$$\varphi(x^*, y^*) = 0$$

we deduce that

$$5 = (4(1 + \lambda))^2$$

whence

$$1 + \lambda = \pm \frac{\sqrt{5}}{4}.$$

As

$$f(x^*, y^*) = 2 \frac{\lambda^2}{(1 + \lambda)^2}$$

we deduce that at a minimiser we have

$$\lambda = -1 + \frac{\sqrt{5}}{4},$$

and this allows to conclude.

Problem 1 (The projected gradient algorithm). We let K be a compact convex subset of $\mathbb{R}^d$ and we let $f \in \mathcal{C}^2(\mathbb{R}^d; \mathbb{R})$. We let Π_K be the orthogonal projection onto K .

(1) Reminders on the orthogonal projection

(a) (1 point) We recall that Π_K is defined as

$\Pi_K : \mathbb{R}^d \ni x \mapsto \Pi_K(x)$ defined as the unique minimiser of $y \in K \|y - x\|$.

Show that Π_K is well-defined (i.e. that the point solving $\min_{y \in K} \|x - y\|$ exists and is unique) and that it is characterised by the variational inequality

$$\forall y \in K, \langle x - \Pi_K(x), y - \Pi_K(x) \rangle \leq 0.$$

(b) (1 point) Show that Π_K is 1-Lipschitz, meaning that

$$\forall x, y, \|\Pi_K(x) - \Pi_K(y)\| \leq \|x - y\|.$$

We now study the optimisation problem

$$(12.11) \quad \min_{x \in K} f(x).$$

We define the projected gradient algorithm as follows: let $\tau > 0$ be given, let $x_0 \in K$ and, for any $k \in \mathbb{N}$, define

$$x_{k+1} := \Pi_K(x_k - \tau \nabla f(x_k)).$$

(2) The strongly convex case In this part, we assume that f has a μ -Lipschitz gradient and that f is α -strongly convex.

(a) (1 point) Show that (12.11) has a unique solution $x^* \in K$.

(b) (2,5 points) Show that

$$\forall k \in \mathbb{N}, \|x_{k+1} - x_k\|^2 \leq (1 + \tau^2 \mu - 2\tau \alpha) \|x_k - x_{k-1}\|^2.$$

(c) (3 points) Show that the series

$$\sum_{k=0}^{\infty} \|x_{k+1} - x_k\|$$

converges, and conclude that the sequence $\{x_k\}_{k \in \mathbb{N}}$ converges. Does it converge to a minimiser?

(3) The convex case We still assume that f has a μ -Lipschitz gradient and that it is convex (but not necessarily α -strongly convex). We want to show that

$$f(x_k) \xrightarrow{k \rightarrow \infty} \min_{x \in K} f(x).$$

(a) (1,5 points) Show that x_{k+1} solves

$$\min_{x \in K} \left(\langle \nabla f(x_k), x - x_k \rangle + \frac{1}{2\tau} \|x - x_k\|^2 \right)$$

and deduce that

$$\forall x' \in K, \left\langle \nabla f(x_k) + \frac{1}{\tau} (x_{k+1} - x_k), x' - x_{k+1} \right\rangle \geq 0.$$

(b) (1 point) We define $y_k := x_k - x_{k+1}$. Show that whenever $\tau \in \left(0; \frac{1}{\mu}\right)$

$$f(x_{k+1}) - f(x_k) \leq -\frac{1}{2\tau} \|y_k\|^2.$$

(c) (1 point) Let x^* be a minimiser of f in K . Show that for any $k \in \mathbb{N}$ there holds

$$f(x_{k+1}) - f(x^*) \leq f(x_{k+1}) - f(x_k) - \langle \nabla f(x_k), x_{k+1} - x_k \rangle + \langle \nabla f(x_k), x_{k+1} - x^* \rangle.$$

(d) (1 point) Show that

$$\langle \nabla f(x_k), x_{k+1} - x^* \rangle \leq \frac{1}{2\tau} (\|x_k - x^*\|^2 - \|x_{k+1} - x^*\|^2 - \|x_{k+1} - x_k\|^2).$$

(e) (2 points) Conclude that

$$f(x_{k+1}) - f(x^*) \leq \frac{1}{2\tau} (\|x_k - x^*\|^2 - \|x_{k+1} - x^*\|^2).$$

(f) (1 point) Conclude.

(g) (2 points) Assume that f is, in addition, strictly convex. Show that $\{x_k\}_{k \in \mathbb{N}}$ converges to x^* .

Solution 4.17. (1) Reminders on the orthogonal projection This was done in the first exercise sheet.

(2) The strongly convex case In this part, we assume that f has a μ -Lipschitz gradient and that f is α -strongly convex.

(a) (1 point) The existence of a solution follows from the compactness of K . Regarding the uniqueness of a solution, argue by contradiction and assume that there are two distinct solutions x_1, x_2 . Consider the map

$$g : t \mapsto f((1-t)x_1 + tx_2).$$

By convexity of K , for any $t \in [0; 1]$ we have $(1-t)x_1 + tx_2 \in K$ and so, by differentiating g we obtain

$$g'(0) \geq 0, g'(1) \leq 0.$$

As g is strictly convex, this entails that g is constant. However,

$$g''(t) = \langle \nabla^2 f((1-t)x_1 + tx_2)(x_2 - x_1), x_2 - x_1 \rangle > 0$$

by strong convexity. The conclusion follows.

(b) (2,5 points) We proceed as in the lecture notes. Recall that Π_K is a 1-Lipschitz function. Consequently, for any $k \in \mathbb{N}$,

$$\begin{aligned} \|x_{k+1} - x_k\|^2 &\leq \|(x_k - x_{k-1} + \tau(\nabla f(x_{k-1}) - \nabla f(x_k)))\|^2 \\ &= \|x_k - x_{k-1}\|^2 + \tau^2 \|\nabla f(x_k) - \nabla f(x_{k-1})\|^2 \\ &\quad - 2\tau \langle x_k - x_{k-1}, \nabla f(x_k) - \nabla f(x_{k-1}) \rangle \\ &\leq \|x_k - x_{k-1}\|^2 (1 + \tau^2 \mu^2 - 2\tau\alpha). \end{aligned}$$

(c) We now assume that

$$(12.12) \quad \alpha(\tau)^2 = |1 + \tau^2 \mu^2 - 2\tau\alpha| < 1.$$

Iterating the previous estimate we deduce that for any $k \in \mathbb{N}$ we have

$$\|x_{k+1} - x_k\| \leq \|x_1 - x_0\| \cdot \alpha(\tau)^k.$$

We thus deduce the absolute convergence of the series and it follows that $\{x_k\}_{k \in \mathbb{N}}$ converges. Let x_∞ be its limit, and let us show that x_∞ solves (12.11). First note that

$$x_\infty = \Pi_K(x_\infty - \tau \nabla f(x_\infty)).$$

Then observe that by convexity of f we have

$$\forall y \in K, \langle \nabla f(x_\infty), y - x_\infty \rangle \leq f(y) - f(x_\infty)$$

while, by property of the orthogonal projection

$$\langle x_\infty - \tau \nabla f(x_\infty) - \Pi_K(x_\infty - \tau \nabla f(x_\infty)), y - \Pi_K(x_\infty - \tau \nabla f(x_\infty)) \rangle \leq 0.$$

This suffices to establish that $f(y) \geq f(x_\infty)$ and, by uniqueness, that x_∞ is the unique solution to (12.11).

- (3) The convex case We still assume that f has a μ -Lipschitz gradient and that it is convex (but not necessarily α -strongly convex). We want to show that

$$f(x_k) \xrightarrow[k \rightarrow \infty]{} \min_{x \in K} f(x).$$

- (a) One needs to complete the square. Indeed,

$$\begin{aligned} \langle \nabla f(x_k), x - x_k \rangle + \frac{1}{2\tau} \|x - x_k\|^2 &= \frac{1}{2\tau} (\|x - x_k\|^2 + 2\tau \langle \nabla f(x_k), x - x_k \rangle) \\ &= \frac{1}{2\tau} (\|x - x_k + \tau \nabla f(x_k)\|^2 - \tau^2 \|\nabla f(x_k)\|^2). \end{aligned}$$

The definition of the orthogonal projection yields the desired result. The last inequality is the optimality condition at x_{k+1} : consider any $x' \in K$ and let $x_t := (1-t)x_{k+1} + tx'$. Define

$$\varphi : t \mapsto \langle \nabla f(x_k), x_t - x_{k+1} \rangle + \frac{1}{2\tau} \|x_t - x_k\|^2.$$

We have $\varphi'(0) \geq 0$ whence

$$\langle \nabla f(x_k), x' - x_{k+1} \rangle + \frac{1}{\tau} \langle x' - x_{k+1}, x_{k+1} - x_k \rangle \geq 0.$$

- (b) By standard estimates, we have

$$\begin{aligned} f(x_{k+1}) - f(x_k) &\leq \langle \nabla f(x_k), x_{k+1} - x_k \rangle + \frac{\mu}{2} \|x_{k+1} - x_k\|^2 \\ &\leq \langle \nabla f(x_k), x_{k+1} - x_k \rangle + \frac{1}{2\tau} \|x_{k+1} - x_k\|^2 \\ &\leq \langle \nabla f(x_k), x_{k+1} - x_k \rangle + \frac{1}{\tau} \|x_{k+1} - x_k\|^2 - \frac{1}{2\tau} \|x_{k+1} - x_k\|^2 \\ &\leq \frac{1}{\tau} \langle \tau \nabla f(x_k) + x_{k+1} - x_k, x_{k+1} - x_k \rangle - \frac{1}{2\tau} \|x_{k+1} - x_k\|^2. \end{aligned}$$

Introduce $z := x_k - \tau \nabla f(x_k)$ and observe that

$$\begin{aligned} \langle \tau \nabla f(x_k) + x_{k+1} - x_k, x_{k+1} - x_k \rangle &= \langle \tau \nabla f(x_k) + \Pi_K(z) - x_k, \Pi_K(z) - x_k \rangle \\ &= \langle \Pi_K(z) - z, \Pi_K(z) - x_k \rangle \\ &\leq 0 \end{aligned}$$

by the fundamental property of orthogonal projections. This concludes the proof.

- (c) Since f is convex, we have

$$\begin{aligned} f(x_{k+1}) - f(x^*) &\leq f(x_{k+1}) - f(x_k) - \langle \nabla f(x_k), x^* - x_k \rangle \\ &\leq f(x_{k+1}) - f(x_k) - \langle \nabla f(x_k), x_{k+1} - x_k \rangle - \langle \nabla f(x_k), x^* - x_{k+1} \rangle. \end{aligned}$$

- (d) Observe that from the first question

$$\begin{aligned} \langle \nabla f(x_k), x_{k+1} - x^* \rangle &\leq \frac{1}{\tau} \langle x_{k+1} - x_k, x^* - x_{k+1} \rangle \\ &= \frac{1}{2\tau} (\|x_k - x^*\|^2 - \|x_{k+1} - x_k\|^2 - \|x_{k+1} - x^*\|^2). \end{aligned}$$

(e) *All the previous estimates lead to*

$$\begin{aligned} f(x_{k+1}) - f(x^*) &\leq f(x_{k+1}) - f(x_k) - \langle \nabla f(x_k), x_{k+1} - x_k \rangle - \frac{1}{2\tau} \|x_{k+1} - x_k\|^2 \\ &\quad + \frac{1}{2\tau} (\|x_k - x^*\|^2 - \|x_{k+1} - x^*\|^2) \\ &\leq \frac{1}{2\tau} (\|x_k - x^*\|^2 - \|x_{k+1} - x^*\|^2). \end{aligned}$$

(f) *We observe that the sequence $\{\|x_k - x^*\|^2\}_{k \in \mathbb{N}}$ is non-increasing, and thus converges, which implies the convergence of $\{f(x_k)\}_{k \in \mathbb{N}}$ to $f(x^*)$.*

(g) *As f is strictly convex, it has at most one minimiser and, from the previous questions, any closure point x_∞ of $\{x_k\}_{k \in \mathbb{N}}$ is a minimiser. Consequently, the sequence has a unique closure point, and thus converges.*

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